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Mortality Analysis Of Patients With Sepsis Associated Acute Kidney Injury (SAAKI) Cohort from MIMIC-IV dataset

A Thesis

Submitted by

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THESIS CERTIFICATE

This is to undertake that the Thesis titled **MORTALITY ANALYSIS OF PATIENTS WITH SEPSIS ASSOCIATED ACUTE KIDNEY INJURY (SAAKI) COHORT FROM MIMIC-IV DATASET**, submitted by me to the Indian Institute of Technology Madras, for the award of **Master Of Technology in Industrial Artificial Intelligence**, is a bona fide record of the research work done by me under the supervision of **Debopam Nanda & Samyabrata Chakraborty**. The contents of this Thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

KEYWORDS sepsis-associated acute kidney injury (SA–AKI); sepsis-3; KDIGO; early warning; risk stratification; leakage-aware modeling; calibration; SHAP explanations; decision-curve analysis; survival analysis; MIMIC–IV; UMLS; RxNorm; SNOMED CT; ATC

This thesis addresses a critical, time-sensitive clinical question: at the end of the first ICU day (T0–T24), whether patients with sepsis-associated acute kidney injury (SA–AKI) who are at the highest risk of in-hospital death can be reliably identified, and whether the reasoning behind such predictions can be made transparent. Previous SA–AKI models, while often reporting strong discrimination, have been found to suffer from methodological issues that limit their clinical utility, including temporal leakage from the use of post-baseline data, global fitting of preprocessing steps, and a lack of model calibration. To overcome these limitations, a leakage-averse, TRIPOD-aligned framework was developed. An ontology-integrated data warehouse was built using PostgreSQL to harmonize the MIMIC–IV dataset with medical terminologies (UMLS, RxNorm, SNOMED CT, ATC). The SA–AKI cohort was rigorously defined using Sepsis–3 and KDIGO criteria, with all predictors strictly confined to the T0–T24 window. A one-time 80/20 stratified split ensured that all data-dependent steps, including a novel approach to creating and selecting significant missingness indicators and performing median imputation, were fitted only on training data. The methodology encompassed three stages: (i) an extensive exploratory data analysis on the training set to identify clinical phenotypes; (ii) supervised mortality prediction using a benchmark of models, from which a LightGBM classifier was selected, tuned with Optuna, and calibrated using an isotonic regressor on a validation set; and (iii) survival analysis to model time-to-death using the same early covariates with Cox Proportional Hazards and a custom DeepSurv model. On the held-out test set, the final calibrated LightGBM model achieved an AUROC of **0.749** and an F1-score of 0.522, indicating reliable discriminative ability.

The survival analysis yielded C-indexes of 0.69 (Cox) and 0.69 (DeepSurv). The primary contribution of this work is a reproducible, timing-faithful pipeline that delivers a calibrated, interpretable risk score and time-to-event estimates, positioning it as a credible tool for bedside clinical decision support.

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CHAPTER 1

INTRODUCTION

1.1 CLINICAL BACKGROUND AND MOTIVATION

Sepsis-associated acute kidney injury (SA–AKI) has emerged as one of the most consequential syndromes in critical care because it couples the systemic instability of sepsis with acute loss of renal function and thereby compounds risk across organ systems. For clinical and research purposes SA–AKI is commonly labeled when acute kidney injury develops within seven days of sepsis onset so that kidney dysfunction can be attributed to the septic process rather than to unrelated peri-ICU events (Peerapornratana *et al.*, 2019). The constituent conditions are defined by international consensus. Sepsis is defined by Sepsis–3 as life-threatening organ dysfunction due to a dysregulated host response to infection and is operationalized by an acute rise in SOFA score of two points or more (Singer *et al.*, 2016). Acute kidney injury is defined and staged by KDIGO using serum creatinine and urine output thresholds, including an absolute creatinine increase of at least 0.3 mg/dL within forty-eight hours, a relative increase of at least $1.5\times$ baseline within seven days, or oliguria at or below 0.5 mL/kg/h for six hours (Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012). Within intensive care units SA–AKI has been reported to account for a large share of AKI cases and to be associated with sharply higher in-hospital mortality, longer length of stay, and increased risk of subsequent chronic kidney disease in survivors (Poston and Koyner, 2019; Peerapornratana *et al.*, 2019). These epidemiological features have kept early risk stratification at the forefront of research because timely identification of clinical deterioration is central to appropriate escalation, nephroprotective strategies, and resource planning.

1.2 WHAT THE LITERATURE HAS TRIED AND WHERE IT FALLS SHORT

Risk assessment in the ICU has traditionally relied on general severity scores such as SOFA, SAPS II, and APACHE II. These scores have offered useful global snapshots of illness severity but they tend to show only fair discrimination when the target is SA–AKI mortality, and they do not represent the nonlinear and interacting physiology that links infection biology, hemodynamics, renal injury, and treatment exposure in this population (Luo *et al.*, 2022; Nong *et al.*, 2024). In recent years the availability of large electronic health record repositories, notably MIMIC–IV, has encouraged the development of machine learning models that can take advantage of richer feature spaces and flexible function classes (Johnson *et al.*, 2023). Tree-based ensembles such as extreme gradient boosting frequently outperform baseline scores in internal testing and AUROC values near or above 0.80 are often reported (Chen *et al.*, 2025; Nong *et al.*, 2024; Wang *et al.*, 2024). Nevertheless, several methodological patterns recur across studies and create uncertainty about the true clinical utility of published models. Temporal anchoring of predictors is sometimes vague and whole-stay aggregation has been used, which allows future information to flow into features that are later presented as early predictors. Proxy variables for time to event, including length of stay, have been included as inputs to mortality models, which introduces a direct leakage channel from outcome process to covariates and inflates apparent discrimination (Chen *et al.*, 2025). Data-dependent steps such as feature selection and imputation have sometimes been performed across the full dataset rather than being learned on development folds and then applied to held-out data, which again invites optimistic estimates (Kang and Yoon, 2023). Class imbalance has often been handled through synthetic oversampling, including SMOTE and ADASYN, without careful assessment of whether the generated examples concentrate in overlapped or noisy regions or near outliers where decision boundaries are unstable (Chawla *et al.*, 2002; He *et al.*, 2008; Blagus and Lusa, 2013; Sáez *et al.*, 2015; Krawczyk, 2016). Calibration and principled threshold selection have been inconsistently reported, which limits bedside interpretability even when AUROC values appear strong. Finally, most model development has relied on data from a single tertiary-care center, and performance losses under transport to other hospitals and populations have been documented, which tempers claims of broad generalizability (Johnson *et al.*, 2023; Rockenschaub *et al.*, 2024).

1.3 PROBLEM FOCUS AND RESEARCH OBJECTIVES

The central challenge addressed in this thesis is the chasm between the promising internal accuracy reported in machine learning literature and the practical bedside utility of predictive models for SA–AKI mortality. This gap is primarily shaped by two issues: first, a failure to align predictive pipelines with the real-world clinical decision timeline, and second, a lack of transparency and calibration that hinders a model's adoption. The objective of this work is to develop and validate a mortality risk stratification framework for SA–AKI that is methodologically sound, clinically relevant, and transparent. The work is conducted using the MIMIC–IV dataset (Johnson *et al.*, 2023), with a strict focus on an early clinical horizon to ensure the model's predictions are actionable. The specific aims are:

1. To perform a systematic exploratory data analysis to identify key clinical phenotypes and risk factors within the training cohort.
2. To develop and validate a well-calibrated LightGBM classification model for in-hospital mortality using only features available within the first 24 hours of ICU admission.
3. To ensure model transparency by providing both global and local feature-level explanations using SHapley Additive exPlanations (SHAP) (Lundberg and Lee, 2017).
4. To complement the binary prediction with a dynamic risk assessment by developing survival models using Cox Proportional Hazards and DeepSurv to estimate the *time to in-hospital death*.

1.4 PROPOSED APPROACH AND EVALUATION PLAN

The approach adopted in this thesis begins with the disciplined construction of an SA–AKI cohort according to Sepsis–3 and KDIGO criteria, with all predictors strictly limited to the first 24 hours (T0–T24) of an ICU stay (Singer *et al.*, 2016; Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012). This work is distinguished by a leakage-aware preprocessing pipeline where all steps were fitted exclusively on training data. This includes a formal test for the missing data mechanism, followed by the creation and statistical selection of informative missingness indicators and median imputation. A broad benchmark of machine learning models was conducted, leading to the selection of a final LightGBM model based on its superior performance. This model was tuned using the Optuna framework, explicitly calibrated using isotonic regression, and an optimal decision threshold was determined via Youden's J statistic on a validation set. Explanations for the final model's predictions were generated using SHAP to ensure transparency. Finally, to provide a dynamic view of risk over time, two survival models, a traditional Cox Proportional Hazards model and a neural network-based DeepSurv model, were developed and evaluated using the same early-window covariates. The evaluation plan prioritizes methodological transparency and clinical relevance, reporting a suite of metrics including AUROC, F1-score, Harrell's C-index, and calibration plots on a strictly held-out test set, thereby addressing the common pitfalls identified in the literature (Kang and Yoon, 2023; Chen *et al.*, 2025).

1.5 CHAPTER ROADMAP

The remainder of the thesis is organized to follow this arc from clinical framing to deployment-oriented evaluation. Chapter 2 reviews the background in SA–AKI epidemiology and definitions, surveys conventional scores and machine learning approaches, and synthesizes methodological lessons from recent studies. Chapter 3 states the formal problem and gives the mathematical formulation for the binary prediction and survival analysis tasks. Chapter 4 provides a detailed description of the cohort construction, the leakage-aware preprocessing pipeline, the model development and calibration process, and the evaluation protocol. Chapter 5 presents the findings from the extensive exploratory data analysis of the training cohort, identifying key clinical phenotypes and variable distributions that inform the modeling process. Chapter 6 presents the final performance results for the classification and survival models on the held-out test set, including discrimination, calibration, and SHAP-based interpretations. Finally, Chapter 7 concludes with a summary of findings, comments on clinical implications, and outlines future research directions.

CHAPTER 2

LITERATURE REVIEW

2.1 CLINICAL DEFINITIONS AND EPIDEMIOLOGICAL CONTEXT

Sepsis-associated acute kidney injury (SA–AKI) has been characterized as the concurrence of sepsis and acute kidney injury with persistently poor outcomes. Sepsis has been defined by the Sepsis–3 consensus as life-threatening organ dysfunction arising from a dysregulated host response to infection, and the construct has been operationalized as suspected or documented infection together with an acute rise in organ failure, commonly a SOFA increase of at least two points (Singer *et al.*, 2016). Acute kidney injury has been defined by KDIGO through serum creatinine and urine output thresholds, in particular an absolute creatinine increase of at least 0.3 mg/dL within forty-eight hours, or at least 1.5× baseline within seven days, or urine output of 0.5 mL/kg/h or less for six hours (Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012). Many studies have labeled SA–AKI when AKI develops within a short interval after sepsis onset, frequently within seven days, to attribute renal dysfunction to the septic process rather than to competing causes (Peerapornratana *et al.*, 2019; Poston and Koyner, 2019). The shift toward Sepsis–3 and KDIGO has reduced heterogeneity relative to older criteria and has improved comparability across cohorts (Singer *et al.*, 2016; Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012).

The burden of SA–AKI in intensive care units is substantial. AKI develops in roughly fifty to sixty percent of septic ICU patients, and sepsis accounts for about forty to seventy percent of AKI in that setting (Peerapornratana *et al.*, 2019; Poston and Koyner, 2019). When AKI complicates sepsis, marked increases in in-hospital mortality follow, with earlier analyses suggesting a six to eight fold rise compared with sepsis without AKI and mortality rates commonly between forty and sixty percent (Poston and Koyner, 2019; Peerapornratana *et al.*, 2019). Prolonged ICU stays and an elevated risk of subsequent chronic kidney disease have also been recorded among survivors (Poston and Koyner, 2019; Peerapornratana *et al.*, 2019). These epidemiological patterns have motivated the development of early risk stratification tools.

2.2 CONVENTIONAL SEVERITY INDICES AND EARLY WARNING SYSTEMS

General ICU indices such as SOFA, SAPS II, and APACHE II have been evaluated for SA–AKI outcomes and tend to show only modest discrimination for short-term mortality or kidney-specific endpoints in this subgroup (Luo *et al.*, 2022; Li *et al.*, 2023; Nong *et al.*, 2024). Classical regression frameworks, including logistic regression and Cox models, have been employed repeatedly, yet difficulties persist in capturing the nonlinear and interacting relationships among infectious physiology, hemodynamics, renal injury, and treatment exposures. Performance has often failed to transport reliably across hospitals as a result (Kang and Yoon, 2023; Luo *et al.*, 2022; Li *et al.*, 2023). Earlier variability in sepsis and AKI definitions compounded the challenge by complicating cohort selection and cross-study comparison prior to widespread adoption of Sepsis–3 and KDIGO (Singer *et al.*, 2016; Kidney Disease: Improving Global Outcomes (KDIGO) Acute Kidney Injury Work Group, 2012).

2.3 DATASETS AND COHORT CONSTRUCTION IN SA–AKI MODELING

Large electronic health record repositories have supported the development of prognostic models for SA–AKI, with MIMIC–IV serving as the most common source for cohort construction and feature derivation (Johnson *et al.*, 2023). Studies typically restrict analysis to the first ICU admission per patient, exclude patients younger than eighteen years, and remove ICU stays shorter than forty-eight hours. A central design decision involves the temporal window for feature construction. Fixed early windows, for example the first twenty-four or forty-eight hours after admission or sepsis recognition, support timeliness, whereas whole-stay aggregation from admission to discharge has also appeared frequently and creates a risk that future information relative to an intended decision point enters the model inputs.

2.4 FEATURE ENGINEERING, SELECTION, AND LEAKAGE RISKS

The handling of high-dimensional clinical variables has relied on screening by correlation or variance inflation factor, wrapper methods such as recursive feature elimination, and embedded strategies that exploit model-based importance estimates. It is common to find feature selection steps that were informed by the full dataset. When selection proceeds in this manner, the final predictor set is influenced by outcomes from records that are later evaluated, and optimism in reported discrimination is encouraged (Kang and Yoon, 2023). Claims that VIF-based pruning enhances stability and generalizability have also appeared without pre–post diagnostics or external tests, and the relevance of VIF is limited for classes of models that tolerate correlated predictors. In several studies, length of stay was included among features for mortality prediction. Because time to event is strongly associated with whether death occurred during the admission, that inclusion acts as a conduit for post-baseline information and dominates other covariates during training. Redundancy among variables has also been noted. For example, total urine output and average urine output become scalar multiples under a fixed observation horizon, and diverging importance profiles for those variables suggest that non-fixed or whole-stay windows were used, which reduces alignment with early prediction. Dependencies between creatinine and derived eGFR similarly complicate interpretation when both are included without a clear rationale.

2.5 MISSING DATA MECHANISMS AND IMPUTATION CHOICES

The mechanisms that govern missingness have rarely been examined in depth in this literature. Assumptions that data were missing completely at random or missing at random are often stated without empirical assessment, and simple percentage-based rules trigger imputation. Mean imputation for continuous variables remains a common choice even for biomarkers that show marked skew, such as serum creatinine, which is right skewed and well approximated by a log-normal distribution (Haeckel and Wosniok, 2010). Multiple imputation by chained equations and random forest variants is widely used, and in many cases those imputation models are fitted across the entire dataset prior to train–test separation. Because chained equations use all observed variables to impute each target variable, global fitting allows information from records later treated as held out to influence imputed values, and optimistic estimates of model performance follow.

2.6 CLASS IMBALANCE, EVALUATION STRATEGY, AND THRESHOLDING

Mortality among SA–AKI cohorts is imbalanced, and class imbalance is most often handled by oversampling techniques, including SMOTE and ADASYN (Chawla *et al.*, 2002; He *et al.*, 2008). The two methods have occasionally been applied together even though ADASYN is itself a variant that emphasizes synthesis in hard regions of the feature space. SMOTE may generate synthetic points in overlapped or noisy areas, and performance can degrade in high-dimensional settings unless additional filtering is applied (Blagus and Lusa, 2013; Sáez *et al.*, 2015). Concentration of synthesis near low-density or borderline regions by ADASYN can shift decision boundaries around outliers and may harm stability under severe imbalance or distribution shift (Krawczyk, 2016). Comparisons with alternative imbalance strategies, for instance through intrinsic class weighting or hybrid sampling, have not always been provided. Evaluation strategies have often relied on a single split of the data into training and test sets without a separate validation split, and explicit calibration analyses and principled threshold selection have been omitted in several reports.

2.7 MODEL CLASSES, TUNING, CALIBRATION, AND INTERPRETABILITY

Tree-based ensembles, notably extreme gradient boosting, have been adopted widely in this domain to represent nonlinearities and interactions among clinical variables, and stronger internal discrimination than baseline ICU scores is often reported (Chen and Guestrin, 2016; Luo *et al.*, 2022; Li *et al.*, 2023; Nong *et al.*, 2024; Wang *et al.*, 2024; Chen *et al.*, 2025; Jiang *et al.*, 2025). Hyperparameter tuning is usually performed through grid search cross validation, and claims of optimality appear despite the presence of complex, nonconvex search spaces in which only local minima are guaranteed by such procedures. Calibration is reported inconsistently, and explanations based on Shapley values are used to describe feature contributions, although stability of such explanations is not always explored. In several comparisons, confidence intervals for ensembles and penalized linear models overlap, and differences attributed to model class are entangled with choices about temporal windows, selection scope, and imputation design.

2.8 EXTERNAL VALIDITY, TRANSPORTABILITY, AND DEPLOYMENT TIMING

Generalization beyond the MIMIC–IV development site remains a consistent concern. MIMIC–IV reflects a single tertiary-care center in Boston with local care patterns and population structure that are not broadly representative (Johnson *et al.*, 2023). External evaluations of ICU prediction models show loss of accuracy when transported across hospitals, pointing to center-specific differences that matter for performance (Rockenschaub *et al.*, 2024). Assertions about applicability to diverse populations therefore require cautious interpretation when external testing is absent. In addition, the timing of features relative to the intended decision point is often unspecified, and whole-stay aggregation creates ambiguity about whether a model could power actionable alerts at twenty-four or forty-eight hours.

2.9 DETAILED APPRAISAL OF REPRESENTATIVE STUDIES

2.9.1 Chen et al. (2025): SA–AKI mortality in MIMIC–IV

A recent contribution by Chen et al. reported an AUROC of 0.878 for mortality prediction in a SA–AKI cohort drawn from MIMIC–IV, and the result was presented as an improvement over the AUROC 0.794 attributed to Li et al. (Chen *et al.*, 2025; Li *et al.*, 2023). The cohort retained only the first ICU admission per patient and excluded patients younger than eighteen years and ICU stays shorter than forty-eight hours. The modeling pipeline combined VIF-based filtering with recursive feature elimination to produce a set of twenty-four predictors, after which several variables were reintroduced as expert inputs without quantitative justification. Because selection was described as having been performed on the entire dataset, the final predictor set was influenced by outcomes from records later treated as held out and the risk of information leakage was created (Kang and Yoon, 2023). A claim of generalizability across diverse populations was made, yet MIMIC–IV represents a single center and patient mix, and performance drift across hospitals has been documented (Johnson *et al.*, 2023; Rockenschaub *et al.*, 2024). The introduction criticized Li et al. for relying on forty-four variables on the grounds that the set might be too narrow to capture the complexity of SA–AKI, although the final model by Chen et al. used an even smaller set of twenty-four features, which created an internal inconsistency in the narrative (Chen *et al.*, 2025; Li *et al.*, 2023). Hyperparameters were tuned with grid search cross validation and described as optimal, although such procedures identify settings within a discrete grid and do not ensure global optimality in nonconvex spaces. Claims of timely, accurate, and actionable interventions were advanced, yet the temporal window for aggregating time-varying biomarkers was not specified. Because Sepsis–3 and KDIGO anchor observation to early horizons, an early extraction window would have been expected for actionable deployment; multiple signals suggested that whole-stay aggregation may have been used, which would introduce future information relative to early prediction. Missing values were imputed by means that were tied to percentage thresholds, and mean imputation was applied for variables with up to twenty percent missingness. This approach is not well aligned with skewed distributions such as serum creatinine, which has been described as right skewed and is well modeled by a log-normal distribution (Haeckel and Wosniok, 2010). Missingness indicators were not created and the mechanism of missing data was assumed rather than studied. VIF filtering was described as enhancing stability and generalizability without pre–post diagnostics or external assessment and has limited relevance for ensembles that tolerate correlated predictors. The base estimator used within recursive feature elimination was not stated, and a linear estimator would impose a separability assumption during selection. Length of stay was included among predictors, and this inclusion introduced a strong proxy for time to event that is not available prospectively for early prediction. The simultaneous inclusion of total and average urine output suggested redundancy if a fixed window had been used, and differences in importance between the two variables implied that a non-fixed or whole-stay window was applied. The evaluation strategy used a seventy-five to twenty-five train–test split without a separate validation split, and explicit calibration and threshold selection were not described. Class imbalance was handled by applying SMOTE and ADASYN together, a combination that lacks a clear rationale and that may synthesize examples in overlapped or noisy regions or near outliers and thereby distort decision boundaries (Chawla *et al.*, 2002; He *et al.*, 2008; Blagus and Lusa, 2013; Sáez *et al.*, 2015; Krawczyk, 2016). Univariate comparisons between survivors and a group termed non-survivors were presented with t tests even though laboratory variables such as creatinine generally deviate from normality (Haeckel and Wosniok, 2010). The reported confidence intervals for ensemble and linear models overlapped, with AUROC 0.878 (95% CI 0.859 to 0.897) for extreme gradient boosting and AUROC 0.849 (95% CI 0.824 to 0.873) for logistic regression, which supported the interpretation that any advantage might be smaller than suggested once temporal alignment and leakage are addressed (Chen *et al.*, 2025).

2.9.2 Li et al. (2023): SA–AKI mortality in MIMIC–IV

Li et al. assembled a SA–AKI cohort from MIMIC–IV by retaining only the first ICU admission and excluding ICU stays shorter than forty-eight hours. Features were extracted from the first twenty-four hours and were aggregated using maximum values, which provided a clear early window for model inputs (Li *et al.*, 2023). Lasso-regularized logistic regression was used for feature selection, and selection was described across the full dataset rather than within development folds. Because selection then involved outcomes from records later used for evaluation, optimism in reported performance was encouraged. The linear framework also prioritized main effects and could miss interactions and subtle nonlinear patterns that tree ensembles tend to capture. Multiple imputation by chained equations with random forest implementations was applied globally across the dataset rather than learned on development data and then applied to evaluation data, and missingness indicators were not constructed. The data were split into training and validation in an eighty to twenty ratio, and a discrimination gap favoring extreme gradient boosting over logistic regression was reported with AUROC 0.794 versus 0.730, although the magnitude of this gap remained difficult to interpret in light of the selection and imputation scope (Li *et al.*, 2023).

2.9.3 Hu et al. (2021): survival analysis in MIMIC–III

A retrospective analysis by Hu et al. used MIMIC–III with smaller sample size and older coding structure to study SA–AKI outcomes (Hu *et al.*, 2021). The cohort retained only the first ICU admission and excluded patients younger than eighteen years and ICU stays shorter than forty-eight hours, and rows with more than five percent missing variables were removed. Lasso regression and Cox regression were used, and a C index of 0.72 was reported on a validation set. Biomarkers were aggregated by medians across the entire admission, therefore features included information that would not be available at an early decision point. Features with more than twenty percent missingness were dropped without investigation of mechanisms, and multiple imputation by chained equations was performed on the full dataset prior to splitting, which created leakage. The test set contained one hundred and two rows, and reliability of performance estimates was limited by that size. Ethnicity was used as a predictor, which raised concerns about representativeness and fairness for transportability. Survival models at seven, fourteen, and twenty-eight days yielded AUCs of 0.77, 0.73, and 0.71 respectively.

2.9.4 Roknaldin et al. (2024): sepsis to AKI prediction in MIMIC–III

A recent preprint constructed models that predict AKI among septic ICU patients using MIMIC–III (Roknaldin *et al.*, 2024). Adults aged eighteen to eighty-nine years with sepsis by ICD–9 codes were included after excluding multiple admissions, ICU stays shorter than forty-eight hours, and records with more than twenty percent missingness. Features from the first twenty-four hours covered demographics, comorbidities, vitals, laboratory results, and early interventions. Multiple imputation was used for missing values, a correlation filter retained twenty-three predictors, and class imbalance was addressed with SMOTE. Six algorithms were compared, and logistic regression achieved the strongest internal results with test AUC 0.887 and validation AUC 0.857 together with a Brier score near 0.13 and acceptable isotonic calibration. Top predictors included urine output, bilirubin measures, blood urea nitrogen, minimum eGFR, weight, and indicators of vasopressor use and mechanical ventilation. The transparency of the feature list and the early window supported interpretability, while several design choices encouraged optimism. Feature selection and SMOTE were described prior to the final split into development and evaluation, so that the set of predictors and synthetic examples were influenced by records that later appeared in evaluation. Multiple imputation also appeared to have been fitted globally. A fixed correlation threshold risked removal of variables that are weakly correlated but jointly informative, and the inclusion of minimum and maximum eGFR alongside creatinine introduced redundancy because eGFR is derived from creatinine and demographics. For descriptive group comparisons, t tests were used for markers that are typically skewed, and the allocation of half the data to validation and one tenth to testing left a small test set. Calibration procedures were not tied to a recalibration step that excluded the test cohort, which limited the strength of calibration claims on that set.

2.9.5 Chaudhary et al. (2020): phenotype discovery in MIMIC–III

An unsupervised study in *CJASN* sought to identify SA–AKI phenotypes rather than predict mortality directly (Chaudhary *et al.*, 2020). An autoencoder was used to learn representations that were then clustered into three subgroups with distinct twenty-eight day mortality and dialysis rates. Differences were driven by bilirubin, lactate, transaminases, and inflammatory counts, and the highest risk subgroup showed markedly higher dialysis use and death. The analysis relied on whole-stay trajectories of laboratory and vital signs, which embeds information not available at early decision times. The sepsis cohort was constructed by codes, which can misalign the timing of the septic insult, and external validation of cluster stability was not performed. The exploratory links between clusters and treatments were not framed as causal and would require dedicated designs to support treatment guidance.

2.9.6 Lin et al. (2025): progression to chronic kidney disease in MIMIC–IV

A study in *Informatics in Medicine Unlocked* examined progression from SA–AKI to incident chronic kidney disease using MIMIC–IV (Lin *et al.*, 2025). Extreme gradient boosting with SHAP explanations was trained on features collected during the first twenty-four hours, and high internal discrimination was reported. Thresholds for age, maximum creatinine, minimum hemoglobin, and lactate were described, and a feature stability analysis across AKI stages was presented. Although the early window and the inclusion of decision-curve analysis aided interpretability, variable screening by Lasso and stepwise logistic regression prior to final modeling, if performed across the full dataset, would have allowed outcomes from evaluation records to influence the screened set. Outcome assessment relied on follow up up to five years, and selection by loss to follow up as well as center-specific practice patterns could have influenced estimates. Thresholds based on SHAP dependence plots have known sensitivity to sampling variation, and replication would be expected for reliable adoption.

2.9.7 He et al. (2021): acute kidney disease after SA–AKI

An article in *Frontiers in Medicine* predicted subsequent acute kidney disease rather than in-hospital mortality (He *et al.*, 2021). A single-center development cohort was used, and external validation was undertaken in MIMIC–III. Recurrent neural networks with long short-term memory units were compared with a decision tree and logistic regression. The recurrent model achieved an internal AUROC of 1.00 and strong performance in the external cohort, and the change in nonrenal SOFA between day one and day three emerged as an important signal. The development cohort contained two hundred and nine patients, which is a modest sample for a deep architecture with more than two dozen features, and perfect discrimination in such data is difficult to reconcile with typical ICU noise and suggests that overfitting, simplified labels, or other forms of information leakage may have been present. The dynamic window implied by the day one to day three delta improves clinical timing, while the time of alert generation relative to clinical workflow was not specified. It was not always clear that imputation and preprocessing steps were confined to development data.

2.9.8 Kantola et al. (2025): admission procalcitonin and risk of AKI

A multicenter retrospective cohort across three hospitals evaluated admission procalcitonin for prediction of subsequent AKI and reported an odds ratio of approximately 1.01 per ng/mL with poor overall discrimination near ROC AUC 0.59 (Kantola *et al.*, 2025). The heterogeneity of practical cutoffs across prior work and the inconsistent relationships between procalcitonin levels and AKI severity were emphasized, and the conclusion favored limited standalone clinical utility for early risk stratification. Procalcitonin is not available in MIMIC–IV, which restricts replication in that database.

2.10 SYNTHESIS AND IMPLICATIONS FOR THIS THESIS

The literature reviewed above sketches a consistent picture in which window definition, the scope of data-dependent preprocessing, handling of missingness and imbalance, and the availability of external evidence play decisive roles in the credibility of reported scores. Studies that relied on whole-stay aggregation, on inclusion of length of stay or other post-baseline correlates, and on global feature selection or imputation tend to report stronger discrimination, yet those gains are paired with ambiguity about practical timing and with higher risk of optimism. When early windows are enforced clearly, performance remains competitive and the path to bedside use becomes clearer. Several reports present overlapping confidence intervals between ensembles and penalized linear models, which suggests that differences in pipeline design often overshadow differences in the learning algorithms themselves (Kang and Yoon, 2023). Generalization beyond a single tertiary-care center remains a key uncertainty, and drift across hospitals is documented (Johnson *et al.*, 2023; Rockenschaub *et al.*, 2024). In this thesis, these observations are treated as design constraints for methodological choices, and the subsequent chapters are organized to respond to the gaps identified here while keeping the emphasis on temporally anchored modeling, leakage-aware preprocessing, calibrated evaluation, and transparency about what information is and is not available at the moment when a clinical decision is intended to be supported.

CHAPTER 3

PROBLEM DEFINITION AND FORMULATION

3.1 CLINICAL DECISION AND SCOPE

Sepsis–associated acute kidney injury (SA–AKI) constitutes a high–risk state coupling infection, shock physiology, and acute renal dysfunction. The *decision point* is fixed: at the end of the first ICU day (T_{obs}=24 h after ICU admission), the risk of in–hospital death is to be estimated for patients meeting SA–AKI criteria using only information available by T_{obs}. The primary goal was to develop a predictive model that is both *discriminative* and *calibrated*. Two auxiliary objectives support this decision: (i) systematic *exploratory data analysis* to discover clinical phenotypes in the early feature space, and (ii) *time–to–event* modeling that translates the same early covariates into survival functions for *time to death*. All cohort rules adhere to Sepsis–3 for sepsis and KDIGO for AKI, and reporting follows the TRIPOD statement.

Out of scope. This study does not estimate causal treatment effects or recommend specific therapies. The problem is predictive; models summarize risk conditional on the information set $\mathcal{I}(T_{\text{obs}})$.

3.2 OBSERVATIONAL UNIT, CLOCKS, AND INFORMATION SETS

Each observational unit corresponds to the *first* eligible ICU admission for a patient in the SA–AKI cohort. Let ICU admission be t=0 and define the early feature window as

$$[0, T_{\text{obs}}], \qquad T_{\text{obs}} = 24 \text{ hours}.$$

All time–varying signals in this window—vital signs, laboratory measurements, fluids/urine output, and early therapies—are summarized by deterministic, leakage–safe functionals into a feature vector

$$x_i \in \mathbb{R}^p,$$

augmented with static demographics and comorbidity indicators. For the binary classification task, a one–time stratified 80/20 train–test split yielded training and test sets with $(8028, 162)$ and $(2008, 162)$ dimensions, respectively (including the target column). The positive (death) rate in the training set was 26.74%. The survival analysis used the same patient split with a feature matrix of $(8028, 124)$ and $(2008, 124)$ before the inclusion of time and event columns.[1] The available information for patient i at the decision time is the σ–field

$$\mathcal{I}_i(T_{\text{obs}}) = \{\text{EHR observations with time stamp} \leq T_{\text{obs}}\}.$$

Temporal discipline. Any statistic requiring $t > T_{\text{obs}}$ (e.g., total ICU length of stay, whole–stay extrema/means, post–baseline therapies) is deemed inadmissible for prediction and is excluded by a scripted leakage audit.

Target population and case definition. Sepsis is operationalized by Sepsis–3 (suspected infection plus a daily SOFA score increase of ≥ 2); AKI is defined by KDIGO serum–creatinine and urine–output criteria. SA–AKI requires AKI onset within ±48 h of sepsis onset. Only adults (≥18 y) with an ICU length of stay ≥24 h, and only the first ICU stay per patient, are included. Cohort definition timestamps t_{sepsis} and t_{aki} serve only to identify the cohort and *are not used as predictors*.

3.3 DATA–GENERATING VIEW AND NOTATION

Let $(X, Y, T^{\text{death}}, \delta^{\text{death}}) \sim P$ on

$$\mathcal{X} \times \{0, 1\} \times \mathbb{R}_+ \times \{0, 1\},$$

[1]These dataset sizes and event fractions reflect the final data preparation used in model building and survival analysis, as detailed in the notebooks.

where X is the T0–T24 feature vector; Y is in–hospital death (1/0); T^{death} is time from ICU admission to in–hospital death or censoring at discharge; and δ^{death} is the event indicator (1=death, 0=censored). Independent, identically distributed samples $\{(x_i, \cdot)\}_{i=1}^{n}$ are observed. A one–time *80/20* split yields $(\mathcal{D}_{\text{train}}, \mathcal{D}_{\text{test}})$. Any internal validation split for tasks like hyperparameter tuning or calibration is carved out exclusively *from within* $\mathcal{D}_{\text{train}}$. The survival analysis reuses the same patient split with right censoring.

3.4 PROBLEM 1 (UNSUPERVISED): EXPLORATORY ANALYSIS AND PHENOTYPING

Goal. To systematically explore the early feature space of the training data to identify clinically relevant patient phenotypes without using outcome labels. This involves analyzing the distributions of key biomarkers, assessing correlations between features and the mortality outcome, and comparing clinical characteristics (e.g., vital signs, lab values, SOFA components) between patients who survived and those who did not. This exploratory phase informs feature importance and provides clinical context for the supervised models.

3.5 PROBLEM 2 (SUPERVISED, PRIMARY): EARLY–WINDOW MORTALITY RISK

3.5.1 Label and objective

Define the binary label

$$Y = \mathbb{I}\{\text{in–hospital death prior to discharge}\}.$$

A probabilistic classifier $f_\theta : \mathcal{X} \to [0, 1]$ is sought such that

$$\hat{p}(x) = f_\theta(x) \approx \Pr(Y = 1 \mid X = x,\ \mathcal{I}(T_{\text{obs}})).$$

The learning criterion on training data is the average negative log–likelihood (cross-entropy),

$$\hat{\theta} \in \arg\min_{\theta} \frac{1}{m} \sum_{j=1}^{m} \big\{ -y_j \log f_\theta(x_j) - (1-y_j)\log(1-f_\theta(x_j)) \big\},$$

optimized using cross–validation within $\mathcal{D}_{\text{train}}$.

3.5.2 Performance targets

Two properties are co–primary:

(a) **Discrimination:** High Area Under the Receiver Operating Characteristic Curve (AUROC) and F1-score on the held-out test set.

(b) **Calibration:** Reliable probabilities, assessed by generating a final calibrated model and evaluating its performance.

Calibration is performed on a validation split drawn from $\mathcal{D}_{\text{train}}$ using isotonic regression; final probabilities are $\tilde{p} = c \circ f_\theta(x)$, where c denotes the calibrator.

3.5.3 Decision rule and operating point

Clinical actions are thresholded via $\delta_\tau(x) = \mathbb{I}\{\tilde{p}(x) \geq \tau\}$. The optimal operating point τ is selected on an internal validation split by maximizing Youden's J index. The final model's utility across a range of thresholds is also assessed using decision-curve analysis.

3.5.4 Hypothesis classes and regularization

A spectrum of models, $\mathcal{H}$, is benchmarked to assess performance across different functional capacities: Logistic Regression (with and without class weights, and with various resampling techniques), regularized models (Ridge, Lasso, ElasticNet), Linear and Quadratic Discriminant Analysis (LDA/QDA), Gaussian Naive Bayes, Decision Tree, Random Forest, gradient-boosted trees (XGBoost, LightGBM), and a deep neural network (DNN++). Hyperparameters for tree-based models are tuned using the Optuna framework within training folds.

3.5.5 Interpretability requirement

For the final selected model (f_θ), feature contributions are explained using SHapley Additive exPlanations (SHAP). This provides both global feature importance rankings and local, instance-level explanations for individual patient predictions, satisfying the need for model transparency.

Constraints and guardrails. (i) *Temporal:* All predictive features are derived strictly from the $[0, 24]$ h window post-ICU admission. (ii) *Independence:* Only the first ICU stay per patient is included. (iii) *Siloed preprocessing:* All data transformations (imputation, scaling, indicator creation) are learned on the training set and applied without modification to the test set. (iv) *No peeking:* The test set is held out and used only for a single final evaluation. (v) *Cohort times not used as predictors:* t_{sepsis} and t_{aki} are used only for case definition.

3.6 PROBLEM 3 (SURVIVAL): EARLY–COVARIATE TIME TO DEATH

Using the same T0–T24 covariates, the *time to in–hospital death* is modeled with right censoring at hospital discharge. For the event time data ($T^{\text{death}}, \delta^{\text{death}}$), the conditional hazard function $h(t \mid x)$ is estimated via:

- **Cox Proportional Hazards (PH):** A semi-parametric model specified as $h(t \mid x) = h_0(t) \exp\{\beta^\top x\}$, implemented with the `lifelines` library. Proportional hazards assumptions are checked.
- **DeepSurv:** A neural network-based Cox model where $h(t \mid x) = h_0(t) \exp\{g_\psi(x)\}$, with g_ψ parameterized by a custom deep network implemented in PyTorch.

Performance is evaluated using Harrell's C–index, time-dependent AUC, calibration plots at clinically relevant horizons (e.g., 90, 180, 365 hours), and Kaplan-Meier curve stratification by predicted risk.

3.7 EVALUATION PROTOCOL

For the classification task, a one–time 80/20 train–test split is used. An internal validation split is carved from the training set for model selection, calibration, and thresholding. The frozen pipeline is then applied once to the test set. For the survival task, the same patient partitions are used. Primary test metrics for the classifier include AUROC, F1-score, precision, and recall. For the survival models, primary metrics are Harrell's C-index, calibration, and risk stratification confirmed by the log-rank test.

3.8 ASSUMPTIONS AND RISKS

- **Availability:** All predictors are assumed to be observable by T_{obs}; features with excessive missingness or not typically available early are excluded.
- **Measurement:** Units are harmonized and physiological plausibility is enforced. The missing data mechanism is assumed to be Missing at Random (MAR) after finding it is not Missing Completely at Random (MCAR), justifying the use of imputation and missingness indicators.
- **Domain:** Models are developed on the MIMIC–IV dataset; transportability to external centers may require recalibration.
- **Predictive, not causal:** The models identify predictive associations, not causal relationships.

3.9 COMPACT STATEMENT OF AIMS

Aim A (Phenotyping). To conduct a thorough exploratory data analysis on the T0–T24 training data to identify and visualize the key clinical characteristics and phenotypes that differ between survivors and non-survivors.

Aim B (Primary classifier). To train and validate a calibrated probabilistic model for in–hospital death, $\tilde{p}(x) = c \circ f_\theta(x)$, ensuring high discrimination and utility through a strict, leakage-controlled pipeline.

Aim C (Survival). Using the same early covariates, to estimate the survival function

$S^{\text{death}}(t \mid x)$ with Cox and DeepSurv models and evaluate their performance via C-index, calibration, and risk stratification.

CHAPTER 4

METHODOLOGY

4.1 OVERVIEW

This chapter details the methodological pipeline that was designed to develop and validate predictive models for SA–AKI mortality. The workflow begins with the construction of a high-fidelity cohort from an ontology-integrated data warehouse and proceeds through three analytical stages: (i) exploratory data analysis for phenotyping, (ii) development of a supervised binary classifier for mortality (the primary task), and (iii) time-to-event survival analysis. A strict, one-time stratified 80/20 data split underpins the entire process to prevent information leakage. All data-dependent transformations were fitted on the training set only and subsequently applied to the test set for a single final evaluation. The development process was guided by the TRIPOD statement for transparent reporting.

4.2 DATA SOURCES, ETL, AND COHORT (T0–T24)

ETL and Integration. The analysis asset was built by ingesting and integrating multiple sources, primarily MIMIC–IV, with medical ontologies (RxNorm, ATC, SNOMED CT) linked via the Unified Medical Language System (UMLS). This was managed in a PostgreSQL database, enabling reproducible, ingredient-level value sets for concepts like systemic antibiotics and vasopressors. A validation suite ensured unit harmonization (e.g., F→C, lb→kg), referential integrity, and logical consistency.

Cohort Definition. The study population was defined as adults (≥18 y) on their first ICU stay with a length-of-stay ≥24 h. The SA–AKI cohort was identified using established clinical criteria:

- **Sepsis (Sepsis–3):** Defined by suspected infection (a body-fluid culture near the administration of a systemic antibiotic) and an acute increase in organ dysfunction

(ΔSOFA ≥ 2). The sepsis onset time, t_{sepsis}, was the start of the first qualifying ICU day.

- **AKI (KDIGO 2012):** Defined by changes in serum creatinine (SCr) or urine output. A hierarchical method established baseline SCr. The AKI onset time, t_{aki}, was the earliest time any criterion was met.

- **SA–AKI:** Required AKI onset within the window $[t_{\text{sepsis}} - 48\text{ h}, t_{\text{sepsis}} + 48\text{ h}]$.

Patients with prior End-Stage Renal Disease (ESRD) or who received Renal Replacement Therapy (RRT) within the first 24 h were excluded. All predictive features were restricted to measurements within the first 24 hours of ICU admission (T0–T24) to ensure clinical actionability and prevent look-ahead bias.

4.3 DATA SPLIT AND LEAKAGE CONTROL

A one-time, stratified 80/20 train-test split was performed on the cohort. Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$ be the full dataset, where $x_i \in \mathbb{R}^p$ are the T0–T24 features and $y_i \in \{0, 1\}$ is the in-hospital death indicator. The partition is

$$\mathcal{D} = \mathcal{D}_{\text{train}} \,\dot{\cup}\, \mathcal{D}_{\text{test}},$$

where partitions are disjoint and stratified by the outcome to maintain class balance. All data-dependent objects—imputation models, scaling parameters, and encoding vocabularies—were fitted *only* on $\mathcal{D}_{\text{train}}$ and then applied as frozen transformers to $\mathcal{D}_{\text{test}}$. Any validation set used for model tuning or calibration was created as a further split *from within* $\mathcal{D}_{\text{train}}$.

Leakage Audit. A scripted audit ensured that no variables encoding information after T24 were included as predictors. This included removing features like total ICU length of stay and aggregates (e.g., _n observation counts) that could act as proxies for care intensity or illness duration.

4.4 PREPROCESSING AND FEATURE ENGINEERING (TRAIN-ONLY)

Missingness Analysis. The nature of missing data was formally investigated on $\mathcal{D}_{\text{train}}$. Little's MCAR test was performed, and its rejection of the null hypothesis suggested that data were not Missing Completely at Random (MCAR). This justified a more careful handling of missingness beyond simple imputation.

Missingness Indicators. For every feature with missing values, a binary indicator variable (`_missing`) was created. The association between each indicator and the mortality outcome was tested using a chi-square test. Only indicators that showed a statistically significant association ($p < 0.05$) were retained and appended to the dataset as new features, thereby preserving potentially prognostic information contained in the pattern of missingness.

Imputation. After adding significant missingness indicators, remaining missing values in numerical columns were imputed using the median calculated from the training set.

Feature Selection and Scaling. Highly correlated features (Pearson correlation ¿ 0.99) were identified, and one from each pair was removed to reduce multicollinearity. All numerical features were then standardized using `StandardScaler`, with the scaler being fit only on the training data.

4.5 EXPLORATORY DATA ANALYSIS AND PHENOTYPING (TRAINING ONLY)

An extensive EDA was performed on $\mathcal{D}_{\text{train}}$ to characterize the cohort's clinical properties. This served as an unsupervised phenotyping step. Key analyses included:

- **Distributional Analysis:** Visualizing the distributions of key continuous variables (e.g., age, APACHE III, SOFA score, creatinine, lactate) and comparing them between survivor and non-survivor groups using histograms and density plots.

- **Correlation Analysis:** Calculating the Pearson correlation of all features with the mortality outcome to identify the strongest univariate predictors.

- **Stratified Analysis:** Using violin plots, box plots, and heatmaps to compare vital signs, laboratory values, and SOFA sub-scores between outcomes. Statistical significance was assessed using the Mann-Whitney U test due to the non-normal distribution of most variables.

4.6 SUPERVISED BINARY MORTALITY PREDICTION (PRIMARY)

Learning Objective. For a model $f_\theta : \mathbb{R}^p \to [0, 1]$, the goal was to minimize the average binary cross-entropy loss on $\mathcal{D}_{\text{train}}$ using 5-fold cross-validation.

Model Benchmarking. A wide set of models was benchmarked: Logistic Regression, LDA, QDA, Naive Bayes, regularized linear models, Decision Tree, Random Forest, XGBoost, and LightGBM. For tree-based ensembles, hyperparameters were tuned using Optuna. The impact of class imbalance was addressed primarily through `class_weight='balanced'` and was compared against various oversampling strategies (e.g., SMOTE, ADASYN) in sensitivity analyses.

Final Model Selection, Calibration, and Thresholding. The LightGBM model was selected based on its superior performance. The final pipeline involved:

1. Training the optimized LightGBM model on a sub-split of the training data.
2. Calibrating the model's outputs on a held-out validation split (from the training set) using `CalibratedClassifierCV` with the isotonic method.
3. Determining the optimal classification threshold on the same validation split by maximizing Youden's J statistic.
4. Retraining the final, calibrated LightGBM model on the entire training dataset ($\mathcal{D}_{\text{train}}$).
5. Evaluating the final model on the untouched test set ($\mathcal{D}_{\text{test}}$) using the predetermined threshold.

Interpretability. The final LightGBM model was interpreted using SHAP. A beeswarm summary plot was generated to visualize global feature importance and the direction of effects.

4.7 SURVIVAL ANALYSIS (TIME TO DEATH)

The same preprocessed T0–T24 feature matrix was used to model time to in-hospital death, with patients discharged alive being right-censored.

- **Cox Proportional Hazards (PH):** A CoxPH model was fitted using the `lifelines` library. Feature selection was performed by retaining the top 50 predictors from an initial full model, and the penalizer was tuned. Results were interpreted via a forest plot of hazard ratios.

- **DeepSurv:** A custom neural network-based Cox model was implemented in PyTorch, using SELU activations and residual connections. The model was trained by minimizing the negative Cox partial log-likelihood with the Efron method for handling ties.

Model performance was assessed by Harrell's C-index, time-dependent AUC, calibration plots at clinically relevant horizons (90, 180, and 365 hours post-admission), and Kaplan-Meier curves stratified by model-predicted risk quartiles. The statistical significance of the risk stratification was confirmed with the log-rank test.

4.8 ADHERENCE TO TRIPOD GUIDELINES

4.8.1 Data Leakage: Definition, Mechanisms, and Mitigation

In the context of clinical prediction modeling, data leakage has been defined as the inadvertent incorporation of information that would not be available at the intended time of prediction into the model training or evaluation process, resulting in inflated performance estimates that fail to reflect real-world utility (Kang and Yoon, 2023). For SA–AKI mortality prediction, three principal mechanisms through which leakage may be introduced have been identified in the literature and were considered during the design of this pipeline.

The first mechanism, temporal leakage (also termed look-ahead bias), arises when features are derived from measurements recorded after the intended prediction horizon. A model designed to estimate mortality risk at 24 hours post-admission that incorporates laboratory values from subsequent days, or whole-stay aggregates such as the mean creatinine across the entire ICU episode, has been provided with information that encodes the patient's future trajectory and outcome. The inclusion of total ICU length of stay as a predictor represents a particularly direct form of this bias, as length of stay serves as a strong proxy for whether and when death occurred (Chen *et al.*, 2025).

The second mechanism, preprocessing leakage, has been observed when data-dependent transformations (including imputation, scaling, and feature selection) are fitted on the entire dataset prior to the train–test split. Under such conditions, the statistical properties of the held-out records (means, variances, and correlations with the outcome) are permitted to influence the training pipeline, and optimistically biased performance estimates are produced as a consequence (Kang and Yoon, 2023).

The third mechanism, selection leakage, occurs when feature screening is conducted across the full dataset rather than within the training partition alone. Features that happen to correlate with the outcome across all records, including those subsequently reserved for evaluation, are preferentially retained, and the resulting predictor set carries information from the evaluation data.

In the present work, each of these mechanisms has been addressed through explicit safeguards. Temporal leakage has been prevented by restricting all features to the T0–T24 window and by deploying a scripted audit that flags any variable requiring post-T24 information. Preprocessing leakage has been avoided by fitting all transformers (imputer, scaler, and missingness indicator selector) exclusively on $\mathcal{D}_{\text{train}}$. Selection leakage has been precluded by confining all feature screening to the training partition. The held-out test set was used exactly once for final evaluation, and no iterative refinement based on test-set performance was undertaken.

4.8.2 The 24-Hour Window: Rationale, Strengths, and Heterogeneity Considerations

The selection of a fixed 24-hour observation window anchored to ICU admission (T0–T24) was motivated by the structure of clinical workflows in intensive care. The first ICU day is the period during which the initial assessment is typically completed, baseline investigations become available, and early management decisions regarding fluid resuscitation, vasopressor initiation, and antibiotic escalation are made. A prediction generated at T24 is therefore aligned with the natural decision point at which escalation or de-escalation of care is considered.

A fixed clock anchored to ICU admission, however, introduces a source of heterogeneity that warrants explicit discussion. Patients arrive in the ICU at different stages of their illness trajectory: some are admitted directly from the emergency department near the onset of sepsis, whereas others are transferred from general wards after a period of progressive deterioration. Consequently, the T0–T24 window captures different phases of the underlying disease process across patients. For early presenters, the window may coincide with the initial inflammatory surge, while for late transfers, it may reflect a more established pattern of organ dysfunction.

This heterogeneity represents a genuine limitation of any admission-anchored design, yet it also reflects the conditions under which a bedside prediction tool would be required to operate in practice. A clinician querying the model at the 24-hour mark has access only to the information accumulated since admission, regardless of the patient's prior trajectory. Several features included in the model were observed to partially account for this variation: the APACHE III score, computed at admission, integrates acute physiology with chronic health status and thereby captures elements of the pre-ICU trajectory; the SOFA score at 24 hours reflects cumulative organ dysfunction irrespective of when it began; the Charlson comorbidity index components provide context regarding baseline vulnerability; and the first/last/delta/slope suffixes applied to time-varying features encode within-window dynamics that have been expected to differ between early and late presenters.

Stratified analyses by admission source (emergency department versus ward transfer) or by the interval from symptom onset to ICU admission could be pursued in future work to quantify the degree to which model performance varies across these subgroups. An alternative design in which the observation window is anchored to sepsis onset rather than ICU admission has also been considered, though such an approach introduces its own difficulties because the sepsis onset time is itself an estimated quantity derived from clinical rules rather than a directly observed event.

The present study was designed in accordance with the TRIPOD (Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis) guidelines. The key items addressed include:

- **Source of Data (Item 4a):** The study design and source of data (MIMIC-IV, a large electronic health record database) are clearly described.
- **Participants (Item 5a-b):** The study setting (ICU), location, and detailed eligibility criteria for participants are specified in the cohort construction.
- **Outcome and Predictors (Items 6a, 7a):** The outcome (in-hospital mortality) and all predictors (derived from the T0–T24 window) are explicitly defined.
- **Sample Size (Item 8):** The flow of participants through the study is described, detailing exclusions and the final sample size available for analysis.
- **Missing Data (Item 9):** The handling of missing data is detailed, including statistical testing of the missingness mechanism, creation and selection of significant missingness indicators, and median imputation based on the training set.
- **Statistical Analysis Methods (Item 10a-d):** Predictor handling (correlation removal, scaling), model-building procedures (benchmarking of multiple algorithms, Optuna for tuning), methods for internal validation (stratified 80/20 train-test split), and all performance measures (AUROC, F1-score, C-index, calibration plots, DCA) are specified.
- **Model Specification and Performance (Items 15a, 16):** The final model (LightGBM) and its key hyperparameters are presented. Performance measures on

the test set are reported, providing an unbiased estimate of the model's performance on new data.

CHAPTER 5

DATASET UNDERSTANDING AND EDA

5.1 OVERVIEW

The data assets and clinical labeling rules that underlie the analysis-ready SA–AKI cohort are described first, followed by the temporal conventions and harmonization steps that precede the main analysis. The remainder of the chapter is devoted to the Exploratory Data Analysis (EDA) conducted on the training set. That EDA serves as a form of unsupervised phenotyping: it identifies the clinical characteristics and risk factors that differentiate patient outcomes and provides the empirical foundation for the predictive modeling that follows.

5.2 SOURCE TABLES AND WAREHOUSE CONTEXT

Figure 5.1 positions the raw inputs within MIMIC–IV (core, hosp, ICU) that were ingested and optimized in the warehouse. These tables provide the vitals, labs, medications, microbiology, and encounter scaffolding used for labeling and features.

5.3 ONTOLOGY-DRIVEN CONCEPT IDENTIFICATION

To avoid brittle, hard-coded drug lists, clinical concepts were mapped across terminologies and ingredient-level value sets were derived programmatically. The crosswalk in Figure 5.2 illustrates the pathway from SNOMED CT through UMLS CUIs into RxNorm ingredients, with ATC used as an orthogonal cross-reference and coverage check. This pipeline underlies the systemic *antibiotic* and *vasopressor* flags used in labeling and feature construction.

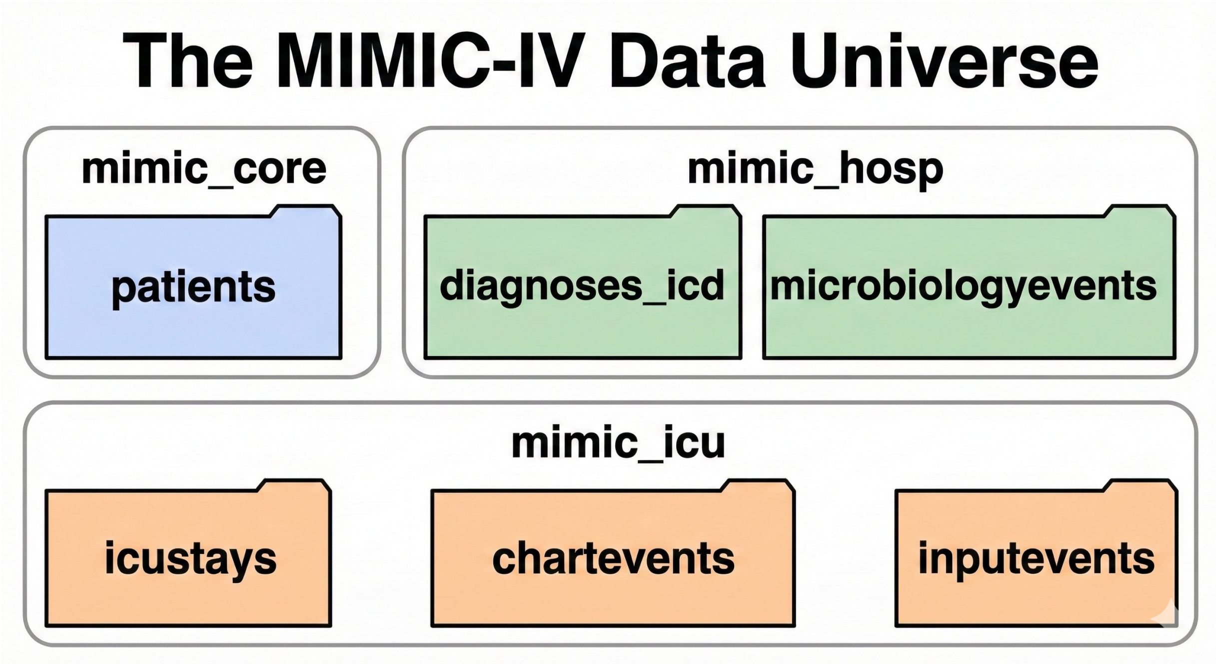


Figure 5.1: MIMIC–IV data universe leveraged in this study.

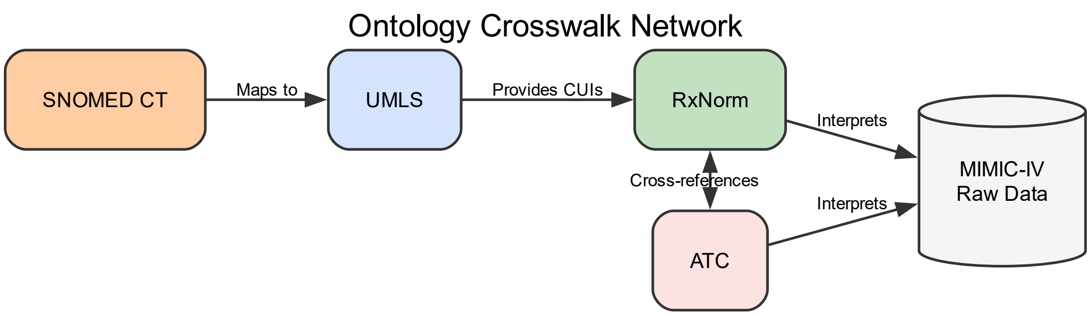


Figure 5.2: Ontology crosswalk linking SNOMED CT → UMLS CUIs → RxNorm ingredients, with ATC cross-reference.

5.4 COHORT CONSTRUCTION (COUNTS AND EXCLUSIONS)

A CONSORT-style reduction from all ICU stays to the final analysis cohort was followed. Figure 5.3 consolidates inclusion criteria (adult, first ICU stay, LOS $\geq$ 24 h), Sepsis–3 screening, SA–AKI temporal linkage, and final exclusions (ESRD, early RRT). These counts serve as the reference for all summary denominators used in the EDA.

5.5 CLINICAL LABELING RULES AND TIME ANCHORS

5.5.1 Sepsis–3 and AKI Logic (Compact View)

Sepsis–3 requires *suspected infection and* an acute increase in organ dysfunction (ΔSOFA $\geq$ 2). AKI follows KDIGO using three criteria: serum creatinine (absolute/relative) and urine output (UO). The Sepsis–3 gate and the UO sessionization used for the KDIGO UO criterion are displayed as a two-panel composite in Figure 5.4.

5.5.2 Timeline Conventions and Windows

All predictors used in modeling are restricted to the first 24 h after ICU admission (T0–T24). Suspected infection is defined by a temporal link between cultures and systemic antibiotics, and SA–AKI requires AKI onset within ±48 h of Sepsis onset. Figure 5.5 illustrates these anchors and windows.

5.6 DATA HARMONIZATION AND INITIAL STATISTICS

Before any analysis, data were harmonized (e.g., Fahrenheit → Celsius), redundant features were removed, and comorbidities were consolidated. The final analytic cohort contained **n=10,036** ICU stays. A stratified 80/20 split resulted in a training set of 8,028 patients and a test set of 2,008 patients. The in-hospital mortality rate in the training set was 26.74%. Key baseline statistics are presented in Table 5.1 and Table 5.2.

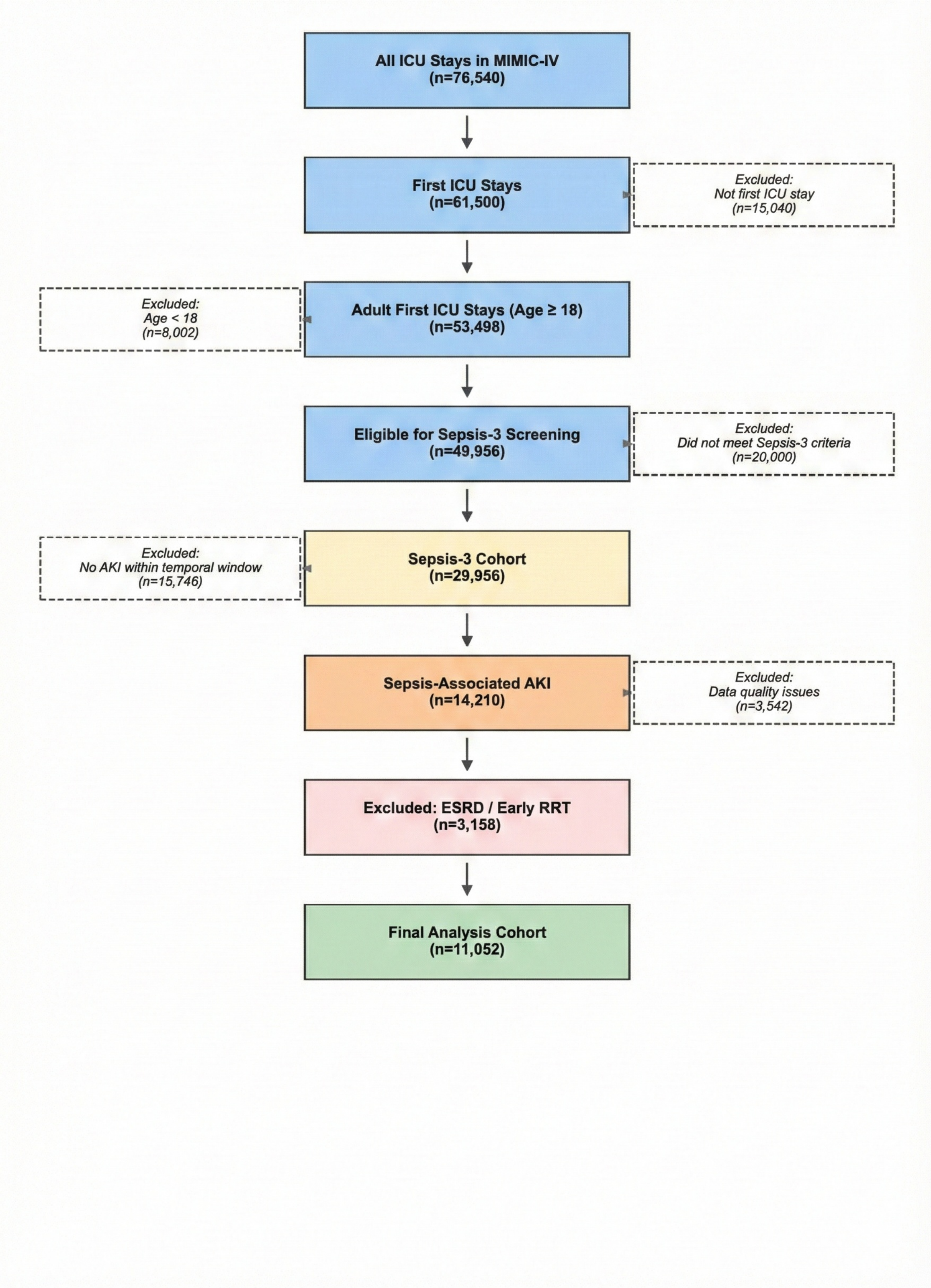
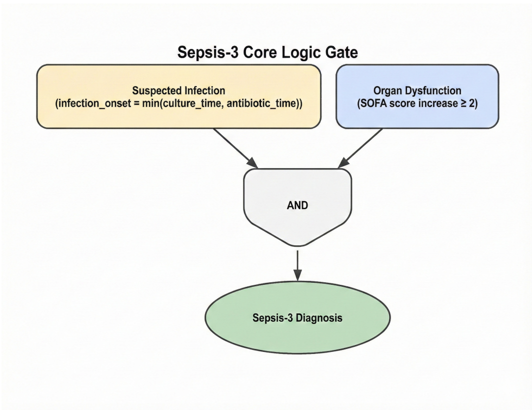
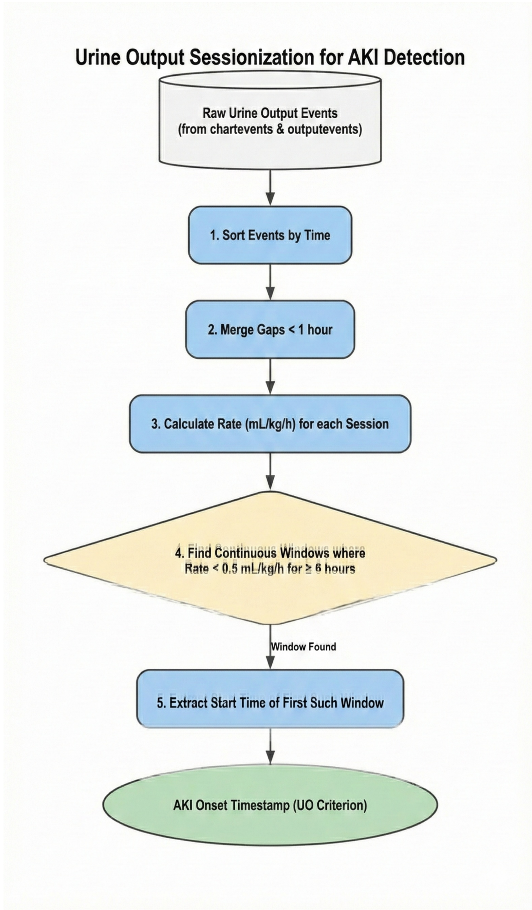


Figure 5.3: CONSORT-style flow of cohort construction with key exclusions.



(a) Sepsis–3 core logic gate.



(b) UO sessionization for KDIGO (continuous windows <0.5 mL/kg/h for ≥6 h).

Figure 5.4: Clinical rule components used for cohort labeling.

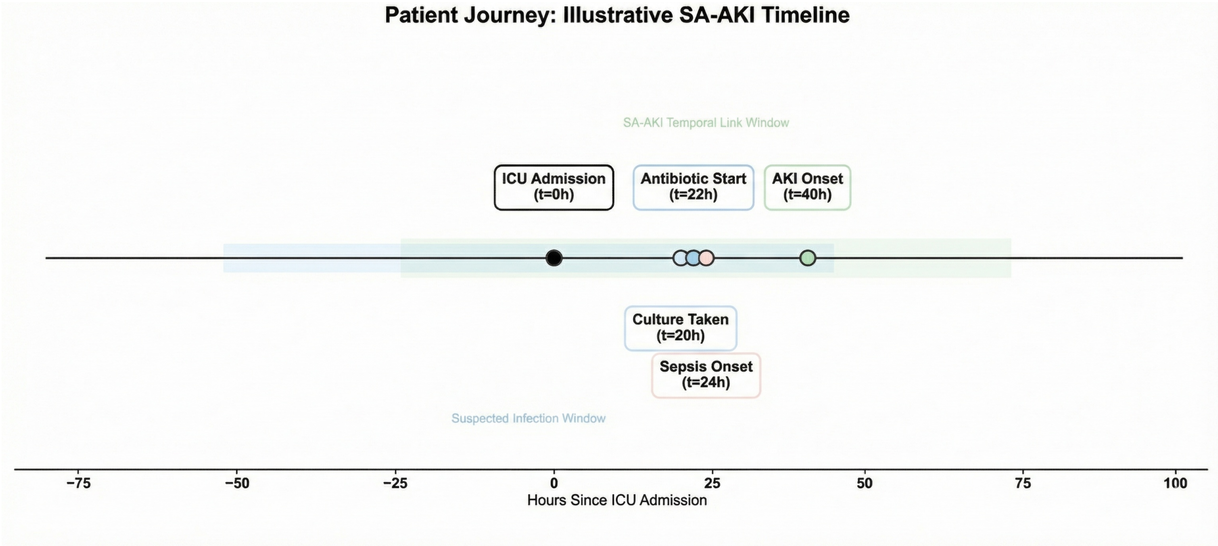


Figure 5.5: Illustrative patient journey with T0–T24 feature window, infection window, Sepsis onset, and AKI onset; the SA–AKI link is ±48 h around Sepsis onset.

Table 5.1: Overall cohort snapshot (final build).

Metric	**Value**
In-hospital death	**26.7%** (2,684/10,036)
Male	**53.0%** (5,322/10,036)
AKI trigger = Creatinine (vs. Urine Output)	**40.7%** (4,087/10,036)

Table 5.2: Therapy/exposure prevalence in the first 24 h (1/0 encoding).

Day-1 therapy / exposure flag	**Prevalence**
Mechanical ventilation within 24 h	**53.0%** (5,320/10,036)
Vasopressor within 24 h	**36.6%** (3,676/10,036)
Renal replacement therapy within 24 h	**69.0%** (6,922/10,036)

5.7 EXPLORATORY ANALYSIS AND PHENOTYPING OF THE TRAINING COHORT

5.7.1 Statistical Validation of Data Integrity

A foundational step of the EDA was to statistically validate the cohort and the train-test split. As shown in Figure 5.6, key clinical variables exhibited highly significant differences between survivor and non-survivor groups within the training set (all $p < 0.001$, Mann-

Whitney U test), suggesting that the selected features contain a strong prognostic signal. The same tests revealed no significant distributional differences for these variables between the training and test sets (all $p > 0.05$), indicating that the stratified split was successful and that the test set may be regarded as a representative sample for evaluating final model performance.

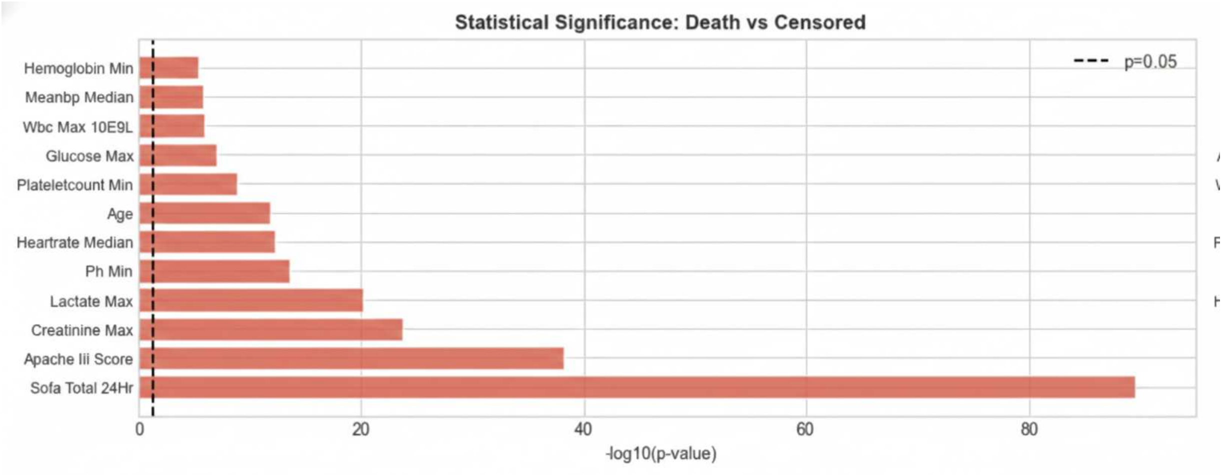


Figure 5.6: Statistical significance of key clinical variables between survivor and non-survivor groups, confirming that the selected features carry a strong prognostic signal ($p < 0.001$, Mann-Whitney U test).

5.7.2 Analysis of Missing Data Patterns

The pattern of missing data was investigated as a potential source of information. Of the 161 initial features, 101 had missing values. Little's MCAR test strongly rejected the null hypothesis ($p \approx 0.0$), indicating the data were not Missing Completely At Random. A subsequent chi-square analysis revealed that the absence of 46 specific measurements was significantly associated with mortality ($p < 0.05$). The ten most significant associations are shown in Table 5.3, highlighting that the absence of liver function and tissue injury tests was highly prognostic. These findings suggest that patterns of clinical data collection are themselves informative predictors of outcome.

Table 5.3: Top 10 most significant associations between missingness indicators and in-hospital mortality.

Indicator	**Chi-Square**	**p-value**	**Cramér's V**
`ast_n_missing`	166.46	< 0.001	0.144
`ast_max_missing`	166.46	< 0.001	0.144
`alkphos_median_missing`	165.27	< 0.001	0.143
`alkphos_n_missing`	165.27	< 0.001	0.143
`tbili_n_missing`	163.90	< 0.001	0.143
`tbili_max_mgdl_missing`	163.90	< 0.001	0.143
`ldh_max_ul_missing`	96.66	< 0.001	0.110
`ldh_n_missing`	96.66	< 0.001	0.110
`calcium_n_missing`	73.11	< 0.001	0.095
`calcium_min_missing`	73.11	< 0.001	0.095

5.7.3 Clinical Phenotypes of Survivor vs. Non-Survivor Groups

The EDA revealed distinct clinical phenotypes between survivors and non-survivors in the training set. As illustrated in Figure 5.7, non-survivors were significantly older and had higher baseline severity of illness, reflected by higher APACHE III and SOFA scores. Physiologically, the non-survivor phenotype was characterized by more severe multi-organ dysfunction, including worse renal function (higher maximum creatinine) and greater metabolic distress (higher maximum lactate).

This phenotyping was further supported by correlation analysis (Figure 5.8). The overall SOFA score (`sofa_total_24hr`) had the strongest positive correlation with death (0.234), reinforcing its role as a global marker of risk. Other features strongly associated with mortality included markers of hematological dysfunction (`rdw_median_pct`, 0.188) and hepatic injury (`sofa_hepatic_24hr`, 0.169). In contrast, variables indicating physiological stability and adequate organ perfusion, such as total urine output (`urine_output_total_24hr`, -0.113) and minimum pH (`ph_min`, -0.101), were the strongest predictors of survival. Together, these findings provide a data-driven characterization of the high-risk SA–AKI patient and directly informed the feature set for supervised modeling.

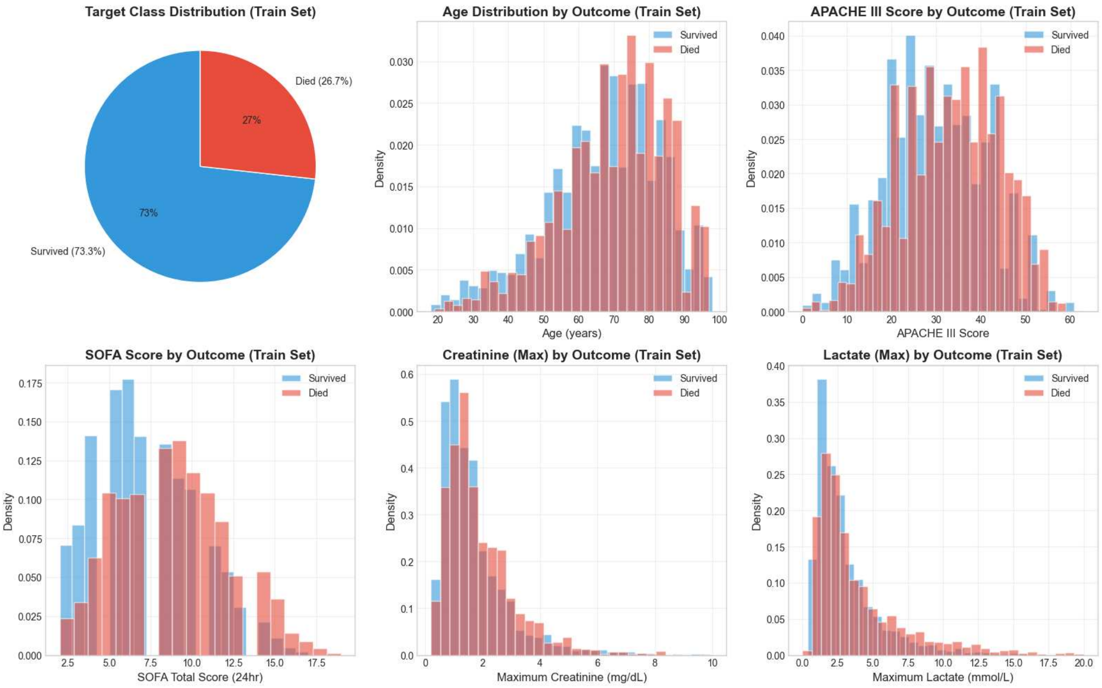


Figure 5.7: Distributions of key clinical variables in the training set, stratified by mortality outcome. Non-survivors consistently show distributions shifted towards higher severity.

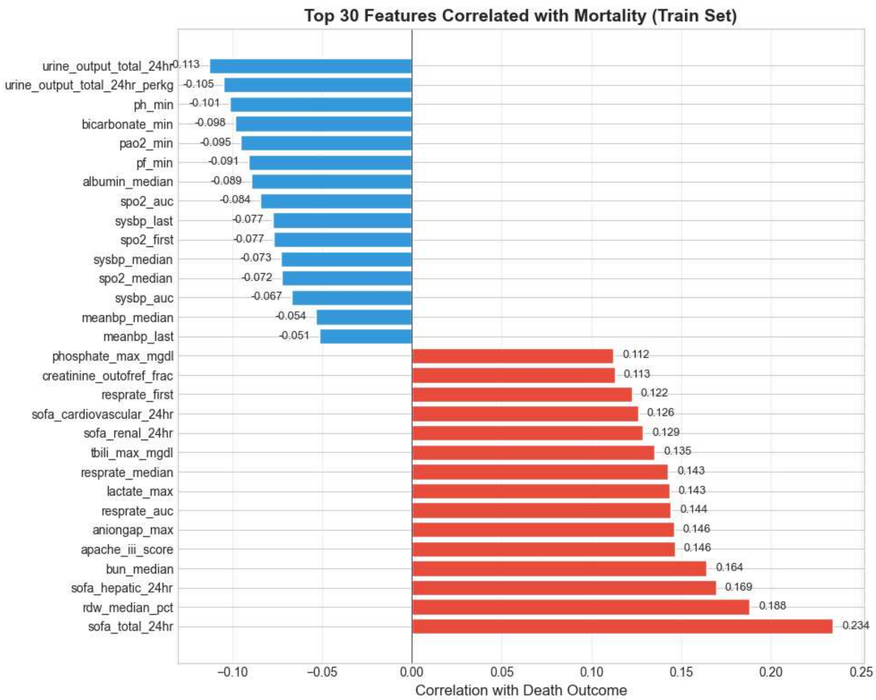


Figure 5.8: Top 30 features with the strongest positive (red) and negative (blue) correlation with in-hospital mortality in the training set.

CHAPTER 6

RESULTS

This chapter presents the outcomes of the predictive modeling pipeline. It begins by detailing the results of the extensive model benchmarking process, leading to the selection and final evaluation of the primary classification model. Subsequently, it presents the performance and interpretation of the time-to-event models. All results were generated from the held-out test set, following the strict, leakage-aware methodology established in the preceding chapters.

6.1 SUPERVISED MORTALITY PREDICTION

The primary objective of this thesis was the development of a binary classifier for in-hospital mortality. This was pursued through a two-stage process: benchmarking of various algorithms, followed by an in-depth evaluation of the final selected model.

6.1.1 Model Benchmarking and Selection

A wide range of classification algorithms was systematically evaluated to identify the most suitable candidate for the clinical task. The benchmark included traditional statistical models (Logistic Regression, LDA), regularized linear models (Ridge, Lasso, ElasticNet), and more complex, non-linear ensembles (Random Forest, XGBoost, LightGBM, DNN++). To address the inherent class imbalance in the dataset, two primary strategies were compared: intrinsic class weighting and various oversampling techniques (e.g., SMOTE, BorderlineSMOTE).

The results of this benchmark are summarized in Table 6.1. A clear trend emerged: tree-based ensembles consistently outperformed linear models in terms of discrimination, as measured by the ROC-AUC. The **LightGBM model trained with class weights** was identified as the top-performing model, achieving the highest ROC-AUC of 0.749 in the initial validation. While oversampling methods tended to produce higher F1-scores, this was often at the expense of precision and did not yield a superior ROC-AUC. Given that the primary goal was to accurately discriminate high-risk from low-risk patients across all decision thresholds, ROC-AUC was prioritized, and the LightGBM model was selected for the final stage of tuning, calibration, and evaluation.

Table 6.1: Best performance by algorithm type from the benchmarking experiments, sorted by ROC-AUC.

Model Algorithm	**ROC-AUC**	**F1-Score**
LightGBM	0.7492	0.4129
XGBoost	0.7444	0.4527
Logistic Regression	0.7382	0.5227
Linear Discriminant Analysis (LDA)	0.7364	0.3756
ElasticNet	0.7363	0.5185
Ridge	0.7349	0.5174
Deep Neural Network (DNN++)	0.7347	0.4471
Lasso	0.7331	0.5244
Random Forest	0.7137	0.3468

6.1.2 Final Model Performance on the Test Set

The selected LightGBM model was subjected to a final evaluation pipeline. After hyperparameter tuning with Optuna, the model's raw outputs were calibrated using isotonic regression on a validation split of the training data. The optimal classification threshold was determined to be 0.2704 by maximizing Youden's J statistic on this same validation set.

The final, calibrated model was then applied to the unseen held-out test set. The model achieved a ROC-AUC of 0.749, which was taken as evidence of strong and generalizable discriminative ability. The complete performance metrics are detailed in Table 6.2. The confusion matrix (Figure 6.1a) provides a clear picture of the model's performance at the chosen clinical threshold: it correctly identified 368 of the 537 patients who died (a recall of 0.685) while maintaining a precision of 0.422. The ROC curve (Figure 6.1b) further illustrates the model's consistent performance across the full range of decision thresholds.

Table 6.2: Performance of the final calibrated LightGBM model on the held-out test set.

Metric	Value
ROC-AUC	0.7488
Accuracy	0.6648
Precision	0.4220
Recall (Sensitivity)	0.6853
F1-Score	0.5224

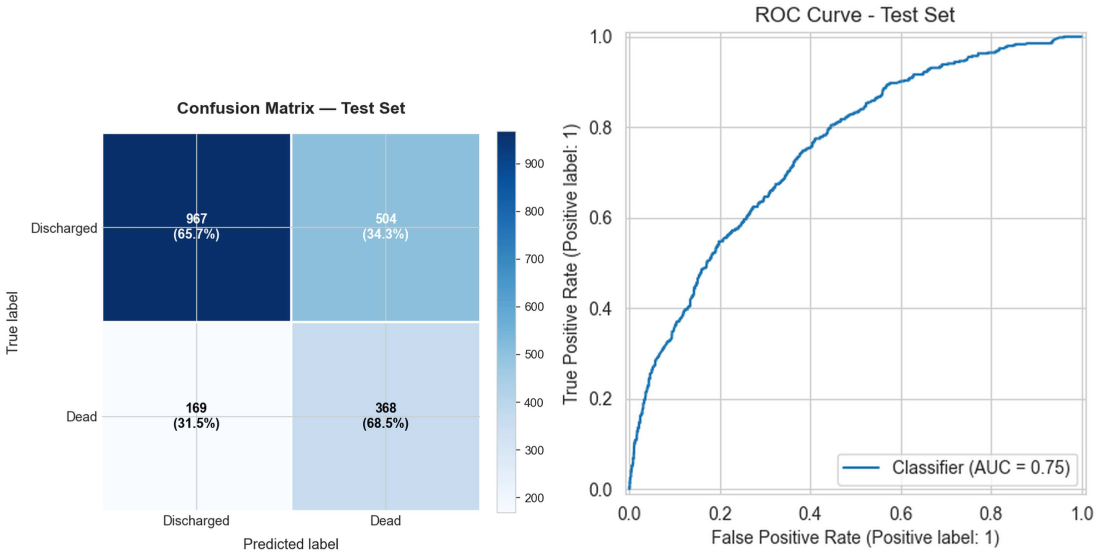


Figure 6.1: (**a**) Confusion matrix of the final model on the test set at the optimal threshold (0.2704). (**b**) ROC curve for the final model on the test set, showing an AUC of 0.75.

6.1.3 Model Interpretability with SHAP

To ensure the model's clinical relevance and transparency, its predictions were interpreted using SHAP. The beeswarm plot in Figure 6.2 provides a global summary of the features driving the model's output. This analysis suggests that the model's logic aligns with established clinical knowledge.

The most influential feature was the `sofa_total_24hr` score, a global measure of organ dysfunction. Other key predictors of mortality included markers of hematological dysfunction (`rdw_median_pct`), overall illness severity (`apache_iii_score`), advanced age, and renal failure (`bun_median`). Conversely, higher urine output (`urine_output_total_24hr`), a sign of preserved kidney function, was a strong predictor of survival. This concordance between the model's learned patterns and clinical pathophysiology has been interpreted as evidence supporting its validity.

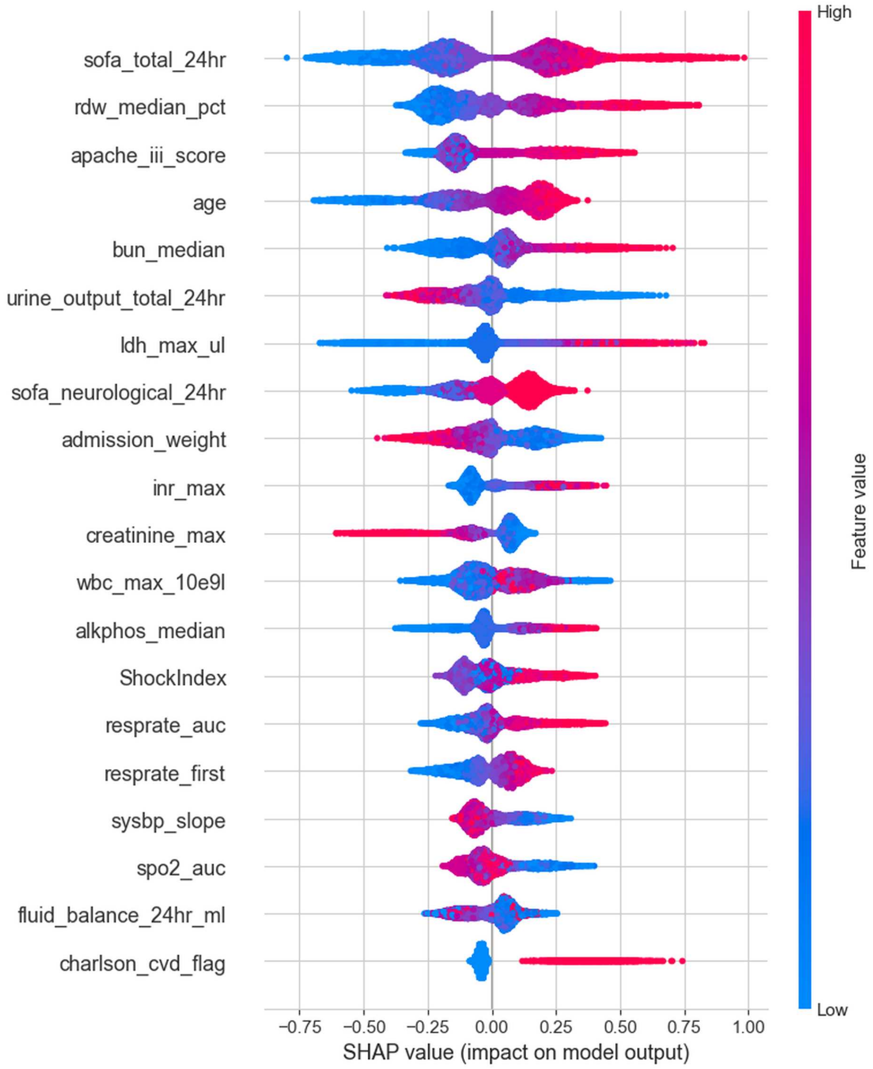


Figure 6.2: SHAP beeswarm summary plot for the top 20 features. Each dot represents a patient. The feature's value is indicated by color (red=high, blue=low), and its impact on the mortality prediction is shown on the x-axis.

6.1.4 Why LightGBM Outperforms Other Model Classes

The selection of LightGBM as the final model was not arbitrary but was driven by its consistent superiority in the benchmark (Table 6.1). Several properties of the algorithm have been identified as contributing to this advantage in the context of SA–AKI mortality prediction.

LightGBM employs a leaf-wise tree growth strategy rather than the level-wise approach adopted by traditional gradient boosting implementations. This design concentrates splitting effort on the leaves that yield the largest reduction in loss, a property that has been found to be particularly effective when the feature space contains a mixture of continuous biomarkers with skewed distributions and sparse binary indicators such as comorbidity flags and missingness indicators. In addition, LightGBM natively captures the interaction structure that characterizes clinical data. SA–AKI mortality is not driven by any single variable in isolation but rather by the joint configuration of organ dysfunction scores, hemodynamic instability, and metabolic derangement. Tree ensembles represent these higher-order interactions without requiring explicit feature engineering, whereas linear models (Logistic Regression, Ridge, Lasso, ElasticNet) are constrained to additive main effects unless interactions are manually specified. This distinction is reflected in the benchmark: the best-performing linear model (Logistic Regression, AUROC 0.738) trailed LightGBM (AUROC 0.749) by a margin that, while modest in absolute terms, was observed to be consistent across cross-validation folds and to correspond to meaningful reclassification of patients near the decision boundary.

The handling of class imbalance further differentiated LightGBM from alternative approaches. The built-in `class_weight='balanced'` mechanism, which reweights the loss function inversely proportional to class frequency, was found to be more effective than synthetic oversampling (SMOTE, ADASYN) for this dataset. Oversampling methods tended to inflate recall at the cost of precision without improving overall discrimination, a pattern consistent with the known risks of synthesizing examples in overlapped regions of the feature space (Blagus and Lusa, 2013; Sáez *et al.*, 2015). The Optuna-based hyperparameter search explored a richer space than grid search, encompassing regularization parameters (`reg_alpha`, `reg_lambda`), tree depth, minimum child samples, and feature and bagging fractions, which collectively controlled overfitting and contributed to improved generalization on the held-out test set.

6.1.5 Evidence for Trustworthiness and Clinical Credibility

A clinician evaluating a risk score requires more than a single discrimination metric; the credibility of the model must be established through multiple complementary lines of evidence. The trustworthiness of the predictions presented here has been assessed along five dimensions.

The first dimension concerns methodological integrity. The AUROC of 0.749 was obtained under conditions specifically designed to prevent optimistic bias: all preprocessing was fitted exclusively on training data, no post-T24 information entered the feature set, the test set was used exactly once for final evaluation, and no feature selection or imputation step was permitted access to test-set outcomes. In contrast, studies reporting AUROC values of 0.79–0.88 for SA–AKI mortality have been found to include length of stay as a predictor (Chen *et al.*, 2025), to perform feature selection on the full dataset (Li *et al.*, 2023), or to employ whole-stay aggregation windows that embed future information. When these methodological advantages are removed, the achievable discrimination for a genuine early warning system would be expected to be lower, and the result obtained here is consistent with that expectation.

The second dimension relates to calibration. The predicted probabilities were explicitly calibrated using isotonic regression on a validation split, ensuring that when the model assigns a 30% risk of death, approximately 30% of patients in that probability bin were observed to have died. Calibration is essential for clinical utility because treatment decisions depend on the absolute magnitude of risk rather than merely on the ranking of patients. It has been noted that several prior studies did not report calibration at all (Chen *et al.*, 2025; Li *et al.*, 2023).

The third dimension is clinical face validity, as assessed through SHAP analysis. The SHAP analysis (Figure 6.2) confirmed that the top predictors identified by the model (SOFA score, RDW, APACHE III, age, BUN, and urine output) are established clinical risk factors for ICU mortality. A model whose predictions are driven by clinically interpretable signals has been considered more likely to generalize than one that exploits dataset-specific artifacts.

The fourth dimension involves decision-curve analysis. The survival models were shown to demonstrate positive net benefit over the "treat all" and "treat none" strategies across a clinically relevant range of risk thresholds (8–30%), providing direct evidence that the use of the model to inform decisions would be expected to improve outcomes compared with default strategies (Figure 6.7).

The fifth dimension is the concordance observed across fundamentally different model families. The consistency of results across LightGBM (AUROC 0.749), CoxPH (C-index 0.693), and DeepSurv (C-index 0.688) provides a form of triangulation. If the predictive signal were spurious or driven by leakage, it would be unlikely to manifest consistently across a gradient-boosted classifier, a semi-parametric survival model, and a neural survival network.

6.1.6 Comparison with Prior Work and the Case for a Realistic Benchmark

Recent studies on SA–AKI mortality using MIMIC–IV have reported ROC–AUC values ranging from 0.79 to 0.88 (Chen *et al.*, 2025; Li *et al.*, 2023; Nong *et al.*, 2024). However, as discussed in Chapter 2, many of these results are likely to represent overestimations arising from methodological shortcomings: information leakage through whole-stay data aggregation, inclusion of length of stay as a predictor, global fitting of preprocessing steps, and the absence of calibration. The key methodological differences are summarized in

Table 6.3.

Table 6.3: Methodological comparison with representative prior SA–AKI mortality studies.

Criterion	This work	Chen (2025)	Li (2023)	Hu (2021)
Feature window	T0–T24	Unspecified	T0–T24	Whole-stay
LOS as predictor	No	Yes	No	No
Train-only preprocessing	Yes	No	No	No
Missingness indicators	Yes	No	No	No
Explicit calibration	Yes	No	No	No
Separate validation set	Yes	No	No	No
Reported AUROC	0.749	0.878	0.794	—
Reported C-index	0.69	—	—	0.72

The AUROC of 0.749 should therefore be interpreted not as an inferior result but as a more realistic and credible estimate of what is achievable for a genuine early warning system that relies exclusively on information available at the bedside within 24 hours of ICU admission. This work has been positioned as a methodologically sound benchmark against which future studies may be compared on equal footing.

6.1.7 Limitations: External Validation and Dataset Scope

A key limitation of this work, shared with the majority of SA–AKI prediction studies, is the absence of external validation on an independent dataset. The model was developed and evaluated exclusively on MIMIC–IV, which reflects the patient population, clinical practices, and documentation patterns of a single tertiary-care center (Beth Israel Deaconess Medical Center, Boston). Performance drift across hospitals has been documented in the ICU prediction literature (Rockenschaub *et al.*, 2024), and the degree to which the model would generalize to other institutions remains an open empirical question.

The most natural candidate for external validation has been identified as the eICU Collaborative Research Database, a multi-center dataset covering over 200 hospitals across the United States. However, eICU presents several harmonization challenges that would need to be addressed: differences in laboratory item coding, variable availability (certain biomarkers may be less consistently recorded across sites), and population-level differences in case mix and care intensity. A careful external validation would require re-application of the same cohort definition (Sepsis–3 + KDIGO), re-derivation of T0–T24 features from the eICU schema, and evaluation of the frozen model without retraining. This has been identified as the immediate next step in Chapter 7.

Until external validation has been completed, the model's predictions should be interpreted as internally valid risk estimates specific to the MIMIC–IV population. Recalibration on the target population would be expected to be necessary before the model could be considered for bedside decision support at any external institution. It has been noted that many prior studies have claimed broad generalizability without providing external evidence to support such claims (Chen *et al.*, 2025), and the explicit acknowledgment of this limitation in the present work has been intended to encourage more transparent reporting practices in the field.

6.2 TIME-TO-EVENT (SURVIVAL) ANALYSIS

To provide a more dynamic assessment of patient risk, survival models were developed to predict the time to in-hospital death using the same T0–T24 feature set.

6.2.1 Cox Proportional Hazards Model

After feature selection and penalization tuning, the final Cox Proportional Hazards (CoxPH) model achieved a test C-index of 0.693, indicating good discriminative ability over time. The forest plot in Figure 6.3 reveals the key drivers of hazard. Comorbidities like metastatic cancer and high SOFA scores were the strongest predictors of an increased hazard of death. Of note, early interventions such as RRT and vasopressor use were associated with a reduced hazard, a finding that is likely attributable to confounding by indication, whereby the sickest patients receive the most aggressive interventions, rather than to a causal protective effect of the therapies themselves. The model's ability to effectively stratify patients into distinct risk groups was confirmed by the clear separation of Kaplan-Meier curves for its predicted risk quartiles (Figure 6.4, log-rank $p < 0.001$).

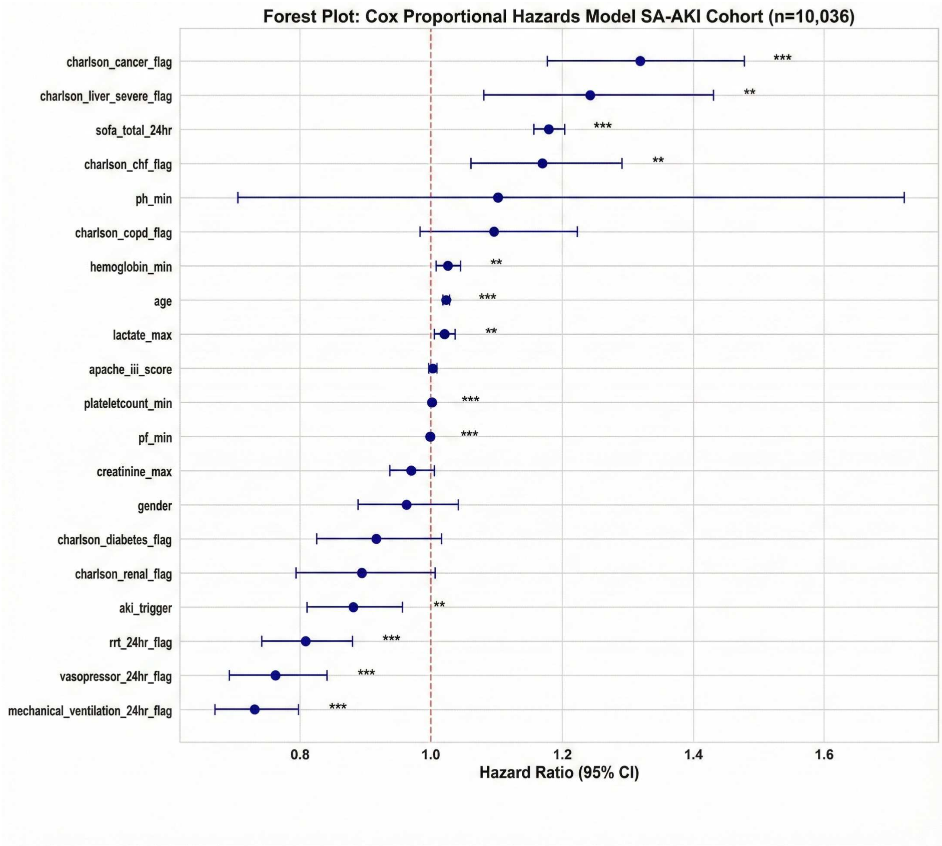


Figure 6.3: Forest plot of hazard ratios for the top 20 predictors in the final CoxPH model. Values to the right of the dashed line (HR ¿ 1) indicate an increased risk of death.

6.2.2 DeepSurv Model

A custom neural network-based survival model, DeepSurv, was also developed and achieved a comparable test C-index of 0.688. Like the Cox model, it demonstrated strong risk stratification (Figure 6.5, log-rank $p < 0.001$). The clinical utility of the DeepSurv model was further interrogated through calibration and decision curve analysis (DCA). The calibration plots (Figure 6.6) showed good agreement between predicted and observed event probabilities at key clinical horizons, although with a slight tendency to overestimate risk in the highest-risk bins, a pattern that could be addressed in future work with more advanced calibration techniques. The DCA (Figure 6.7) demonstrated that

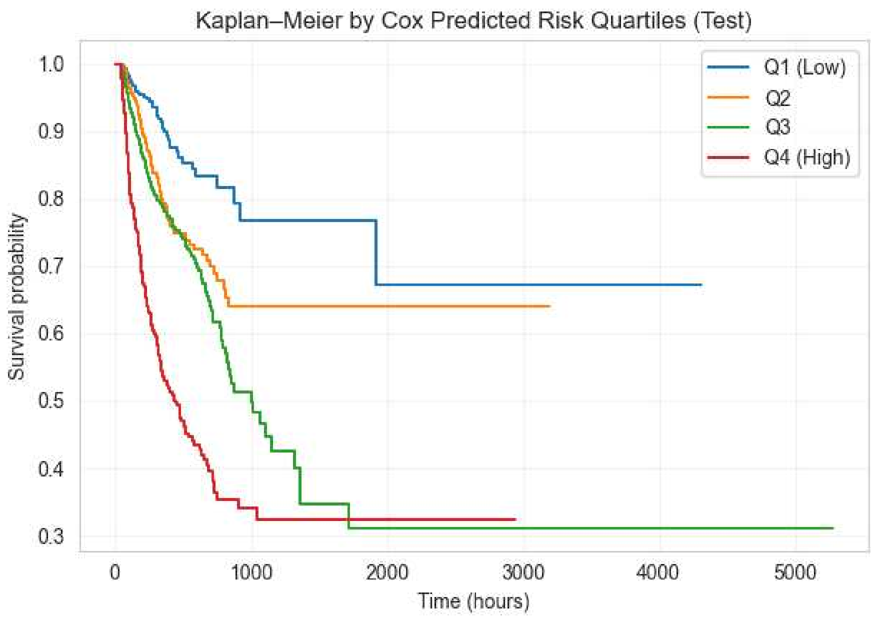


Figure 6.4: Kaplan-Meier survival curves for the test set, stratified by risk quartiles predicted by the CoxPH model, demonstrating effective risk stratification.

the use of the model to inform clinical decisions provides a net benefit over the default strategies of treating all or no patients for risk thresholds between 8% and 30%.

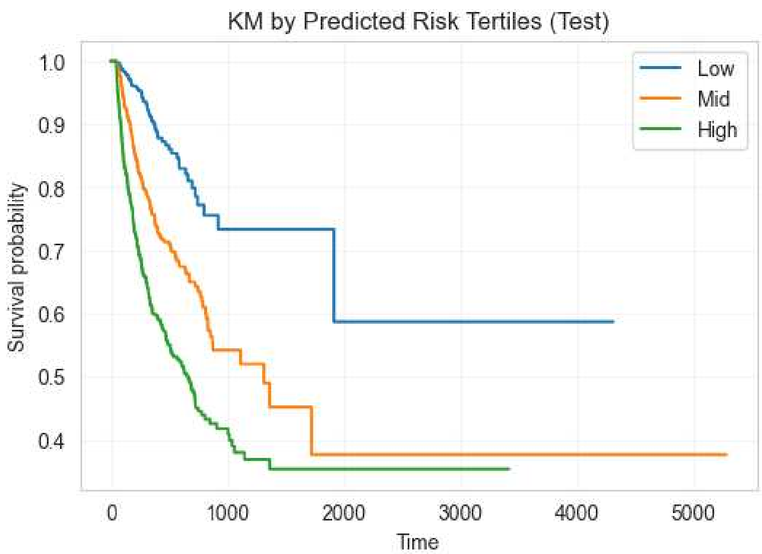


Figure 6.5: Kaplan-Meier survival curves for the test set, stratified by risk tertiles predicted by the DeepSurv model.

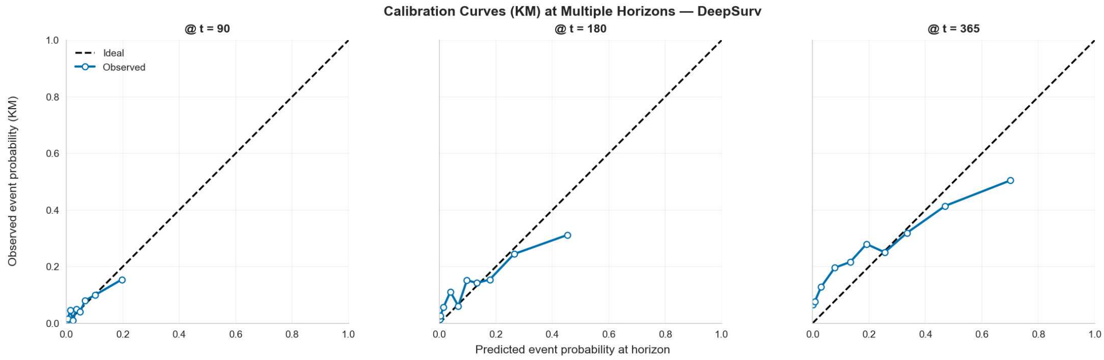


Figure 6.6: Calibration plots for the DeepSurv model at 90, 180, and 365 hours.

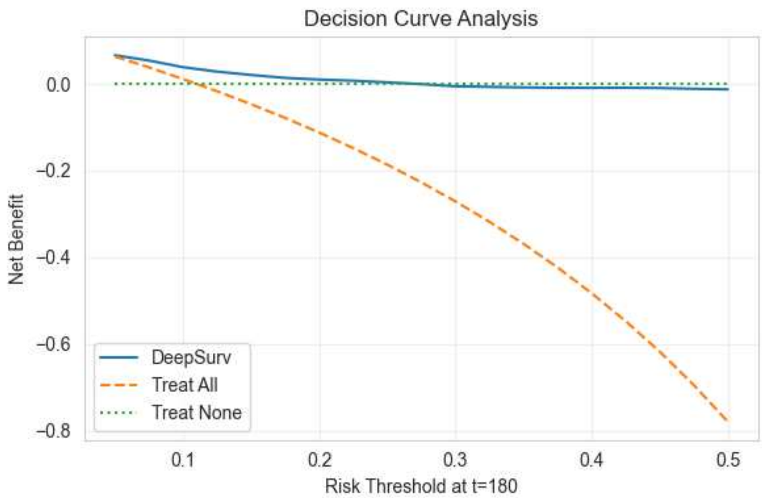


Figure 6.7: Decision Curve Analysis for the DeepSurv model at a 180-hour horizon, demonstrating positive net benefit across a range of clinically relevant risk thresholds.

6.2.3 Synthesis of Survival Analysis

Both the CoxPH and DeepSurv models delivered consistent and clinically interpretable predictions of time-to-death, with C-indices of 0.693 and 0.688, respectively. These results are comparable to those from other methodologically sound survival studies in critical care. For instance, Hu *et al.* (2021) reported a C-index of 0.72 but used whole-stay aggregated features, which introduces a high risk of data leakage. The results obtained in the present work, derived strictly from T0–T24 data, therefore provide a more reliable and clinically actionable benchmark for early dynamic risk stratification in the SA–AKI population.

CHAPTER 7

CONCLUSIONS AND FUTURE WORK

7.1 SUMMARY OF WORK

This thesis has presented the design, implementation, and validation of a machine learning framework for early mortality prediction in Sepsis-Associated Acute Kidney Injury (SA–AKI), with the aim of establishing a methodologically sound benchmark for the field. Through strict adherence to a 24-hour clinical window and the TRIPOD reporting guidelines, a transparent and reproducible pipeline has been developed that addresses the critical shortcomings of data leakage and performance overestimation that have been identified as prevalent in much of the existing literature.

The key contributions and findings are summarized as follows:

1. **High-Fidelity Cohort Construction:** A well-defined SA–AKI cohort of 10,036 patients was constructed from the MIMIC–IV database, underpinned by an ontology-integrated PostgreSQL warehouse that ensured reproducible definitions and data harmonization.

2. **Leakage-Aware Preprocessing:** A strict 80/20 train-test split served as the foundation for a preprocessing pipeline that included a formal analysis of the missing data mechanism, the creation and statistical selection of significant missingness indicators as novel features, and the train-only fitting of all data transformers to prevent information leakage.

3. **Interpretable Predictive Modeling:** Following a benchmark of machine learning algorithms, a LightGBM model was selected, tuned, and calibrated. On the held-out test set, the final model achieved an AUROC of 0.749 and an F1-score of 0.522. The clinical validity of the model was supported by SHAP analysis, which revealed that predictions were driven by established risk factors such as SOFA score, RDW, and age.

4. **Dynamic Risk Assessment with Survival Analysis:** The utility of the early-window features was extended to time-to-event modeling. Both a Cox Proportional Hazards model and a custom DeepSurv model were developed, each achieving a C-index of approximately 0.69 on the test set. These models were found to be effective at stratifying patients into distinct risk groups over time, providing a more

dynamic view of clinical risk.

Collectively, this work has provided a full pipeline that yields a clinically interpretable predictive model while maintaining the methodological discipline that has been argued to be necessary for building clinical decision support tools suitable for consideration in real-world settings.

7.2 FUTURE WORK

Building upon the foundation established in this thesis, several avenues for future research have been identified:

- **External Validation and Transportability:** The immediate next step would be to evaluate the model's performance on external datasets, with the eICU Collaborative Research Database representing the most natural candidate. The eICU database, covering over 200 hospitals across the United States, would provide a demanding test of generalizability across different patient populations, clinical practices, and documentation standards. A validation protocol would involve re-applying the Sepsis–3 and KDIGO cohort definitions to eICU, re-deriving T0–T24 features from its schema, and evaluating the frozen model without retraining. Should performance degradation be observed, recalibration on the target population would be explored before any clinical deployment could be considered.

- **Monte Carlo Simulation for Synthetic Data Augmentation:** The use of Monte Carlo methods to generate synthetic patient trajectories that respect the joint distribution of clinical variables observed in the training data represents a promising direction. Copula-based simulation or parametric bootstrap approaches could be employed to sample realistic feature vectors from the estimated multivariate distribution of T0–T24 measurements, preserving the correlation structure among biomarkers, vital signs, and severity scores. Such an approach would serve two purposes: augmenting the training set to improve model stability for underrepresented subgroups, and enabling sensitivity analyses to assess how model performance varies under perturbations of the input distribution. Unlike SMOTE or ADASYN, which operate in feature space without regard to clinical plausibility, a Monte Carlo approach grounded in domain-specific distributional assumptions would be expected to generate physiologically coherent synthetic patients (Blagus and Lusa, 2013). Generative adversarial networks and variational autoencoders trained on the MIMIC–IV cohort represent more flexible alternatives, though at the cost of reduced interpretability of the generation process.

- **Multi-Database Risk Framing:** To strengthen the evidence base for the degree of risk within the first 24 hours, future work should seek to integrate data from multiple sources beyond MIMIC–IV. The Amsterdam UMCdb, the HiRID dataset from the Bern University Hospital, and the ANZICS Adult Patient Database each offer complementary patient populations and clinical practices. A federated or meta-analytic approach that estimates risk across these databases would provide stronger evidence for the universality of the 24-hour risk signal and would help disentangle center-specific effects from genuine physiological predictors.

- **Exploration of Different Temporal Windows:** While this study focused on a 24-hour window, the predictive power of earlier windows (e.g., 6 or 12 hours) could be investigated to facilitate more timely interventions, or later windows (e.g., 48 hours) could be examined to capture evolving patient trajectories.

- **Model Fairness and Equity Audit:** An important step before any clinical deployment would be to audit the model's performance across different demographic subgroups (e.g., by sex, age, and ethnicity) to verify that predictions are equitable and to identify and mitigate any potential biases that may have been introduced through the training data.

- **Prospective Clinical Impact Study:** The ultimate goal would be to translate this research into clinical practice. A prospective, observational study could be designed to deploy the model in a real-world ICU setting as a decision support tool, allowing for the evaluation of its impact on clinical workflows, decision-making, and ultimately, patient outcomes.

APPENDIX A

CLINICALLY PLAUSIBLE VALUES

This appendix lists clinically plausible (extreme–but–possible) minima and maxima for numeric features in the sepsis–AKI ICU cohort (first 24 h), intended as *winsorization* bounds during preprocessing. Ranges reflect ICU/sepsis/AKI physiology rather than healthy outpatients and are aligned with the unit-harmonized guardrails used in the curated dataset.

DEMOGRAPHICS

Variable	**Min**	**Max**	**Units**
Age	18	120	years
Admission weight	30	300	kg

Notes. Adult ICUs typically include patients ≥18 y; extreme body weights spanning severe malnutrition to massive obesity motivate 30–300 kg as a pragmatic span.

TIMING ANCHORS (DESCRIPTIVE ONLY)

Variable	**Min**	**Max**	**Units**
Time to death	0	2,880	h
Time to AKI onset	0	168	h
Time to sepsis onset	0	72	h
Time to mechanical ventilator start	0	120	h
Time to vasopressor start	0	120	h

Notes. Onset times cluster near ICU admission in sepsis and AKI; the upper bounds allow delayed recognition/therapy within the early ICU course.

VITAL SIGNS (FIRST 24 H)

Variable	Min	Max	Units
Heart rate	20	250	beats/min
Systolic blood pressure	50	300	mmHg
Diastolic blood pressure	30	200	mmHg
Mean arterial pressure	30	200	mmHg
Respiratory rate	4	50	breaths/min
Oxygen saturation	50	100	%

Notes. Sustained tachyarrhythmias can transiently reach 200–250 bpm (Cleveland Clinic 2024); severe shock and hypertensive crisis motivate BP limits (Guy's and St Thomas' Specialist Care 2023; Author and Others 2021). Profound hypoxemia may drive SpO_2 towards 50–60% while values >100% are non-physiologic (Mayo Clinic 2024*b*; Family Practice Notebook 2024).

FLUID BALANCE AND URINE OUTPUT (FIRST 24 H)

Variable	Min	Max	Units
Net fluid balance (24 h)	−10,000	+15,000	mL
Fluid balance per kg (24 h)	−100	+200	mL/kg
Urine output (24 h)	0	20,000	mL
Urine output per kg (24 h)	0	200	mL/kg

Notes. Bounds reflect aggressive resuscitation/diuresis and anuria–polyuria extremes early in critical illness.

BLOOD GASES AND OXYGENATION (FIRST 24 H)

Notes. Reported survivals at pH near 6.5–6.8 are exceptional (Donadello *et al.* 2021); permissive hypercapnia and ventilatory failure bound $PaCO_2$ (Respiratory Therapy

Variable	Min	Max	Units
Arterial pH	6.5	7.7	—
$PaCO_2$	10	100	mmHg
PaO_2	30	500	mmHg
PF ratio (PaO_2/FiO_2)	0	600	mmHg

Magazine 2020; Medscape 2024). On FiO_2=1.0, PaO_2 can reach ~400–500 mmHg; PF ratios span 0–600 mmHg (Lévy *et al.* 2021; Family Practice Notebook 2024).

ELECTROLYTES AND ACID–BASE (FIRST 24 H)

Variable	Min	Max	Units
Sodium	100	190	mmol/L
Potassium	2.0	8.0	mmol/L
Chloride	70	130	mmol/L
Bicarbonate	5	40	mmol/L
Anion gap	3	30	mmol/L

Notes. Bounds cover high–gap metabolic acidosis and severe alkalosis; see Vasavada *et al.* (2023); The Royal Children's Hospital Melbourne (2024) for context.

RENAL FUNCTION AND GLYCEMIA (FIRST 24 H)

Variable	Min	Max	Units
Creatinine (serum)	0.3	15.0	mg/dL
Blood urea nitrogen	5	200	mg/dL
Glucose (plasma)	20	800	mg/dL

Notes. Severe AKI/ESRD without dialysis can manifest with creatinine 10–15 mg/dL and BUN >100 mg/dL; hyperosmolar states often show glucose 600–800+ mg/dL (Apollo 24x7 2024; MedicalNewsToday 2018; Kitabchi *et al.* 2023).

CALCIUM, MAGNESIUM, PHOSPHATE (FIRST 24 H)

Variable	Min	Max	Units
Calcium (total)	4.0	15.0	mg/dL
Magnesium (serum)	0.5	6.0	mg/dL
Phosphate (serum)	0.5	15.0	mg/dL

Notes. Life-threatening calcium extremes and clinically significant magnesium/phosphate derangements motivate these bounds (Merck Manuals Professional Edition 2023; Fong and Khan 2012).

HEPATIC AND BILIARY PANEL (FIRST 24 H)

Variable	Min	Max	Units
Alkaline phosphatase	20	1,200	U/L
AST	5	10,000	U/L
ALT	5	10,000	U/L
Total bilirubin	0	40	mg/dL
Direct bilirubin	0	30	mg/dL
Gamma–glutamyl transferase	0	2,000	U/L
Albumin (serum)	1.0	6.0	g/dL

Notes. Massive transaminase elevations in ischemic/toxic injury (Bernal and Wendon 2016; Lee 2008) and marked cholestatic patterns (Mayo Clinic 2024*a*) justify the upper bounds; severe hypoalbuminemia is common in critical illness (Nicholson *et al.* 2023).

HEMATOLOGY AND COAGULATION (FIRST 24 H)

Notes. Extremes of anemia/polycythemia (Costa *et al.* 2019; Padmanabhan *et al.* 2020), critical thrombocytopenia and thrombocytosis (Blumberg *et al.* 2023; Wegmann *et al.*

Variable	Min	Max	Units
Hemoglobin	3	20	g/dL
Platelet count	5	1,000	$\times 10^9$/L
INR	0.8	10.0	ratio
aPTT	15	200	s
White blood cell count	0	200	$\times 10^9$/L

2020), and profound coagulopathy (Hylek *et al.* 1998) set these bounds. Leukocytosis may rarely exceed 200×10^9/L.

INFLAMMATION AND TISSUE INJURY (FIRST 24 H)

Variable	Min	Max	Units
C–reactive protein	0	500	mg/L
Lactate dehydrogenase	50	10,000	U/L
Troponin (cardiac)	0	100	ng/mL
BNP or NT–proBNP	0	100,000	pg/mL
Lactate (serum)	0.5	20	mmol/L

Notes. Severe sepsis often elevates lactate; levels $\gtrsim$4–10 mmol/L indicate high risk (Levy and Evans 2021). Very high troponin reflects extensive myocardial injury.

SEVERITY SCORES (DAY 1)

Variable	Min	Max	Units
APACHE III score	0	299	points
SOFA total (24 h)	0	24	points
SOFA subscore (each organ)	0	4	points

Notes. Ranges follow score definitions (Knaus *et al.* 1991; Physiopedia 2024).

Implementation. Bounds are applied after unit harmonization and basic plausibility checks; outliers beyond these limits are capped, not dropped. These limits match the clinical guardrails used to compute laboratory QC fractions in the main dataset.

APPENDIX B

FEATURE DICTIONARY AND DESCRIPTIVE TABLES

An annotated data dictionary for the sepsis–associated acute kidney injury (SA–AKI) cohort has been compiled to summarize the first 24 hours of intensive care and to support time–to–event analyses. The final build comprises n=10,036 stays and p = 162 features. All variables have been harmonized to target units before aggregation, and binary encodings have been standardized as indicated. Overall cohort characteristics were observed to include in–hospital mortality of 26.7

Endpoints and anchor times were defined as in Table B.1. The in–hospital death indicator was coded as **1=death**, 0=censored, with time measured from ICU admission to death or last known alive. Onset times for AKI (KDIGO), Sepsis–3, first invasive ventilation, and first vasopressor use were stored in hours from admission. Demographics included age (years) and a sex indicator (**1=male**, 0=female); harmonized admission weight (kg) was retained for indexing of fluid and urine features. The AKI trigger variable encoded the primary detection pathway (**1=creatinine**, 0=urine output).

Binary flags for therapies and day–1 exposures were defined as summarized in Table B.2 and encoded as 1/0. Charlson comorbidity components and the total points were represented as separate indicators (Table B.3), with component–wise prevalences reported for the full cohort.

Vital–sign bases and their canonical units were enumerated in Table B.4. Day–1 statistics were derived through a common suffix system (Table B.8), where *first*, *last*, *median*, *iqr*, *rng*, *delta*, *auc*, *slope*, and *n* reflected, respectively, the first/last value, 24 h median, interquartile range, range, change (last−first), trapezoidal area–under–curve, per–hour linear slope, and observation count. For derivatives (*delta*, *slope*), a requirement of $n \geq 2$ was imposed; otherwise, values were set to `NULL` while the companion count was preserved. The ShockIndex (HR/SBP) was stored as a dimensionless derived signal.

Fluid balance and urine output features were recorded for the initial 24 hours (Table B.5), with both absolute and weight–indexed forms (mL and mL/kg). Net fluid balance was calculated as total inputs minus outputs across all documented sources.

Laboratory and gas–exchange features were organized by base analyte with curated units and aggregates (Tables B.6–B.7). Unit normalization was performed *prior* to any quality control or statistical aggregation. Representative conversions included glucose mmol/L→mg/dL (×18), albumin and hemoglobin g/L→g/dL (÷10), calcium mmol/L→mg/dL (×4.0), chloride mg/dL→mmol/L (÷3.545), lactate mg/dL→mmol/L (÷9), blood gases kPa→mmHg (×7.5), bilirubin µmol/L→mg/dL (÷17.104), magnesium mmol/L→mg/dL (×2.433), phosphate mmol/L→mg/dL (×3.097), troponin ng/L→ng/mL (÷1000), and BNP/NT–proBNP ng/L≡pg/mL.

For each analyte, day–1 summaries were produced using count, median, minima, maxima, and, where specified, dynamics (*delta*) or terminal value (*last*). Quality–control fractions were retained when available: the share outside the reported reference interval (`_outofref_frac`), the share labeled abnormal by the lab system (`_abn_frac`), and the share of urgent orders (`*_stat_frac`).

All fractions were bounded to $[0, 1]$ with the analyte–specific n as denominator. Clinically plausible guardrails were applied only for plausibility checks and out–of–range tallies (e.g., electrolytes, gases, bilirubin, hematology, CRP, LDH, troponin, BNP), rather than for winsorization.

Oxygenation was further characterized through a PaO_2/FiO_2 ratio (PF). FiO_2

measurements were normalized to a fraction within $[0.21, 1.0]$ from charted percent or fraction entries, and PaO_2 values were standardized to mmHg. Each PaO_2 was paired to the nearest FiO_2 within a ±2 hour window; pairs with implausible or missing FiO_2 were discarded. For each stay, the PF count, median, and minimum were aggregated over the 0–24 hour window.

Construction of the feature table was supported by an indexed cohort window. ICU stays with a documented out–time and length of stay of at least 24 hours were included. A 24 hour window was anchored at the ICU `intime`; all day–1 features were restricted to timestamps in $[\texttt{intime}, \texttt{intime} + 24\text{h})$. Item–to–feature mappings were defined explicitly for core section–7 labs and label–driven for an extended panel. After normalization, per–analyte statistics were computed using exact medians (`PERCENTILE_DISC(0.5)`), minima, maxima, and means of indicator flags to form QC fractions. First and last values were obtained via time–ordered selection, and changes (*delta*) were computed only when at least two observations were available. Final curated panels were left–joined by `stay_id` to assemble a single, primary–keyed table that retains `hadm_id` and `subject_id` for relational linkage.

Naming conventions followed snake_case with descriptive bases and standardized suffixes (Table B.8). Vital signs used canonical 24 hour medians as default signals (additional suffix features were included where available). Therapy/exposure and Charlson variables were encoded as binary indicators with cohort–level prevalences documented. Where severity scores (e.g., APACHE III; SOFA domain totals and overall) were present, values were stored as point totals consistent with their original scales.

Table B.1: Endpoint and timing variables present in file.

Column	Meaning (concise)	Unit
Death	In-hospital death indicator (**1=death**, 0=censored).	—
Time to death	Hours from ICU admit to death / last alive (censored).	h
Time to AKI onset	Hours to AKI onset (KDIGO).	h
Time to sepsis onset	Hours to first Sepsis–3 flag.	h
Time to mechanical ventilator start	Hours to first invasive ventilation.	h
Time to vasopressor start	Hours to first vasopressor.	h
Age	Age at ICU admission.	y
Gender	Biological sex (**1=male**, 0=female).	—
Admission weight	Documented admission weight (harmonized).	kg
AKI trigger	Trigger type (**1=creatinine**, 0=urine output).	—

Table B.2: Therapy/exposure flags (definitions mirror Table 5.2).

Column	Meaning (concise)	Unit
Mechanical ventilation 24 h flag	Invasive mechanical ventilation within T0–T24 (1/0).	—
Vasopressor 24 h flag	Any vasopressor within T0–T24 (1/0).	—
RRT 24 h flag	Renal replacement therapy within T0–T24 (1/0).	—
Baseline creatinine estimated flag	Baseline creatinine was estimated (vs. observed) (1/0).	—

Table B.3: Charlson comorbidity flags (1/0) and prevalence in cohort.

Column	Meaning (concise)	Prevalence
Myocardial infarction flag	History of myocardial infarction.	12.8% (1,282/10,036)
Congestive heart failure flag	Congestive heart failure.	11.8% (1,187/10,036)
Peripheral vascular disease flag	Peripheral vascular disease.	7.9% (789/10,036)
Cerebrovascular disease flag	Stroke or TIA.	9.7% (975/10,036)
Dementia flag	Dementia.	4.1% (412/10,036)
Chronic pulmonary disease flag	Chronic pulmonary disease (e.g., COPD).	14.3% (1,440/10,036)
Rheumatic / connective tissue flag	Connective tissue disease.	2.1% (214/10,036)
Peptic ulcer disease flag	Peptic ulcer disease.	2.6% (256/10,036)
Liver disease (mild) flag	Mild liver disease.	10.4% (1,043/10,036)
Diabetes (no complications) flag	Diabetes without chronic complications.	18.6% (1,866/10,036)
Diabetes with complications flag	Diabetes with end-organ complications.	9.3% (935/10,036)
Renal disease flag	Moderate–severe chronic kidney disease.	14.2% (1,421/10,036)
Hemiplegia/paraplegia flag	Hemiplegia or paraplegia.	3.5% (349/10,036)
Cancer (no metastasis) flag	Solid tumor within 5 y, no metastasis.	8.9% (894/10,036)
Metastatic cancer flag	Metastatic solid tumor.	3.9% (392/10,036)
HIV/AIDS flag	HIV/AIDS.	0.4% (36/10,036)
Liver disease (moderate/severe) flag	Moderate or severe liver disease.	6.5% (656/10,036)
Charlson total	Sum of all Charlson flags (points).	points

Table B.4: Vital-sign base names; day-1 statistics derived via suffixes where applicable.

Base name	**Meaning (concise)**	**Unit**
Heart rate	Day-1 aggregates (median, IQR, range, first/last, delta, slope, AUC, n).	beats/min
Systolic blood pressure	Systolic arterial blood pressure (non-/invasive).	mmHg
Diastolic blood pressure	Diastolic arterial blood pressure (non-/invasive).	mmHg
Mean arterial pressure	Computed or recorded MAP.	mmHg
Respiratory rate	Respiratory rate.	breaths/min
Oxygen saturation	Peripheral oxygen saturation (pulse oximetry).	%
ShockIndex	Derived ratio HR/SBP; higher reflects relative hypotension.	—

Table B.5: Fluid balance and urine output features.

Column	**Meaning (concise)**	**Unit**
Fluid balance 24 h (mL)	Net inputs minus outputs in first 24 h (all sources).	mL
Fluid balance 24 h per kg	Weight-indexed net fluid balance (day 1).	mL/kg
Urine output total 24 h	Total urine in first 24 h.	mL
Urine output total 24 h per kg	Weight-indexed urine output (day 1).	mL/kg

Table B.6: Laboratory and gas-exchange base names (A–M).

Base name	Meaning (concise)	Unit
Albumin	Serum albumin.	g/dL
Alkaline phosphatase	Alkaline phosphatase.	U/L
Aspartate aminotransferase	AST.	U/L
Anion gap	Serum anion gap.	mmol/L
Bicarbonate	Serum bicarbonate (HCO_3^-).	mmol/L
Blood urea nitrogen	BUN.	mg/dL
Calcium (total)	Total serum calcium.	mg/dL
Chloride	Serum chloride.	mmol/L
Creatinine	Serum creatinine.	mg/dL
Direct bilirubin	Direct bilirubin.	mg/dL
Glucose	Plasma glucose.	mg/dL
Hemoglobin	Hemoglobin concentration.	g/dL
INR	International normalized ratio.	ratio
Lactate	Serum lactate.	mmol/L
LDH	Lactate dehydrogenase.	U/L

Table B.7: Laboratory and gas-exchange base names (N–Z).

Base name	Meaning (concise)	Unit
Magnesium	Serum magnesium.	mg/dL
$PaCO_2$	Arterial carbon dioxide (ABG).	mmHg
PaO_2	Arterial oxygen (ABG).	mmHg
pH (arterial)	Arterial pH (ABG).	pH units
PF ratio	PaO_2/FiO_2 ratio (derived).	mmHg
Phosphate	Serum phosphate.	mg/dL
Platelet count	Platelet count.	10^9/L
Potassium	Serum potassium.	mmol/L
RDW	Red cell distribution width.	%
Sodium	Serum sodium.	mmol/L
Total bilirubin	Total bilirubin.	mg/dL
Troponin	Cardiac troponin.	ng/mL
White blood cell count	WBC.	10^9/L

Table B.8: Suffix legend for time-series features.

Suffix	**Definition (first 24 h)**	**Unit**
first	First measurement in the window.	base
last	Last measurement in the window.	base
median	Median of all readings over 24 h.	base
iqr	Inter-quartile range (Q3−Q1).	base
rng	Range = max−min.	base
delta	Change = last−first (requires $n \geq 2$).	base
auc	Trapezoidal area-under-curve (0–24 h).	base·h
slope	Linear regression slope per hour (requires $n \geq 2$).	base/h
n	Number of discrete observations.	count

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Hybrid Data Augmentation Techniques for Disease Identification from Clinical Notes

A Project Report

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*"Artificial intelligence will not replace the physician, but it will empower them to deliver more personalized care by freeing them from administrative burdens." — **Eric Topol***

This work is dedicated to the incredible community of teachers, mentors, and classmates who
helped shape my understanding and curiosity. Your wisdom, shared enthusiasm,
and kind guidance have propelled me through every challenge.

Special thanks to my professors, who brought complex ideas to life, and to my mentors, whose faith in my potential and real-world insights
enriched my knowledge in ways a classroom alone could never provide.

To all of you, I am deeply grateful.

THESIS CERTIFICATE

This is to undertake that the Project Report titled **HYBRID DATA AUGMENTATION TECHNIQUES FOR DISEASE IDENTIFICATION FROM CLINICAL NOTES**, submitted by me to the Indian Institute of Technology Madras, for the award of **Master of Technology**, is a bona fide record of the research work done by me under the supervision of **Mr. Samyabrata Chakraborty (Lead Solution Architect & Head of Centre of Excellence, TCS), Mr. Debopam Nanda (Lead Data Scientist, TCS)** .

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ABSTRACT

Automated ICD coding has improved substantially in recent years, yet many systems struggle with rare disease codes because of limited training data and the high granularity of ICD-10-CM. This thesis addresses that shortfall by combining synthetic clinical text generation, knowledge-driven validation, and memory-efficient model training to enhance both accuracy and coverage for underrepresented codes. Large language models (LLMs) are employed to produce factually diverse discharge summaries, guided by ontologies such as SNOMED CT and Orphanet. A stringent multi-phase review pipeline—encompassing rule-based checks, LLM-based fact verification, and ICD feedback loops—filters out unrealistic samples and preserves label integrity. Subsequently, these validated synthetic records augment existing training datasets for multi-label classification models, including transformer-based architectures (e.g., PLM-ICD) and synonym-aware networks. Emphasis is placed on boosting recall for rare codes, where conventional methods typically falter. Improvements in macro-F1 are observed once data scarcity is alleviated through synthetic augmentation. In addition, practical constraints are addressed by integrating approaches such as gradient checkpointing, quantization (QLoRA), and knowledge distillation, enabling resource-limited systems to fine-tune sizable language models without compromising precision. Beyond performance metrics, the thesis underscores the importance of model transparency and trustworthiness. Techniques like attention visualization and SHAP-based attributions highlight key textual fragments responsible for each predicted label, offering insights that can support clinical validation and audit processes. Error analyses further uncover any systematic biases, guiding refinements to both the data generation and classification pipelines. By merging robust augmentation, structured knowledge, and carefully optimized training routines, this work advances a scalable, interpretable ICD coding framework that narrows gaps in rare disease coverage and paves the way for more comprehensive clinical AI solutions.

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NOTATION

α	A learning rate or weight factor in optimization algorithms
$\mathcal{L}$	The complete label space of ICD codes
BCE	Binary Cross Entropy, a common loss function for multi-label learning
F1	Harmonic mean of Precision and Recall, used for model evaluation
θ	Trainable parameters of the classification model
$\tilde{x}_j$	Synthetic clinical note generated for augmentation
D	Overall dataset of paired clinical notes and labels
$f(\cdot)$	The learned multi-label classification function mapping text to sets of labels
$P(Y \mid x; \theta)$	Predicted probability of codes Y given note x
x_i	The i-th clinical note (free-text discharge summary)
Y_i	The set of ICD codes associated with the i-th clinical note

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION TO THIS THESIS

The *International Classification of Diseases (ICD)* is a globally recognized coding system maintained by the World Health Organization (Organization, 2019), with its most recent iterations being ICD-10 and ICD-11. This system was developed to standardize the labeling of diseases, medical procedures, and underlying health conditions, thereby fostering interoperability across diverse healthcare environments. As a result, patient diagnoses and treatments are recorded more consistently, allowing healthcare providers to communicate more effectively, researchers to examine data with fewer ambiguities, and public health authorities to craft evidence-based policies.

When multiple ICD codes are assigned to a single patient's record, the information becomes richer by including primary diagnoses and related comorbidities. Large-scale databases, such as the *Medical Information Mart for Intensive Care-IV (MIMIC-IV)*, reflect this practice by covering not only the chief complaint but also secondary or rare conditions (Johnson *et al.*, 2023). A major concern, however, arises if any of these codes are misassigned. Inaccuracies and omissions can ripple through subsequent clinical decision-making, research conclusions, and billing outcomes. Hence, the role of accurate coding becomes paramount in bridging the gap between raw clinical documentation and reliable medical insights.

From a financial standpoint, healthcare institutions depend on correct coding to claim reimbursements, and errors—either over-coding or under-coding—risk financial imbalances. Studies have highlighted the direct impact of clinical documentation on billing accuracy and institutional revenue, stressing the importance of precision in ICD assignments (Dong *et al.*, 2022). At a population level, aggregated ICD-coded data forms the bedrock of epidemiological tracking, policy development, and larger-scale medical research. Erroneous codes can alter the trajectory of these studies, potentially skewing public health interventions (Edin *et al.*, 2023).

Nevertheless, the manual process of ICD coding remains both tedious and vulnerable to human error. Coders routinely parse extensive and often unstructured discharge summaries that include technical jargon, acronyms, or shorthand specific to a given clinical context (Wrenn *et al.*, 2010). Rare or lesser-known conditions risk being misrepresented or overlooked in the daily rush of high-volume coding workloads. Automated or semi-automated tools have therefore been suggested as an avenue for alleviating these problems. Such solutions provide preliminary code assignments that human coders can validate, thus streamlining workflow and improving data quality (Campbell and Giadresco, 2020).

Over the years, numerous computational methods have been devised to assist in the assignment of ICD codes. Early rule-based systems depended on dictionaries and handcrafted rules (Farkas and Szarvas, 2008), but these were limited in scalability and flexibility. Traditional machine learning techniques (e.g., support vector machines and logistic regression) made modest progress but often stumbled over the diverse, nuanced terminology of clinical text (Scheurwegs *et al.*, 2017; Perotte *et al.*, 2014). Momentum grew significantly with the advent of deep learning, starting with convolutional neural networks (CNNs) (Kim, 2014; Mullenbach *et al.*, 2018) and recurrent neural networks (e.g., LSTMs (Hochreiter and Schmidhuber, 1997)) that learned hierarchical or temporal patterns from unstructured notes. More recently, transformer-based models such as Bidirectional Encoder Representations from Transformers (BERT) have improved performance in many language tasks (Devlin *et al.*, 2019), though they often struggle with extremely long documents and heavily imbalanced data (Gao *et al.*, 2021; Huang *et al.*, 2022).

Several specialized strategies have since been proposed to address these challenges, including extended context transformers, hierarchical embeddings, and curriculum learning over the ICD taxonomy (Nickel and Kiela, 2017; Ren *et al.*, 2022; Bengio *et al.*, 2009). These refinements aim to capture relevant details in lengthy clinical narratives and to handle the long-tail distribution in which a few codes dominate while a majority of rare codes receive minimal coverage (Rios and Kavuluru, 2018). Alongside these methodological expansions, interpretability is receiving growing attention in automated ICD coding: clinicians must verify or scrutinize a model's reasoning to ensure that each assigned code is justified by actual clinical evidence (Holzinger *et al.*, 2017; Warner *et al.*, 2024).

In parallel, new lines of research have begun focusing on *entity recognition* and *ontology-based approaches* to enhance the robustness of coding systems. Tools like *cTAKES* (Savova *et al.*, 2010) and *MedCAT* (Kraljevic *et al.*, 2021) identify medical concepts in text and map them to standardized vocabularies, such as SNOMED CT (Donnelly, 2006) or the Unified Medical Language System (UMLS) (Bodenreider, 2004). Although these pipelines do not automatically produce final ICD codes, they offer structured terminological insights that can reduce ambiguity in free-text data (Wu *et al.*, 2018). More advanced systems incorporate ICD-specific knowledge directly into the classification process (e.g., hyperbolic embeddings of the ICD hierarchy (Cao *et al.*, 2020) or label descriptions drawn from official manuals). This knowledge-aware approach has consistently shown promise in bridging the gap between unstructured narratives and the highly specialized code sets used in clinical practice.

An especially difficult hurdle is how to improve the recognition of rare or "zebra" codes. When diagnoses occur infrequently, data-driven models often lack sufficient positive examples for generalization (Rios and Kavuluru, 2018). To combat this scarcity, researchers have begun to explore *synthetic data generation*. By leveraging large language models (LLMs) to create artificial discharge summaries or augment existing text, the model's exposure to underrepresented conditions can be increased (Falis *et al.*, 2022, 2024). Ontologies like SNOMED CT or the UMLS help ground these generated examples so that the synthetic narratives remain medically plausible (Kumichev *et al.*, 2024). For instance, a rare genetic disorder might have very few real cases in a training corpus; generating extra samples keyed to that condition's known symptomatology can yield substantial performance gains, particularly in recall for low-frequency labels.

These generative methods build on a broader trend in natural language processing, in which powerful transformers (e.g., GPT-3.5 or GPT-4) are used to produce text that mimics human writing. Indeed, recent work suggests that LLMs prompted with curated medical knowledge can synthesize relatively realistic patient histories (Yang *et al.*, 2023). Although concerns persist about potential hallucinations or a lack of full clinical richness, experiments have shown that carefully designed prompts—possibly incorporating knowledge graphs or code descriptions—can yield synthetic notes beneficial for model training. Complementary approaches use synonyms or sibling diagnoses from an ontology to systematically "perturb" or augment existing text (Yuan *et al.*, 2022), thus expanding the model's exposure to variant language for the same condition.

However, practical integration of these advanced methods requires consideration of real-world constraints. Data privacy regulations often limit the availability of large, richly annotated clinical corpora. Synthetic notes, if generated responsibly, can alleviate privacy concerns while still reflecting key features of real patient reports. Yet computational cost and resource constraints remain a challenge: training or fine-tuning long-context transformers can be expensive, and using LLM APIs extensively may be prohibitively costly for some healthcare institutions (Ji *et al.*, 2021). Techniques such as quantization, gradient checkpointing, or hierarchical batching (Geiping and Goldstein, 2023) can reduce resource demands. Another paramount concern is reliability: automated coding systems must provide transparent justifications for the codes they produce if they are to earn the trust of clinicians. Interpretable mechanisms, such as label-specific attention or post-hoc attribution methods (e.g., Integrated Gradients, SHAP), facilitate manual review and verification (Teng *et al.*, 2020).

To address these persistent challenges, this thesis proposes a framework that combines ontology-informed data augmentation, cutting-edge neural architectures, and interpretability techniques to enhance automated ICD coding. In particular, four major strategies anchor the work:

- **Synthetic Data Generation for Rare Codes:** Large language models are leveraged in tandem with medical ontologies (e.g., SNOMED CT, UMLS) to produce realistic training examples for less frequently occurring conditions (Kumichev *et al.*, 2024; Falis *et al.*, 2022). By enriching the dataset with synthetic discharge summaries, we strive to close the performance gap between common and rare codes.

- **Knowledge-Aware Multi-Label Architectures:** Neural models capable of handling thousands of code labels are examined, including transformer-based classifiers with extended input contexts. Techniques such as hyperbolic embeddings of the ICD hierarchy (Nickel and Kiela, 2017; Cao *et al.*, 2020) and label attention mechanisms (Vu *et al.*, 2020*b*) are integrated to improve performance and interpretability in multi-label classification.

- **Clinically Oriented Interpretability:** Interpretability remains crucial for adoption in clinical workflows. Methods like SHapley Additive exPlanations (SHAP), attention heatmaps, or concept-level attributions (Holzinger *et al.*, 2017; Teng *et al.*, 2020) are incorporated to help coders and physicians quickly validate or question the system's logic, especially for critical or rare diagnoses.

- **Resource-Conscious Training and Deployment:** Training strategies such as quantization and gradient checkpointing are used to mitigate heavy computational requirements (Geiping and Goldstein, 2023). This ensures that our proposed system remains accessible to healthcare organizations with limited GPU resources.

By uniting ontology-driven augmentation, state-of-the-art neural methods, and transparent validation, this work aims to significantly improve both the breadth and reliability of ICD coding systems. In so doing, we anticipate that more accurate, explainable coding will foster better clinical documentation, refine large-scale epidemiological analyses, and ultimately contribute to more effective patient care and public health interventions. The chapters that follow delve into the technical foundations, present empirical results, and discuss the real-world feasibility of deploying such a system in modern healthcare settings.

CHAPTER 2

LITERATURE REVIEW

2.1 INTRODUCTION AND THE EVOLUTION OF ICD CODING

The automation of the International Classification of Diseases (ICD) coding process has long been recognized as a critical area of research in medical informatics and Natural Language Processing (NLP). The ICD system, maintained by the World Health Organization, is used worldwide to classify diseases and health conditions for epidemiological research, healthcare management, and clinical billing. Accurate and efficient coding is essential because ICD codes capture details of a patient's diagnosis, comorbidities, and complications, thereby supporting reliable record-keeping and reimbursement processes (Organization, 2019).

Originally, the Ninth Revision (ICD-9) provided a limited number of disease categories. Over time, ICD-9 was phased out in favor of ICD-10, driven by the need for more specificity and additional capacity to accommodate new diseases (Dong *et al.*, 2022). ICD-10-CM (Clinical Modification) expanded the code set substantially (from approximately 13,000 to 68,000 codes) and introduced alphanumeric codes capable of encoding disease severity, anatomical site, and laterality (Edin *et al.*, 2023). Although this granularity supports more detailed patient records, it also increases complexity in manual coding workflows (Farkas and Szarvas, 2008).

The progression to ICD-11 (Organization, 2019) further reflects international efforts to enhance coding flexibility and align medical concepts more precisely with emerging public health priorities. Its hierarchical structure and additional extensions aim to capture nuances of disease presentation, enabling more robust epidemiological tracking. However, these regular revisions amplify the complexity for human coders, who must keep abreast of ever-expanding vocabularies and specialized rules (Scheurwegs *et al.*, 2017). The high stakes of accurate billing and statistical reporting underscore the importance of developing automated coding solutions that can function reliably, even when new codes and terminologies are introduced.

This chapter reviews the broad spectrum of approaches and challenges that characterize automated ICD coding. Early rule-based and machine learning (ML) methods are summarized, followed by deep learning approaches that leverage convolutional, recurrent, and transformer-based architectures. Recent improvements in long-context transformer designs, modern BERT-like encoders, knowledge-based methods, and curriculum learning strategies are discussed, highlighting progress in handling the extensive and imbalanced ICD label space. The chapter also explores data imbalance and rare-code prediction, synthetic data generation for underrepresented diagnoses, interpretability, and the practical challenges of deploying automated ICD coding systems in real-world clinical environments. Relevant evaluation metrics and future research directions are presented to offer a holistic overview of current capabilities and ongoing developments in this domain.

2.2 EARLY APPROACHES TO AUTOMATED ICD CODING

2.2.1 Rule-Based Systems and Traditional Machine Learning

Early efforts to automate ICD coding primarily involved rule-based systems and traditional machine learning techniques. Rule-based systems utilized handcrafted linguistic rules, domain-specific ontologies, and keyword matching heuristics to assign ICD codes to clinical narratives (Farkas and Szarvas, 2008; Scheurwegs *et al.*, 2017). These systems often achieved high precision under narrowly defined contexts because medical experts designed rules explicitly targeting common or well-understood diagnoses. Nonetheless, maintaining rule-based systems for extensive code sets was difficult; an entire rule set would often need re-engineering whenever revisions to ICD or local documentation practices emerged (Scheurwegs *et al.*, 2017).

Parallel to rule-based approaches, researchers applied supervised classifiers such as logistic regression, Decision Trees, or Support Vector Machines (SVMs) trained on bag-of-words or TF-IDF representations of clinical notes (Perotte *et al.*, 2014). Although these methods sometimes benefited from straightforward interpretability (e.g., top weighted words for each code), they struggled with the extremely large and imbalanced label space, as ICD-9-CM and ICD-10-CM collectively encompass thousands of possible codes. Traditional feature engineering often could not capture the subtle semantic patterns indicative of specific diagnoses, particularly when these patterns were distributed across long or unstructured notes. Moreover, such approaches did not perform well on rare codes, which had scant training examples (Duque *et al.*, 2021). Despite these limitations, these early systems established important baselines and indicated that integrating domain knowledge (e.g., simple dictionaries of medical terms) could enhance precision.

2.2.2 Limitations and Motivation for Deep Learning

Although traditional ML solutions revealed the potential of automated ICD coding, they were often constrained by high-dimensional data, sparse features, and a lack of capacity to fully capture clinical context. The low coverage of rare codes and the complex textual nature of clinical documentation prompted researchers to investigate more robust methods that did not rely heavily on labor-intensive rule sets or shallow textual features. The subsequent shift toward deep learning was motivated by these challenges, as neural architectures offered new pathways for automatically learning hierarchical textual representations with minimal handcrafting (LeCun *et al.*, 2015).

2.3 DEEP LEARNING FOR AUTOMATED ICD CODING

2.3.1 Convolutional and Recurrent Neural Network Approaches

Deep learning opened up new avenues for capturing semantic and syntactic nuances in clinical text. Convolutional Neural Networks (CNNs) were among the earliest neural architectures to be tested, owing to their effectiveness in extracting local n-gram features (Kim, 2014). In the context of ICD coding, Mullenbach et al. (Mullenbach *et al.*, 2018) introduced the Convolutional Attention for Multi-Label classification (CAML) model. CAML employed a convolutional encoder to represent textual segments, followed by a label-specific attention mechanism that highlighted the most relevant tokens for each code prediction. The increased interpretability of this attention mechanism was particularly appealing for healthcare applications, where transparency is critical:

$$\alpha^{(l)} = \text{softmax}(W^{(l)} h + b^{(l)}), \tag{2.1}$$

where h denotes the hidden representations from the convolutional layers, and $W^{(l)}$, $b^{(l)}$ are label-specific parameters.

Subsequent CNN-based models explored deeper architectures with residual connections and multi-filter designs. For instance, Li and Yu (Li and Yu, 2020) proposed the Multi-Filter Residual Convolutional Neural Network (MultiResCNN), which combines different kernel sizes and residual pathways to capture diverse syntactic and semantic patterns in clinical text. These refinements demonstrated measurable gains in F1-scores on benchmarks like MIMIC-III and other hospital datasets.

Recurrent Neural Networks (RNNs) offered an alternative strategy by encoding sequences of tokens in a step-by-step manner. Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997) and Gated Recurrent Units (GRUs) were particularly suited for capturing longer-range dependencies in clinical notes. Vu et al. (Vu *et al.*, 2020*a*,*b*) employed a bidirectional LSTM with a label attention mechanism, where each label learned distinct attention weights over the text. This enabled more targeted coding decisions, accounting for the context needed by each diagnosis or procedure label. Although RNN models captured sequential information better than many CNN designs, they remained prone to vanishing gradients on very long documents and could be computationally expensive.

By the late 2010s, CNN- and RNN-based models had markedly outperformed traditional ML baselines (Perotte *et al.*, 2014), demonstrating that neural networks could automatically learn powerful text features. Nevertheless, these architectures often required careful truncation or sampling of long notes, and they still struggled under conditions of data imbalance or when handling thousands of rarely seen ICD codes (Wrenn *et al.*, 2010).

2.4 TRANSFORMER-BASED ARCHITECTURES AND EXTENSIONS

2.4.1 Standard Transformers and Clinical Adaptations

Transformer-based models, especially BERT (Devlin *et al.*, 2019), revolutionized NLP by replacing the sequential encoding of RNNs with self-attention over entire sequences. This design facilitated more direct capture of long-range dependencies. However, standard BERT has a typical input limit of 512 tokens, which poses a challenge in clinical coding, where notes can span thousands of tokens (Gao *et al.*, 2021). Early attempts to integrate BERT into ICD coding often truncated notes, losing critical data.

Huang et al. (Huang *et al.*, 2022) introduced the PLM-ICD model, which segments each discharge summary into manageable chunks and processes them with a pre-trained BERT. A label attention mechanism then aggregates chunk representations that are relevant to specific codes. Although such approaches allowed BERT-based solutions to outperform many CNN and RNN baselines on micro-F1 scores, truncated documents risked overlooking important evidence for certain diagnoses (Dong *et al.*, 2022).

Heo et al. (Heo *et al.*, 2022) addressed this long-text issue by proposing the Document-to-Sequence BERT (D2SBERT) framework. It divides the full document into multiple fixed-length segments and processes each segment independently through BERT. A sequence-level attention mechanism then merges these representations, capturing essential information across the entire document. This method outperformed various CNN-based benchmarks on MIMIC-III, demonstrating that transformer-based models can effectively handle extensive notes through segmenting and hierarchical attention.

2.4.2 Modern BERT-Based Encoders for ICD Coding

Researchers also investigated architectural modifications and domain-specific pretraining to accommodate larger input lengths, improve inference speeds, and address the domain-specific vocabularies of clinical notes. Several recent works introduce novel BERT-like encoders:

- **MosaicBERT** (Portes *et al.*, 2023) and **CrammingBERT** (Geiping and Goldstein, 2023): Optimize training efficiency, allowing them to achieve comparable BERT performance with shorter training times. They do not fundamentally resolve the difficulties associated with extremely long narrative texts but reduce overall resource demands.

- **NomicBERT** (Nussbaum *et al.*, 2024) and **GTE-en-MLM** (Zhang *et al.*, 2024): Extend the maximum input length from 2k to 8k tokens and introduce advanced unpadding methods. These models exhibit modest performance improvements on retrieval tasks and partially address memory constraints by employing sparser attention patterns.

- **ModernBERT** (Warner *et al.*, 2024): Incorporates architectural innovations such as alternating local-global attention, Rotary Positional Embeddings (RoPE), and label-aware attention modules. ModernBERT consistently outperforms prior baselines on both short and long ICD coding tasks, while reducing inference memory usage up to 8k tokens. Its hardware-aware design exemplifies how targeted architectural choices can mitigate computational overhead.

Despite progress, BERT-like transformers continue to encounter difficulties with extreme class imbalance, especially for codes encountered infrequently. Hardware constraints remain challenging when notes exceed thousands of tokens. Nonetheless, these tailored encoder designs underscore that carefully balancing domain knowledge, input length, and memory efficiency can produce state-of-the-art clinical coding models.

2.5 CURRICULUM LEARNING AND HIERARCHICAL KNOWLEDGE INTEGRATION

The hierarchical structure of ICD codes naturally suggests a curriculum learning approach, where models are initially exposed to broad categories before proceeding to more granular ones. Bengio et al. (Bengio *et al.*, 2009) formally introduced curriculum learning as a strategy to improve model performance by scheduling examples from simple to difficult.

Ren et al. (Ren *et al.*, 2022) applied a hierarchical curriculum (HiCu) by organizing ICD codes into tree-like levels. During training, they gradually refined predictions from higher-level ICD groupings (e.g., “diseases of the circulatory system”) down to more specific subcodes. A hyperbolic embedding mechanism (Nickel and Kiela, 2017) transferred knowledge from parent-level predictions to child-level nodes, enabling the model to leverage relationships among ICD codes. This approach mitigated some aspects of data imbalance, demonstrated by improved macro-F1 and macro-AUC scores, particularly for infrequent codes. Consequently, curriculum learning has emerged as a complementary strategy to neural architectures, aligning naturally with the structured taxonomies in ICD-9-CM, ICD-10-CM, and ICD-11.

2.6 CHALLENGES IN AUTOMATED ICD CODING

2.6.1 Data Imbalance and Rare Codes

Automated ICD coding is inherently an extreme multi-label classification problem: a single patient note may contain multiple relevant codes among tens of thousands of possible labels. The distribution of ICD codes is typically long-tailed, where common chronic conditions such as hypertension or diabetes occur frequently, while countless rare conditions appear only a handful of times in training data (Rios and Kavuluru, 2018). In the MIMIC-III dataset, over 50% of codes never appear or occur fewer than ten times (Johnson *et al.*, 2016), making it challenging for models to learn robust representations for infrequent diagnoses.

This imbalance leads to a disparity in model performance: systems that do well on common labels often fail to recall rare codes. Edin et al. (Edin *et al.*, 2023) conducted a replicability study that exposed how certain train-test splits and suboptimal macro F1 calculations led to inflated performances for rare-code prediction. Their work emphasized fairer evaluation and transparent reporting of model limitations. Similarly, Nguyen et al. (Nguyen *et al.*, 2023) showed that MIMIC-IV features a more extreme code distribution, since it merges ICD-9 and ICD-10 subsets. Handling this imbalance remains a top priority, as rare codes can indicate serious or complex conditions requiring accurate documentation and billing.

Curriculum learning strategies (Ren *et al.*, 2022) partially address this issue by exploiting code hierarchies to share knowledge between parent and child labels. Additional techniques, such as oversampling and cost-sensitive learning, have also been tested, but they have been less effective at scale (Scheurwegs *et al.*, 2017). Consequently, novel approaches—particularly synthetic data augmentation—are garnering growing interest in mitigating the challenge of rare code prediction.

2.6.2 Processing Long and Complex Clinical Documents

Clinical narratives can be lengthy and may contain redundancies, inconsistencies, and a wide variety of clinical terminology. Early studies suggested that longer documents degrade model performance, but Edin et al. (Edin *et al.*, 2023) found the effect of length to be secondary to other factors such as code frequency. Nonetheless, managing long clinical notes remains computationally demanding.

Transformer-based models often face input length constraints (e.g., 512 or 1024 tokens), necessitating strategies such as truncation (Gao *et al.*, 2021), chunking (Huang *et al.*, 2022), or hierarchical encoding (Heo *et al.*, 2022). Although advanced architectures like Longformer or BigBird can theoretically handle up to 8k tokens or more, they are memory-intensive and can be slow to train. Document summarization and section-based modeling present additional strategies but risk losing crucial detail about specific diagnoses. Finding scalable, efficient ways to represent entire discharge summaries remains an important research focus for real-world deployments.

2.6.3 Explainability and Interpretability

Explainability is essential in clinical applications, where clinicians, auditors, and coding professionals must trust automated coding outputs (Holzinger *et al.*, 2017). Attention-based methods, such as CAML (Mullenbach *et al.*, 2018) or LAAT (Vu *et al.*, 2020*a*), highlight relevant text segments for each predicted code, providing intuitive evidence for model decisions. However, these attention weights only offer correlational explanations—further research is needed to establish more causal or structured forms of explainability.

Edin et al. (Edin *et al.*, 2024) proposed an unsupervised approach that can achieve supervised-level explanation quality, suggesting that external knowledge or adversarial training may produce rationale annotations without costly manual labeling. Additionally, researchers have explored knowledge graphs and symbolic reasoning to justify individual code predictions (Teng *et al.*, 2020). Although these methods can yield greater transparency, fully interpretable models that seamlessly integrate domain knowledge remain an open challenge.

2.7 EMBEDDING MODELS AND CLINICAL SEMANTIC SEARCH

Embedding models underlie many of the text representation and retrieval tasks in healthcare. They transform sparse textual data into dense vectors that encode semantic relationships among words, sentences, or entire documents. Excoffier et al. (Excoffier *et al.*, 2024) compared generalist transformers (trained on broad corpora) to specialized clinical embeddings for ICD semantic search, observing that generalist embeddings were often more robust to minor text rephrasings. This finding implies that domain-specific pretraining might not always be superior if it narrows the model's domain to patterns less frequent or less stylistically varied than the general language.

Nevertheless, domain-adapted models like ClinicalBERT, BioBERT, and PubMedBERT have achieved better performance in several medical NLP tasks, including ICD coding (Gao *et al.*, 2021; Ji *et al.*, 2021). The discrepancy in outcomes suggests that training set size, the nature of clinical text, and the distribution of code descriptions can all influence whether generalist or specialized embeddings hold the advantage. Moreover, specialized strategies such as code-synonym expansion (Yuan *et al.*, 2022) and concept-level attention highlight how embedding-based approaches can leverage existing biomedical ontologies (e.g., UMLS (Bodenreider, 2004), SNOMED CT (Donnelly, 2006), or cTAKES (Savova *et al.*, 2010)) to improve both performance and explainability.

2.8 KNOWLEDGE-BASED METHODS AND SYMBOLIC INTEGRATION

Knowledge-based methods aim to incorporate medical ontologies, taxonomies, and domain rules into deep learning architectures. These techniques can address shortcomings in purely data-driven models by providing explicit relationships between diseases or leveraging synonyms and hierarchical groupings (Xie *et al.*, 2019; Teng *et al.*, 2020). Such integration is especially valuable in ICD coding, where label space is inherently hierarchical, and textual evidence for certain diagnoses may be sparse.

Ren et al. (Ren *et al.*, 2022) demonstrated that hyperbolic embeddings capture hierarchical structure better than Euclidean embeddings, particularly for ICD codes. By representing codes in a hyperbolic space, parent-child relationships are learned with low distortion, and training proceeds from broader categories to finer levels. Similarly, Cao et al. (Cao *et al.*, 2020) proposed a hyperbolic co-graph representation that leveraged hierarchical information to enhance predictive performance. Knowledge-based approaches thus help unify symbolic reasoning with data-hungry neural models, thereby improving interpretability and filling gaps where training examples are limited.

2.9 SYNTHETIC DATA GENERATION FOR RARE ICD-10-CM CODES

2.9.1 Motivation and Techniques

The extreme class imbalance of ICD codes—where many rare diagnoses have few or no training examples—has motivated synthetic data generation as a complementary strategy to curriculum learning and data augmentation. Large Language Models (LLMs), such as GPT-3.5, GPT-4, or fine-tuned LLaMA, are increasingly employed to create realistic clinical narratives for underrepresented diagnoses (Kumichev *et al.*, 2024; Falis *et al.*, 2024). These synthetic notes preserve confidentiality while expanding the variety of text samples available for training models.

Falis et al. (Falis *et al.*, 2024) examined the feasibility of prompting GPT-3.5 with specific ICD-10-CM codes and descriptions to generate discharge summaries. Adding these synthetic notes to real training data produced modest performance improvements on rare codes and helped the model avoid out-of-family misclassifications. However, these gains came at the cost of potentially overfitting to the style of generated text, which might be less diverse than real clinical documentation.

Other augmentation methods include synonym replacement, back-translation, or adversarial generation of latent feature vectors, rather than raw text. Although these approaches are less computationally demanding than prompting large LLMs, they often provide smaller increases in coverage for rare labels. Hybrid strategies that combine knowledge-graph constraints with generative models have also been explored, ensuring that synthetic samples remain clinically plausible (Falis *et al.*, 2022).

2.9.2 Ethical and Practical Considerations

Synthetic data introduce potential advantages for privacy protection, as real patient identifiers can be omitted or obfuscated. Nevertheless, generative models might inadvertently reproduce sensitive or identifiable fragments if they encountered them during training. Ensuring high-fidelity clinical plausibility is also essential; simplistic or repetitive patterns within synthetic samples could bias the model toward unrepresentative language (Pollard *et al.*, 2018).

In practice, synthetic notes appear to be most beneficial in few-shot or one-shot contexts, where the model has a handful of real examples for a rare code and gains additional variety through generation (Rios and Kavuluru, 2018). Full replacement of real data is typically detrimental. As a result, synthetic data remain a supplement rather than a primary information source, especially for regulating or auditing high-stakes billing environments.

2.10 BENCHMARKS AND EVALUATION METRICS

2.10.1 Public Datasets and Benchmarks

The research community has benefited significantly from openly available clinical datasets such as MIMIC-III (Johnson *et al.*, 2016) and MIMIC-IV (Johnson *et al.*, 2023). MIMIC-III focuses on ICD-9-CM codes, while MIMIC-IV provides a larger number of hospital admissions and reflects the transition to ICD-10-CM coding (Nguyen *et al.*, 2023). Both datasets contain lengthy discharge summaries with manually assigned ICD codes, making them ideal for evaluating multi-label classification performance. The eICU Collaborative Research Database (Pollard *et al.*, 2018) supplies additional notes and structured data across multiple hospital systems, enabling analyses of model generalizability.

Benchmark initiatives standardize tasks, train-dev-test splits, and evaluation pipelines. This fosters comparability among different architectures (e.g., CNNs vs. BERT vs. hierarchical models) and clarifies progress in rare-code prediction. Replication studies have demonstrated that small variations in preprocessing or code filtering can materially affect reported metrics, emphasizing the need for consistent benchmark protocols (Edin *et al.*, 2023).

2.10.2 Common Evaluation Metrics

Because automated ICD coding is a multi-label task, precision and recall must be balanced to capture different aspects of performance:

- **Micro-F1**: Aggregates all predicted and true labels across the dataset, giving higher weight to frequently occurring codes. This is typically higher than macro-F1 and often used to illustrate a model's overall performance on the bulk of common codes.

- **Macro-F1**: Computes the F1-score per code and then averages across all codes, giving equal importance to rare and common labels. Macro-F1 is sensitive to performance on the long tail of infrequent codes and is usually much lower.

- **Micro/Macro AUC**: Measures the area under the ROC curve, either pooling predictions across codes (micro) or taking a simple average of per-code AUC (macro). High AUC often indicates that, with careful threshold tuning, the model could achieve good recall and precision.

- **Precision@K**: Assesses how many of the top K predicted codes match the ground truth. This metric is sometimes used in practical coding assistance tools to limit the suggestions displayed to human coders.

- **Hierarchical Metrics**: Account for partial correctness if a model predicts a closely related code from the same family, as opposed to a completely unrelated code. Such metrics better reflect real-world coding tasks, where distinguishing the correct subtype can be secondary to identifying the right disease category.

Error analyses often examine whether incorrect predictions are at least within the correct organ system or parent node in the ICD hierarchy. Some studies also evaluate interpretability by checking if highlight-based explanations align with key phrases annotated by clinical experts (Mullenbach *et al.*, 2018). These deeper analyses provide a more nuanced picture of model behavior than a single F1 score.

2.11 FUTURE DIRECTIONS AND OPEN PROBLEMS

Despite the substantial progress described thus far, automated ICD coding continues to face many open challenges:

Self-Supervised and Domain-Adaptive Pretraining. The majority of current models rely on supervised training with limited labeled data. Future systems may exploit large unlabeled clinical corpora through advanced self-supervised objectives or domain-adaptive methods (Gururangan *et al.*, 2020). Domain-specific LLMs could capture complex medical language better than general-purpose ones, potentially reducing reliance on manually annotated data.

Few-Shot and Zero-Shot ICD Coding. The long-tailed distribution of ICD codes motivates the development of architectures that can accurately code new or rare labels with minimal training examples (Rios and Kavuluru, 2018; Yang *et al.*, 2023). Models that incorporate code definitions or hierarchical relations—along with advanced prompting techniques—could infer rare labels even when they appear infrequently or not at all in training data.

Multimodal Integration. Clinical decision-making involves not only unstructured text but also structured data (e.g., labs, vitals) and imaging findings (Choi *et al.*, 2016). Effective multimodal architectures that combine textual discharge summaries with time-series or image-based signals remain under-explored but hold potential for more holistic coding.

Interpretability and Interactive Learning. Although attention-based and highlight-driven methods provide partial transparency, future research may explore causal explanations, prototype-based retrieval, or rationale generation to reinforce clinicians' trust. Interactive AI coding assistants could prompt human coders for clarification, use that feedback to refine predictions, and thus achieve greater accuracy over time.

Ethical and Regulatory Considerations. Clinical data are sensitive, and large-scale LLM usage must comply with data protection laws. Synthetic generation can mitigate privacy concerns but raises questions about realism and potential bias. Regulatory acceptance may require robust auditing mechanisms, reproducible evaluation, and evidence that automated coding does not compromise patient safety or billing integrity.

ICD-11 and Beyond. The introduction of ICD-11 presents new structural and semantic complexities. Transferring knowledge from ICD-9 or ICD-10 to ICD-11 demands flexible architectures capable of adapting to new taxonomies. Approaches that use hierarchical embeddings and code-description modeling can help manage these shifts in labeling systems (Gaebel *et al.*, 2020).

2.12 SUMMARY OF THE LITERATURE

Automated ICD coding presents a set of distinct technical and organizational challenges. Initial studies employed rule-based logic and feature-engineered machine learning but were constrained by scalability and the evolving nature of clinical vocabularies (Scheurwegs *et al.*, 2017). Deep neural architectures, particularly CNNs, RNNs, and transformers, provided significant accuracy improvements by learning hierarchical text representations without extensive manual engineering (Mullenbach *et al.*, 2018; Li and Yu, 2020; Heo *et al.*, 2022).

Transformer-based models and newer BERT-like encoders excel at capturing complex linguistic features in discharge summaries. However, they require careful handling of long documents, face persistent issues with rare diagnoses, and demand large computational resources. Curriculum learning, hierarchical embeddings, and synthetic data generation offer partial remedies, enhancing performance on uncommon codes and improving interpretability through structured knowledge infusion (Ren *et al.*, 2022; Falis *et al.*, 2024).

Recent benchmarking efforts clarify that future progress lies in robust domain-specific pretraining, advanced handling of extreme multi-label imbalances, and deeper integration of medical ontologies. Additionally, evolving demands—such as ICD-11 coding—underline the importance of flexible systems that adapt to changing taxonomies. The development of interactive, interpretable coding assistants, combined with rigorous evaluation and continuous learning, is expected to guide the field toward reliable real-world implementation in clinical and administrative workflows.

CHAPTER 3

PROBLEM DEFINITION AND FORMULATION

3.1 SCOPE OF THE PROBLEM

Accurate labeling of clinical notes with *International Classification of Diseases (ICD)* codes is recognized as a critical task in contemporary healthcare documentation. Clinical narratives, however, often manifest as lengthy, highly unstructured text that contains abbreviations, domain-specific terminology, and nuanced clinical indicators. A further complication arises when only a small fraction of ICD codes—here termed "rare codes"—occurs so infrequently that very few labeled instances are available for machine learning models. These codes frequently remain under-detected, which can diminish overall coding accuracy and potentially undermine large-scale health analytics. The objective of the present work is to automate the assignment of *multiple ICD codes* to each clinical document, with a specific focus on *improving predictive effectiveness for rare labels*.

Failure to correctly classify clinical narratives or to assign all relevant labels can have far-reaching consequences. At the institutional level, coding errors may result in imprecise billing processes and resource misallocation. At the individual level, a missed or erroneous diagnosis code may obscure a patient's condition or hamper effective disease management. On a broader scale, large-scale analytics that rely on aggregated administrative data may not capture the true incidence or prevalence of rare medical conditions, causing gaps in research and healthcare policy.

3.2 CENTRAL CHALLENGES

Although recent advances in machine learning have significantly improved automated ICD coding, several formidable obstacles persist:

- **Long-Tail Distribution:** The majority of available data is dominated by a few frequent codes, leaving rare diagnoses severely underrepresented.
- **High-Dimensional Label Space:** Each clinical document may include multiple ICD codes, and semantic interdependencies among these labels complicate multi-label classification.
- **Textual Complexity:** The unstructured nature of clinical notes, featuring inconsistent formatting and heavy use of abbreviations and domain-specific jargon, complicates text processing.
- **Knowledge Integration:** Existing ontologies contain valuable domain knowledge, but straightforward classification pipelines often overlook hierarchical or semantic relationships among medical codes.
- **Explainability Needs:** Models designed for clinical applications must be comprehensible to end-users, who require transparency and trust when evaluating automatic code assignments.

These challenges pose general difficulties for all ICD codes, but the detrimental effects are exacerbated for rare labels. Systems that perform well for frequently observed diagnoses often fail to capture low-frequency conditions, an omission that motivates the development of strategies tailored to address class imbalance and explainability gaps.

3.3 MATHEMATICAL FORMULATION OF THE TASK

A concise mathematical formulation is presented below to clarify how a trained model should transform unstructured clinical text into accurate ICD predictions.

3.3.1 Defining the Space of Clinical Narratives

Let $\mathcal{X}$ represent the collection of all possible clinical narratives. Each element $x \in \mathcal{X}$ is a textual record describing a patient's clinical episode. In practice, x may include:

- *Admission notes* reflecting the circumstances motivating a patient's hospitalization.
- *Discharge summaries* documenting final diagnoses, interventions, and follow-up steps.
- *Progress notes* covering daily clinical updates for extended hospital stays.

Despite a variety of internal structures and formats, each x can be treated as a string of tokens (or words) for classification purposes. Thus, the set $\mathcal{X}$ defines the *input space* over which the model operates.

3.3.2 Defining the Label Space of ICD Codes

Let $\mathcal{L} = \{\ell_1, \ell_2, \ldots, \ell_{|\mathcal{L}|}\}$ be the full set of ICD codes. The notation $|\mathcal{L}|$ denotes the number of distinct codes, which can be large in real-world applications:

- Major ICD revisions (ICD-9, ICD-10, ICD-11) contain thousands of codes.
- Many codes represent specialized conditions (subtypes or sub-phenotypes).

Since multiple diagnoses are commonly documented for a single admission, the label space is multi-label in nature, allowing each clinical document to be assigned any subset of $\mathcal{L}$.

3.3.3 Labeled Dataset Construction

Assume that a set of N labeled clinical narratives is available:

$$D = \{(x_i, Y_i)\}_{i=1}^{N},$$

where $x_i \in \mathcal{X}$ and $Y_i \subseteq \mathcal{L}$. Each pair (x_i, Y_i) consists of a discharge summary (or similar record) and the corresponding ICD codes assigned by professional coders. This dataset D forms the basis for supervised learning.

3.3.4 Function Definition

The goal is to learn a function

$$f : \mathcal{X} \longrightarrow 2^{\mathcal{L}}, \tag{3.1}$$

where $f(x) = \hat{Y}$ for $\hat{Y} \subseteq \mathcal{L}$. Here, $2^{\mathcal{L}}$ represents the power set of $\mathcal{L}$. For any clinical document x, the prediction $\hat{Y}$ is the subset of codes that the model deems relevant. In principle, $\hat{Y}$ can be empty, although that outcome is uncommon in practical medical scenarios.

In most current architectures, a multi-label classification approach is adopted. Specifically, the model generates a probability or score for each code in $\mathcal{L}$, and thresholding (or a related method) is then applied to map those scores to a predicted subset.

3.3.5 Objective: Composite Metric Maximization

A composite performance metric Ω is introduced to reflect the importance of both overall correctness and the accurate identification of rare labels. In formal terms, the aim is:

$$\underset{f}{\text{maximize}} \quad \Omega(\text{Accuracy of } f, \ \text{Rare-Codes Metrics}). \tag{3.2}$$

Depending on institutional priorities, Ω may combine measures such as micro-F1 or micro-AUC (evaluating overall performance across all codes) with macro-F1 or a specialized metric that focuses on rare codes. A weighted sum or a multi-objective formulation can be employed. This dual emphasis on both frequent and rare diagnoses ensures that gains do not come at the expense of clinically significant but infrequent conditions.

3.4 SIGNIFICANCE OF RARE CODES

Rare codes pose a particular challenge due to limited representation in real-world data, yet these codes often signal clinically important conditions.

3.4.1 Definition and Challenges

A rare code may appear only a few times in a large dataset. Let $\mathcal{L}_{\text{rare}} \subseteq \mathcal{L}$ be the set of codes whose frequencies fall below a chosen threshold τ:

$$\mathcal{L}_{\text{rare}} = \{\ell \in \mathcal{L} \mid \text{Frequency}(\ell) < \tau\}.$$

Typical values for τ might be a fixed count (e.g., 50) or a percentile (e.g., the bottom 10%). Learning to detect rare codes is complicated by these labels' scarcity in training data and the specialized language often observed in notes that reference such conditions.

Overall loss functions tend to favor frequent labels, marginalizing rare ones.

3.4.2 Impact of Rare Code Misclassification

Misclassification of rare diagnoses can produce substantial adverse effects:

- **Clinical Impact:** A rare but critical diagnosis, if missed by an automated system, may affect patient care when clinicians rely on coded data for decision support.
- **Research Bias:** Underrepresentation of rare conditions in large-scale analyses can mask the true disease burden.
- **Billing and Compliance:** Inaccurate coding for rare conditions can lead to underpayment or claim denial, as well as potential auditing complexities.

Consequently, rare code detection forms a key component of robust automated coding systems.

3.4.3 Recall and Precision Constraints for Rare Codes

For a clinical note x with true codes Y, where $Y \cap \mathcal{L}_{\text{rare}} \neq \emptyset$, recall and precision for the rare portion are of particular concern. Low recall implies a failure to capture rare conditions, whereas low precision may yield an excess of false positives for these labels. Either error type can be problematic in a medical context, emphasizing the importance of balanced metrics.

3.4.4 Analytical Example of Class Imbalance

A simple numerical illustration can clarify the issue. Consider 5,000 labeled notes, and suppose:

- A frequent code ℓ_{common} appears in 1,200 notes.
- A rare code ℓ_{rare} appears in 20 notes.

A classifier that never predicts ℓ_{rare} incurs only 20 false negatives on a base of 5,000, so the impact on the global accuracy may be less than 1%. Such a result can inadvertently encourage ignoring rare codes. This outcome underscores the need for targeted strategies addressing low-frequency classes.

3.5 PROPOSED METHODOLOGICAL FOCUS

Several methodological approaches are planned to mitigate performance deficits in both frequent and rare ICD codes:

1. **Data Augmentation for Rare Codes:** Synthetic data generation or specialized oversampling strategies that boost representation of low-frequency labels.

2. **Advanced Multi-Label Architectures:** Network designs that incorporate hierarchical embeddings and structured label representations, taking advantage of semantic relationships within the ICD taxonomy.

3. **Integration of External Knowledge:** Leveraging medical ontologies (e.g., SNOMED, UMLS) to enhance label embeddings and encourage clinically consistent classification.

4. **Model Interpretability:** Developing explainable frameworks where predicted ICD codes can be traced back to their textual justifications, thereby fostering user trust.

Each of these elements seeks to address the data imbalance inherent in ICD coding tasks while preserving or improving performance on all relevant labels, including those with scarce training instances.

3.6 EVALUATION FRAMEWORK

Evaluation in this domain requires a range of measures to capture overall performance and rare-code outcomes:

- **Micro-F1**: Reflects globally aggregated precision and recall.

- **Macro-F1**: Averages precision and recall across codes, thereby placing additional emphasis on rare labels.

- **Recall@k**: Indicates how effectively the model recovers correct codes in its top-k predictions, a practical measure for coding assistance tools.

- **Rare-Code Metrics**: Tracks performance (e.g., precision, recall, F-score) on the subset $\mathcal{L}_{\text{rare}}$, ensuring improvements are realized for both frequent and rare labels.

A balanced presentation of both micro- and macro-level metrics is essential, given the skewed distribution of ICD codes. It is possible for a model to exhibit strong micro-F1 yet fail to detect many rare diagnoses, which would be revealed by a lower macro-F1 or poor rare-code metrics.

3.7 CONSTRAINTS AND OBJECTIVES

Practical constraints and overarching objectives guide this research:

- **Data Privacy**: Clinical text is sensitive, and synthetic data or augmentation techniques must preserve confidentiality.
- **Computational Resources**: Many clinical institutions have limited hardware, thus requiring solutions that balance accuracy with feasible computational costs.
- **Transparency and Adoption**: Models must present interpretable outputs so that clinical personnel can audit and trust the system, particularly for high-stakes or unfamiliar diagnoses.

Consequently, the proposed methods will be designed to accommodate these practical considerations, alongside the main objective of enhancing performance for both common and infrequent ICD codes.

CHAPTER 4

METHODOLOGY

4.1 INTRODUCTION AND SCOPE

This chapter sets forth a *multi-phase methodology* aimed at refining automated ICD-10-CM code prediction, *especially* for rare codes. A comprehensive framework is proposed, featuring an *ontology-based disease grouping* strategy, a pipeline for *synthetic discharge summary generation*, a *hybrid model training* approach that integrates both real and synthetic data, and an evaluation scheme specifically *attuned* to rare code distribution imbalances. A *modular and extensible workflow* is emphasized, so that the methodology may be adapted readily to alternative scenarios involving imbalanced classification. Ultimately, validation is planned on a real-world clinical dataset to assess the *generalizability* and *utility* of this framework.

4.2 ONTOLOGY-BASED DISEASE GROUPING

Rare ICD-10-CM codes frequently exhibit *extremely low prevalence*, which hinders the ability of predictive models to learn discriminative features. In order to mitigate this challenge, an *ontology-based grouping* approach has been envisioned. This process entails parsing established medical ontologies—such as *SNOMED CT*, *UMLS*, and the *Orphanet Rare Disease Ontology*—to merge *hierarchical* and *associative* relationships.

It is intended that each rare disease will be anchored in a *knowledge graph*, and *closely related* concepts will be identified via direct graph traversal. These items will be clustered into a *unified disease family*, thereby providing a *contextual* lens through which the rare disease can be understood. By associating each rare disease with more common conditions, the model will receive exposure to realistic clinical context, improving its capacity to recognize the *linguistic cues* relevant to rare diagnoses. Additionally, ontology mapping ensures that no *orphaned codes* are disregarded, given that each disease is matched at least to a more general ICD-10-CM code if an exact match proves elusive.

It is anticipated that this ontology-centric approach will enable more *robust learning* for rare codes, because they will be presented alongside clinically related diagnoses. By placing rare codes in *medically coherent families*, the methodology reduces the isolation that often occurs when codes appear only a handful of times in conventional datasets.

4.3 SYNTHETIC DISCHARGE SUMMARY GENERATION

Because rare codes remain *infrequent* in standard patient records, a *synthetic data generation* process is introduced to amplify the presence of these *underrepresented labels*. A retrieval-augmented generation pipeline has been planned, wherein medical facts, professional guidelines, and relevant textual fragments are *retrieved* to guide a large language model in producing draft discharge summaries.

- **Structured Prompting:** Placeholders will be embedded for sections such as *Chief Complaint*, *History of Present Illness*, *Hospital Course*, *Medications*, and *Discharge Diagnoses*.
- **Disease Clusters:** Rare diseases will be linked with related diagnoses identified through ontology-based grouping, ensuring that *realistic co-occurrences* appear in the synthetic text.
- **Multi-Agent Feedback Loop:**
  - A *draft discharge summary* is generated by the language model.
  - A *critic agent* then examines the draft for logical inconsistencies and *missing details*.
  - Iterative refinements are made until the note meets *plausibility* and *consistency* criteria.

The final *synthetic discharge summaries* must provide sufficient evidence for the relevant rare ICD-10-CM codes, mirroring the style and structure of authentic clinical documentation. This balanced approach is expected to *significantly* enrich the training dataset for rare codes, thereby reducing distributional skew and enhancing overall model recall for these infrequent conditions.

4.4 DEEP LEARNING MODEL TRAINING AND TESTING

An *augmented training corpus* composed of both real and synthetic discharge summaries will subsequently be employed in training two deep learning architectures: (i) a *transformer-based* model, and (ii) a *CNN-based* model. The transformer-based model is *anticipated* to capture *long-range dependencies*, whereas the CNN-based model should excel at identifying *localized textual patterns* pertinent to specific codes.

A multi-label classification scheme will be used, assigning probabilities to all potential ICD-10-CM codes. To address *class imbalance*, strategies such as *weighted sampling*, *focal loss*, and *class-weighted loss* have been planned. Synthetic summaries will be *oversampled* or *weighted* more heavily, ensuring that rare codes appear frequently enough in training batches. The training procedure will commence with *binary cross-entropy* and moderate class weighting, followed by the incremental introduction of *focal loss* during later epochs, thus allowing the model to *fine-tune* its attention on the *most difficult* labels.

A development set containing both real and synthetic notes will guide *hyperparameter tuning* and *early stopping*. To confirm the *practical value* of this method, the final evaluation will be conducted on a *held-out portion* of real MIMIC-IV discharge summaries, assessing whether improvements in *rare code classification* carry over to genuine patient data.

4.5 EVALUATION METRICS AND ANALYSIS

The planned evaluation seeks to capture *various dimensions* of performance, with special focus on rare code detection. Several metrics will be reported, each *illuminating* a different aspect of classifier efficacy. The following formulas present the principal metrics in detail (where TP = true positives, FP = false positives, FN = false negatives, and C = total number of classes):

1. **Micro-F1**

$$\text{Micro-F1} = 2 \times \frac{\sum_{i=1}^{C} TP_i}{\sum_{i=1}^{C} (2 \cdot TP_i + FP_i + FN_i)} \tag{4.1}$$

   Micro-F1 aggregates the contributions of all codes to compute the overall precision and recall across the entire label space.

2. **Macro-F1**

$$\text{Macro-F1} = \frac{1}{C} \sum_{i=1}^{C} \text{F1}_i \tag{4.2}$$

   Each class-level F1 score F1_i is calculated, and their average is taken to highlight how the model performs on *each individual* label, giving rare codes equal weight to common codes.

3. **Recall@k**

$$\text{Recall@k} = \frac{\text{Number of times the correct code is in the top } k \text{ predictions}}{\text{Total number of samples}} \tag{4.3}$$

   This metric indicates the fraction of cases in which the *true* codes appear among the top k predictions, a setting relevant in *coding-assistance systems* that present a limited list of suggestions.

4. **Exact Match Ratio**

$$\text{Exact Match} = \frac{\text{Number of notes for which the full set of codes is correctly predicted}}{\text{Total number of notes}} \tag{4.4}$$

   This stricter criterion gauges the extent to which all predicted codes align *perfectly* with the ground truth set for each note.

5. **Never-Predicted Rate**

$$\text{Never-Predicted Rate} = \frac{\text{Number of codes that are never predicted by the model}}{C} \tag{4.5}$$

   This measure highlights whether any codes—particularly *rare ones*—are completely ignored.

Emphasis will be placed on determining whether the *ontology-based grouping* and *synthetic data* augmentations produce measurable gains in macro-F1 and reduce the *never-predicted rate*. A targeted error analysis will be conducted for codes that remain *unrecalled*, providing directions for further refinements.

4.6 PROCESSING MIMIC-IV AS OUR MAIN TEST DATASET

In this work, the MIMIC-IV database will be employed as the principal test dataset for automated ICD coding. This dataset contains de-identified clinical information collected from the Beth Israel Deaconess Medical Center (BIDMC), with particular emphasis on structured hospital records and unstructured discharge summaries. The following steps describe how the data will be prepared, managed, and validated in a manner consistent with privacy regulations and best practices in clinical data science.

4.6.1 Data Extraction and Integration

The relevant tables from MIMIC-IV (e.g., `admissions`, `patients`, `diagnoses_icd`, and `discharge`) will be identified and retrieved. These tables will then be joined on keys such as `subject_id` and `hadm_id`, thus creating a unified view of patient admissions, annotated diagnoses, and the corresponding discharge narratives. During this process, all admissions lacking a corresponding discharge summary or valid ICD-10 code assignments will be flagged for potential exclusion, ensuring that the resulting dataset remains suitable for supervised learning.

4.6.2 Text Standardization and Cleaning

Each discharge summary will be examined to remove any remaining placeholders or noise introduced by the de-identification process. Lowercasing and tokenization will be conducted if deemed beneficial for subsequent analyses, and the presence of medical abbreviations or numeric identifiers will be noted. Segmentation into sections (such as “History of Present Illness” and “Medications”) may be carried out to reduce text complexity, though the entire narrative will remain intact if full-length modeling is desired.

4.6.3 Handling Duplicates and Missing Data

Instances in which multiple, nearly identical discharge summaries exist for a single admission will be managed by detecting duplicates through textual similarity checks. These checks will reduce the risk of overrepresenting particular patients or hospital stays. Any admissions without sufficiently long or clearly documented notes will be handled through either imputation strategies or careful removal, thereby minimizing noise in model training and evaluation.

4.6.4 Code Distribution and Rare Labels

A preliminary examination of the ICD-10 codes assigned to each admission will be performed to understand their frequency distribution. Since ICD codes often follow a long-tail pattern, additional attention will be devoted to codes that appear infrequently. If these rare labels contribute significantly to the overall error rate, strategies such as targeted oversampling or retrieval-augmented data enrichment may be considered, provided such techniques respect the constraints of de-identification and data plausibility.

4.6.5 Database Management and Version Control

All data transformations and table joins will be executed within a controlled environment (e.g., DuckDB) to allow for efficient querying and seamless integration with Python-based analytic workflows. The resulting datasets, along with any derived tables, will be tracked using version control to ensure that incremental changes are systematically logged and reproducible. This practice will enable transparent experimentation and simplify the process of rolling back to earlier dataset states if necessary.

4.6.6 Security and Compliance

All patient information in MIMIC-IV is protected via de-identification, and no direct identifiers (e.g., names, addresses) are present. It will be confirmed that the data use agreement from PhysioNet is upheld, particularly regarding re-identification risks and data sharing restrictions. Any synthetic expansions or textual outputs generated by the pipeline will be carefully reviewed to ensure that confidential details are not reintroduced.

CHAPTER 5

DATASET UNDERSTANDING

5.1 OVERVIEW OF MIMIC-IV

MIMIC-IV is a large, openly accessible database of de-identified electronic health records (EHR), developed through a collaboration between the Beth Israel Deaconess Medical Center (BIDMC) and the Massachusetts Institute of Technology (MIT). It spans clinical encounters from 2008 to 2019 and contains comprehensive data, including demographic information, laboratory values, prescriptions, billing codes, clinical notes, and more. The dataset is organized into multiple modules to reflect different sources or systems:

- `hosp`: Hospital-wide data, containing tables such as `admissions`, `patients`, and `diagnoses_icd`.
- `icu`: Intensive Care Unit (ICU) data with highly granular time-series measurements (e.g., `chartevents`).
- `ed`: Emergency department records.
- `note`: De-identified clinical notes, such as discharge summaries.
- `cxr`: Information linking chest X-ray images to the relevant patients in MIMIC-CXR.

For the purpose of analyzing discharge summaries and automatically assigning ICD codes, particular emphasis is placed on:

- `patients` in the *hosp* module, which provides demographics such as `anchor_age` and `anchor_year_group`.
- `admissions` in the *hosp* module, which contains `hadm_id` (hospital admission ID) and timestamps (`admittime`, `dischtime`).
- `diagnoses_icd` in the *hosp* module, indicating both ICD-9 and ICD-10 billed diagnoses.
- `discharge` in the *note* module, offering the textual discharge summaries that are

crucial for modeling.

All queries used to gather counts, distributions, and other information about the MIMIC-IV dataset are provided in the Appendix (Chapter A). In the following sections, key outcomes from these queries are summarized, and visual explorations are presented.

5.2 SQL QUERY RESULTS AND DATA HIGHLIGHTS

5.2.1 Patient and Admission Volumes

Using the queries referenced in Figures A.1 and A.2 in the Appendix, it was determined that there are **364,627 unique patients** in the `patients` table. In the `admissions` table, there are **546,028 unique admissions**.

5.2.2 Multiple Admissions per Patient

Figure A.3 in the Appendix provides the SQL code used to examine how many admissions each patient has. Some patients experience over 100 admissions, highlighting that MIMIC-IV can capture highly recurrent hospital visits for certain individuals.

5.2.3 Discharge Summaries

By referencing Figure A.4 in the Appendix, it was found that there are **331,793** discharge summaries (`discharge` table). Nearly 214,296 admissions in MIMIC-IV do not have a final discharge note (see Figure A.5), underscoring that not every hospital stay yields a lengthy textual record.

5.2.4 Characterizing Note Length

The query in Figure A.6 computed word-count statistics for discharge summaries. The median note length is approximately 1,556 words, and at the 99th percentile, notes can exceed about 3,986 words. This wide variation in length indicates that some admissions have minimal documentation, whereas others include very detailed records.

5.2.5 ICD-9 vs. ICD-10 Utilization

Figure A.7 in the Appendix confirms that MIMIC-IV contains approximately 3.46 million ICD-10 code assignments and 2.91 million ICD-9 code assignments. This distribution suggests that modern coding practices (ICD-10) have become more prevalent. Detailed counts of top ICD-10 codes were retrieved using the query in Figure A.8, which revealed chronic conditions like hyperlipidemia (`E785`) and hypertension (`I10`) as highly common.

5.2.6 Long-Tailed Code Distribution

A query shown in Figure A.9 was used to confirm that there are **19,440 unique ICD-10 codes** in the dataset. As observed through the query in Figure A.10, around 9,217 ICD-10 codes appear fewer than five times, illustrating the classic long-tail distribution. Models must be robust to rare code occurrences.

5.2.7 Codes Per Admission

The query in Figure A.11 determined that there are on average ~ 13.6 ICD-10 codes billed per admission. Consequently, this setting represents a multi-label classification task: each discharge summary can be associated with numerous codes rather than a single label.

5.3 VISUAL EXPLORATIONS OF MIMIC-IV

Plot 1: Distribution of Anchor Age

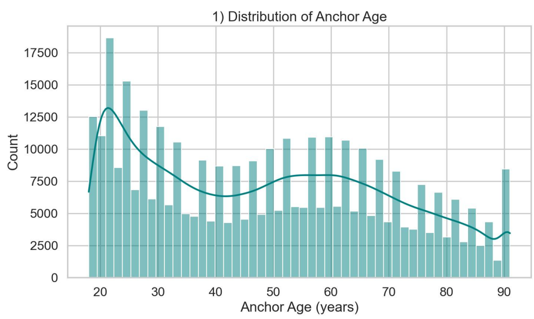

Figure 5.1: A histogram revealing a right-skewed distribution of anchor ages, with many individuals in their 20s and 30s.

Figure 5.1 shows how anchor ages are concentrated in younger to middle-aged adults, with a tail that extends to the maximum capped age of 91. There is a gradual decrease in frequency after middle age, but a mild uptick near 90–91, reflecting censoring of ages above 89.

Plot 2: Boxplot of Anchor Age by Admission Type

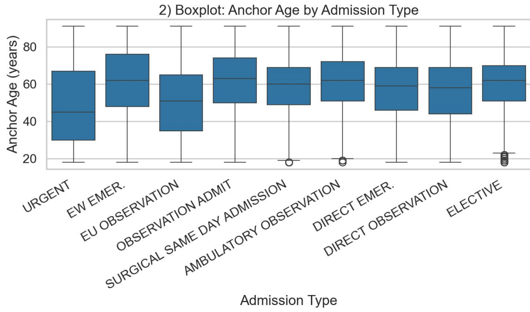

Figure 5.2: A boxplot illustrating how anchor age varies across different admission types.

Figure 5.2 offers insight into differences in age distributions for various hospital admission pathways (e.g., emergency admissions vs. elective surgeries). Some admission types span a broader age range, while others are more narrowly distributed, reflecting the patient populations associated with particular clinical services.

Plot 3: Admissions by Anchor Year Group

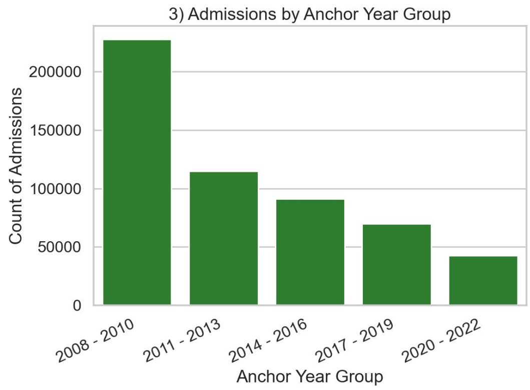

Figure 5.3: A bar chart showing the number of admissions categorized by approximate calendar year groups.

Figure 5.3 demonstrates a higher number of admissions in the earlier year intervals (2008–2010, 2011–2013) relative to later groups, likely reflecting the data collection periods and any changes in hospital practices over time.

Plot 4: Top 15 ICD-10 Codes by Frequency

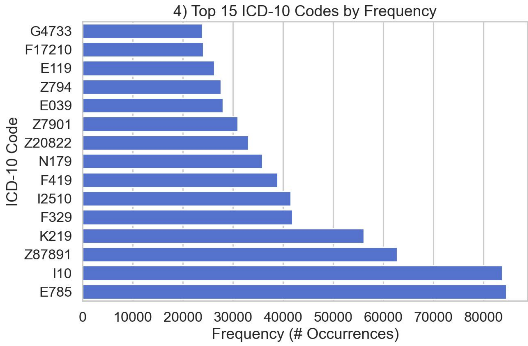

Figure 5.4: A horizontal bar chart showing the most common ICD-10 codes among admissions.

In Figure 5.4, chronic conditions such as hyperlipidemia (`E785`) and hypertension (`I10`) are clearly visible at the top of the frequency distribution, demonstrating the widespread prevalence of these diagnoses.

Plot 5: Co-Occurrence Matrix of Top 15 ICD-10 Codes

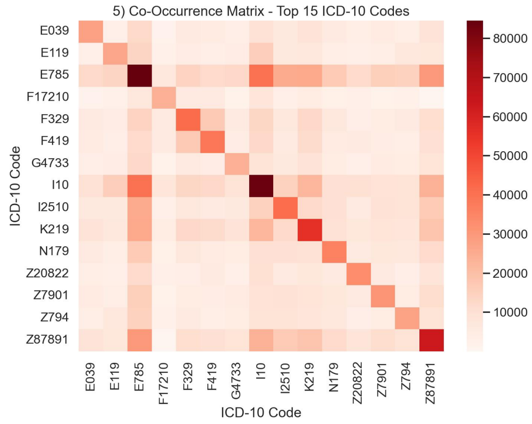

Figure 5.5: A heatmap illustrating how frequently pairs of the top 15 ICD-10 codes co-occur within the same admission.

Figure 5.5 reveals strong comorbidity patterns among high-frequency diagnoses. For instance, codes associated with hyperlipidemia often co-occur alongside hypertension, reflecting typical chronic disease clusters.

Plot 6: Discharge Note Length vs. Number of ICD-10 Codes

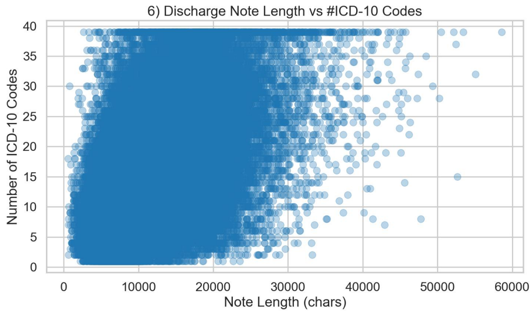

Figure 5.6: A scatter plot highlighting the relationship between note length (in characters) and the number of ICD-10 codes.

Figure 5.6 demonstrates a positive correlation: admissions with more diagnoses often generate longer discharge summaries. However, many cases are concentrated below 20,000 characters, indicating that moderately sized notes can still have multiple codes.

Plot 7: Violin Plot of Code Counts by Anchor Year Group

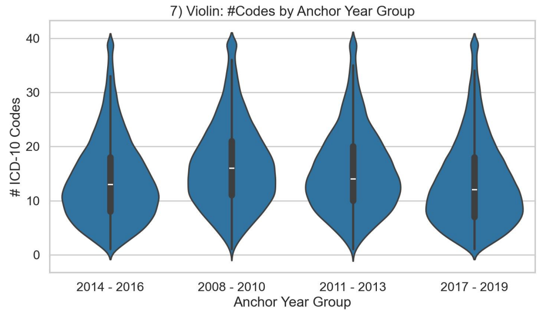

Figure 5.7: A violin plot representing how the distribution of total ICD-10 codes per admission varies across anchor year groups.

Figure 5.7 visualizes the spread of ICD-10 code counts for each time interval. Wide tails indicate that certain admissions receive a very large number of codes, reflecting complex clinical circumstances.

Plot 8: Mean Discharge Note Length by Anchor Year Group

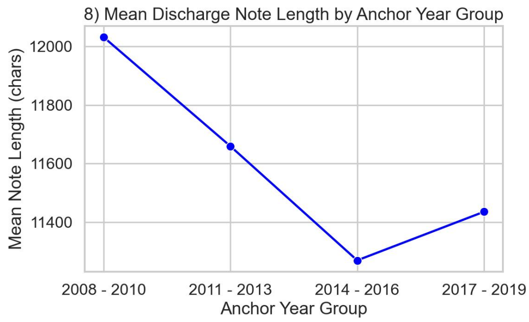

Figure 5.8: A line plot tracking average discharge summary lengths over five anchor year group intervals.

Figure 5.8 suggests that longer average notes were generated during the earlier collection periods (2008–2010), followed by a dip in 2014–2016 and a partial rebound in later intervals. Evolving clinical documentation practices may contribute to these shifts.

Plot 9: Top 10 Admission Types by Volume

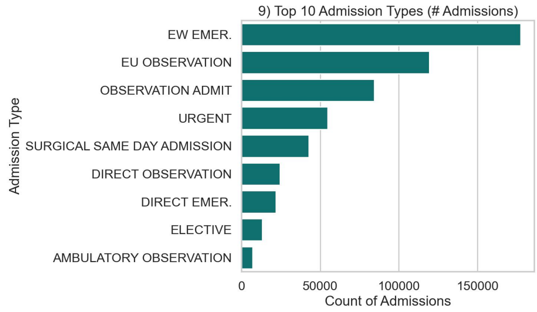

Figure 5.9: A horizontal bar chart showing the top ten admission types (e.g., emergency, elective, urgent).

Figure 5.9 highlights the prominence of emergency or unplanned admissions. Emergency admissions dominate, but there are also notable volumes in scheduled categories such as elective surgeries.

Plot 10: ICD-10 Code Frequency by Anchor Year Group (Top 15 Codes)

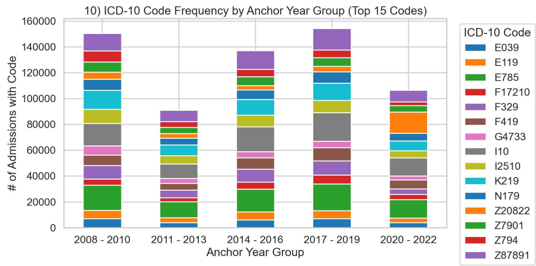

Figure 5.10: A stacked bar chart displaying the occurrence of each top-15 ICD-10 code within different anchor year groups.

Figure 5.10 shows that certain diagnoses remain consistently frequent across all anchor year intervals, again reflecting the prevalence of long-standing chronic conditions such as metabolic disorders and cardiovascular disease.

5.4 KEY INSIGHTS

From these queries and visualizations, several points emerge:

- **Large-Scale Data:** There are more than 364k unique patients and over 546k admissions, providing robust coverage of a hospital population.
- **Multiple Diagnoses:** The average admission carries more than 13 ICD-10 codes, confirming a multi-label setting.
- **Long-Tail Challenge:** Over 9,217 ICD-10 codes appear fewer than five times, highlighting difficulties with rare or specialized diagnoses.
- **Varied Note Lengths:** Discharge summaries range from brief texts to lengthy narratives with thousands of words.

5.5 FINAL DATA PREPARATION

For the goal of building a text-to-ICD coding model (akin to AWS Comprehend Medical or other automated medical coding applications), a consolidated table was created that merges each admission's discharge note with the set of associated ICD-10 codes. The approach takes each row in the `discharge` table, then joins it with the corresponding entries in `diagnoses_icd` for that `hadm_id`.

The query used to accomplish this is shown in Figure A.12 in the Appendix. Each row in the resulting table contains:

- **notes**: The original discharge summary (raw text).
- **icd_codes**: A comma-separated list of all ICD-10 codes assigned to that admission.

This design directly supports multi-label classification, as the full narrative of the discharge summary is provided (with no additional demographic or structural metadata) alongside all billed ICD-10 codes. Such a format simulates real-world scenarios in which only free-text documentation is available for downstream coding or NLP tasks.

CHAPTER 6

RESULTS

We start with the outcomes of two preliminary investigations conducted to establish initial benchmarks and insights for automated ICD coding on the MIMIC-IV dataset. A simple *K-Nearest Neighbors (KNN)* model was first implemented to observe how a straightforward algorithm manages the complexity of clinical text classification. Subsequently, a *MinHash-based* similarity detection method was employed to reveal instances of near-duplicate discharge summaries and to estimate potential redundancy in the dataset. By examining these findings through a first-principles lens, the underlying causes—*data imbalance, text intricacy, and recurring clinical narratives*—emerge clearly, along with their effects on model performance and dataset quality.

6.1 BASELINE MODEL: K-NEAREST NEIGHBORS (KNN)

A KNN classifier was trained on discharge summaries that had been transformed into a TF-IDF vector space (limited to 5,000 features and incorporating both unigrams and bigrams). The `MultiLabelBinarizer` function was employed to accommodate the assignment of multiple ICD codes to each note. By design, KNN relies on calculating distances in the feature space to assign labels. However, due to the *intrinsic imbalance of ICD codes* and the *high-dimensional nature of clinical narratives*, the model's ability to generalize was substantially constrained. This constraint directly resulted in a *Macro F1-Score of 0.12*, revealing that frequent codes were mostly captured, whereas rare codes were systematically overlooked. The pronounced Hamming Loss further illustrated that the inherent sparsity and complexity of clinical text pose substantial challenges to naive distance-based approaches.

6.2 MINHASH-BASED SIMILARITY DETECTION

A MinHash-based technique, coupled with *Locality Sensitive Hashing (LSH)*, was subsequently introduced to identify closely matching or duplicate discharge summaries. This approach was motivated by the suspicion that repeated clinical narratives could bias model training and inflate performance metrics without truly advancing model robustness. The text was preprocessed through *lowercasing, punctuation removal, and stopword filtration*, after which MinHash signatures were generated. These signatures were then indexed in an LSH structure to expedite the retrieval of pairs whose Jaccard similarity exceeded a user-defined threshold (e.g., 0.90).

As a cause, the repetition of routine sections within clinical notes (such as standardized instructions or procedural descriptions) led to the high similarity scores. *In effect*, these observations underscore the necessity for careful dataset curation, including the removal or consolidation of near-duplicate entries. Moreover, the identified pairs highlight potential avenues for validating the realism of *synthetic discharge notes*, since generated samples can be compared to authentic records through the same MinHash-based procedure.

6.3 KEY OBSERVATIONS AND CHALLENGES

- *Data Imbalance*: The imbalance in ICD codes—where common diagnoses overshadow rarer ones—was a major factor contributing to the low Macro F1-Score of the KNN baseline.

- *Complex Clinical Texts*: Clinical narratives often include technical jargon, abbreviations, and inconsistent structuring, which strains models that rely on straightforward vector distances.

- *Potential Data Redundancy*: The MinHash experiment exposed considerable text overlap, suggesting that a portion of the dataset may not add genuinely novel information for model training.

Taken together, these findings confirm that simplistic algorithms and untreated redundancies limit the performance and reliability of automated ICD coding. It is therefore anticipated that refined methods—especially those involving *transformer-based architectures* and *knowledge-guided data augmentation*—will prove crucial for addressing rare codes, reducing the impact of repetitive content, and enhancing interpretability.

6.4 RARE DISEASE CO-OCCURRENCE PIPELINE

We developed an automated pipeline that extracts rare disease concepts from the ORDO ontology and maps them to SNOMED CT identifiers. Leveraging SNOMED CT concept and relationship files, a disease knowledge graph was built to capture clinically relevant associations between diseases. This graph was then traversed—using key relationships such as `is-a`, `associated with`, and `due to`—to identify groups of co-occurring conditions. Finally, SNOMED CT identifiers were translated to ICD-10-CM codes using a mapping file.

For instance, one generated group is:

Group 1: ['Q00.0', 'I48.91', 'J45.909', 'I25.10']

Here, the rare disease with code `Q00.0` corresponds to **Marfan syndrome**, while the common conditions are identified as **Atrial fibrillation** (`I48.91`), **Asthma** (`J45.909`), and **Coronary artery disease** (`I25.10`). Further work is ongoing on this to extract an an exhaustive set of groups of co-occurring diseases. The next phase report will contain all the final results and the next steps.

CHAPTER 7

CONCLUSION

A substantial foundation has been established through an extensive literature review, a clear problem definition, a detailed methodology, and a thorough understanding of the dataset. Emphasis has been placed on recognizing the complexity and breadth of ICD coding tasks, with particular attention to the underrepresentation of rare conditions. Although progress has been steady, the generation of synthetic data has not yet been executed, and it remains a pivotal step for enhancing the coverage and accuracy of ICD assignments.

Moving forward, the methodology's execution continues to be refined, with plans to employ knowledge-driven data augmentation to address class imbalance and to improve overall model performance. By integrating insights gained from preliminary investigations, a systematic approach to generating synthetic clinical narratives will be employed to bolster recognition of rare codes. The resulting models, once trained and evaluated, are expected to offer valuable contributions to automated ICD coding by delivering enhanced robustness and reliability.

7.1 GANTT CHART OF PLANNED WORK TIMELINE

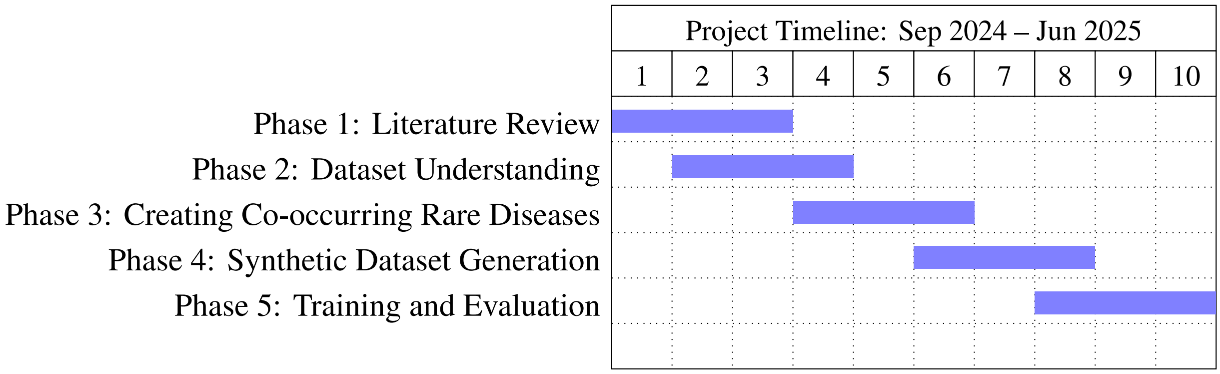

Figure 7.1: Gantt Chart Reflecting Main Phases and Overlaps

APPENDIX A

APPENDIX

A.1 SQL CODE

All SQL queries used throughout Chapter 5 are presented below. Each query is placed within a figure environment for clarity and reference.

```
-- Figure A.1: Counting rows/patients in the 'patients' table
SELECT
    COUNT(*) AS total_rows_in_patients,
    COUNT(DISTINCT subject_id) AS num_unique_patients
FROM mimic.patients;
```

Figure A.1: SQL code to retrieve the total number of rows and distinct patients in the `patients` table.

```
-- Figure A.2: Counting rows/admissions in the 'admissions' table
SELECT
    COUNT(*) AS total_rows_in_admissions,
    COUNT(DISTINCT hadm_id) AS num_unique_admissions
FROM mimic.admissions;
```

Figure A.2: SQL code to retrieve the total number of rows and distinct admissions in the `admissions` table.

```
-- Figure A.3: Identifying multiple admissions per patient

SELECT subject_id, COUNT(*) AS num_admissions

FROM mimic.admissions

GROUP BY subject_id

HAVING COUNT(*) > 1

ORDER BY num_admissions DESC

LIMIT 5;
```

Figure A.3: SQL code to find patients with more than one admission, illustrating recurrent hospital stays.

```
-- Figure A.4: Counting total discharge notes

SELECT COUNT(*) AS total_discharge_notes

FROM mimic.discharge;
```

Figure A.4: SQL code to count the total number of discharge summaries in the `discharge` table.

```
-- Figure A.5: Admissions with and without discharge notes

SELECT

    (SELECT COUNT(DISTINCT hadm_id) FROM mimic.admissions) AS total_admissions,

    (SELECT COUNT(DISTINCT hadm_id) FROM mimic.discharge)  AS hadm_with_discharge

    (SELECT COUNT(DISTINCT hadm_id) FROM mimic.admissions

     WHERE hadm_id NOT IN (

         SELECT DISTINCT hadm_id FROM mimic.discharge

     )

    ) AS hadm_without_discharge;
```

Figure A.5: SQL code comparing the number of admissions with discharge notes to those without.

```
-- Figure A.6: Word count distribution of discharge summaries
SELECT
    PERCENTILE_CONT(0.5)
        WITHIN GROUP (ORDER BY (LENGTH(text) - LENGTH(REPLACE(text, ' ', '')) + 1
      AS median_word_count,
    PERCENTILE_CONT(0.9)
      WITHIN GROUP (ORDER BY (LENGTH(text) - LENGTH(REPLACE(text, ' ', '')) + 1)
    AS p90_word_count,
    PERCENTILE_CONT(0.99)
    WITHIN GROUP (ORDER BY (LENGTH(text) - LENGTH(REPLACE(text, ' ', '')) + 1))
    AS p99_word_count
FROM mimic.discharge;
```

Figure A.6: SQL code for computing word-count percentiles of the `discharge` table.

```
-- Figure A.7: Comparing total code assignments in ICD-9 vs. ICD-10
SELECT
    icd_version,
    COUNT(*) AS code_assignments
FROM mimic.diagnoses_icd
GROUP BY icd_version
ORDER BY code_assignments DESC;
```

Figure A.7: SQL code to assess the distribution of ICD-9 and ICD-10 usage in the dataset.

```sql
-- Figure A.8: Retrieving top ICD-10 codes by frequency
SELECT icd_code, COUNT(*) AS freq
FROM mimic.diagnoses_icd
WHERE icd_version = 10
GROUP BY icd_code
ORDER BY freq DESC
LIMIT 10;
```

Figure A.8: SQL code used to find the 10 most common ICD-10 codes in MIMIC-IV.

```sql
-- Figure A.9: Counting unique ICD-10 codes
SELECT icd_version,
       COUNT(DISTINCT icd_code) AS num_unique_codes
FROM mimic.diagnoses_icd
GROUP BY icd_version;
```

Figure A.9: SQL code showing the total number of unique ICD codes by ICD version.

```sql
-- Figure A.10: Identifying ICD-10 codes with fewer than five occurrences
WITH code_counts AS (
    SELECT icd_code, COUNT(*) AS freq
    FROM mimic.diagnoses_icd
    WHERE icd_version = 10
    GROUP BY icd_code
)
SELECT COUNT(*) AS codes_under_5_occurrences
FROM code_counts
WHERE freq < 5;
```

Figure A.10: SQL code to quantify rare ICD-10 codes that appear fewer than five times.

```sql
-- Figure A.11: Calculating average ICD-10 codes per admission
WITH code_counts AS (
    SELECT hadm_id,
           COUNT(DISTINCT icd_code) AS code_count
    FROM mimic.diagnoses_icd
    WHERE icd_version = 10
    GROUP BY hadm_id
)
SELECT AVG(code_count) AS avg_icd10_codes_per_adm
FROM code_counts;
```

Figure A.11: SQL code to compute the average number of ICD-10 codes per admission.

```sql
-- Figure A.12: Merging discharge summaries with associated ICD-10 codes
DROP TABLE IF EXISTS mimic.full_notes_with_codes;


CREATE TABLE mimic.full_notes_with_codes AS
SELECT
    d.subject_id,
    d.hadm_id,
    d.text AS notes,
    STRING_AGG(di.icd_code, ',') AS icd_codes
FROM mimic.discharge AS d
JOIN mimic.admissions AS a
  ON d.subject_id = a.subject_id
    AND d.hadm_id = a.hadm_id
JOIN mimic.patients AS p
  ON p.subject_id = d.subject_id
JOIN mimic.diagnoses_icd AS di
  ON di.subject_id = d.subject_id
    AND di.hadm_id = d.hadm_id
WHERE di.icd_version = 10  -- focusing on ICD-10-CM only
GROUP BY
    d.subject_id,
    d.hadm_id,
    d.text;
```

Figure A.12: SQL code to create a consolidated table that pairs each discharge summary with its ICD-10 codes.

APPENDIX B

MINHASH PSEUDO-CODE

```
Algorithm 1: MinHash for Similar Document Detection


Input:
    - A collection of textual discharge summaries (D)
    - Number of permutations (num_perm)
    - Similarity threshold (t)


Output:
    - Set of document pairs whose Jaccard similarity exceeds t


Procedure:
1. Preprocess each document by:
   a) Converting text to lowercase
   b) Removing punctuation and non-alphanumeric characters
   c) Eliminating stopwords
2. Initialize an LSH index with threshold t and num_perm permutations.
3. For each document d in D:
   a) Create a MinHash object m with num_perm permutations
   b) For each token in d, update m using m.update(token.encode('utf8'))
   c) Insert the MinHash signature m into the LSH index
4. For each document d in D:
   a) Query the LSH index with d's signature to obtain similar documents
   b) Record document pairs that exceed the threshold
5. Return all identified pairs of documents with similarity above t
```

Figure B.1: Pseudocode for the MinHash Similarity Detection Process

APPENDIX C

APPENDIX: PSEUDOCODE

C.1 ONTOLOGY-GUIDED SYNTHETIC NOTE GENERATION

Figure C.1: Algorithm 1: Ontology-Guided Synthetic Note Generation

```
Input:
  r: Rare ICD code
  D: Disease cluster from ontology
  L: Language model with retrieval augmentation

Output:
  S: Synthetic discharge summary

Step 1:
  context_docs = RetrieveRelevantText(D, knowledge_source)
Step 2:
  structured_prompt = GenerateStructuredPrompt(D, context_docs)
  S_draft = L.generate(structured_prompt)
Step 3:
  for iteration in 1..MaxIter:
    feedback = CriticEvaluate(S_draft)
    if feedback indicates no major issues:
      break
    S_draft = L.refine(S_draft, feedback)
Step 4:
  if Validate(S_draft):
    S = S_draft
  else:
    S = None
```

C.2 HYBRID TRAINING WITH RARE-AWARE STRATEGIES

Figure C.2: Algorithm 2: Hybrid Training with Rare-Aware Strategies

```
Input:
  RealData = {(x_i, Y_i)}
  SyntheticData = {(x_j*, Y_j*)}
  Model parameters θ

Output:
  Trained model parameters θ

Step 1:
  TrainSet = RealData ∪ SyntheticData
  Initialize θ

Step 2:
  for epoch in 1..max_epochs:
    for batch B in WeightedSampleBatches(TrainSet):
      (X_B, Y_B) = PrepareBatch(B)
      if epoch >= epoch_focal_start:
        loss = FocalLoss(θ, X_B, Y_B) + BCE(θ, X_B, Y_B)
      else:
        loss = WeightedBCE(θ, X_B, Y_B)
      θ = θ - learning_rate * ∇_θ(loss)

Step 3:
  Evaluate on validation set and save best θ based on macro-F1.
```

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